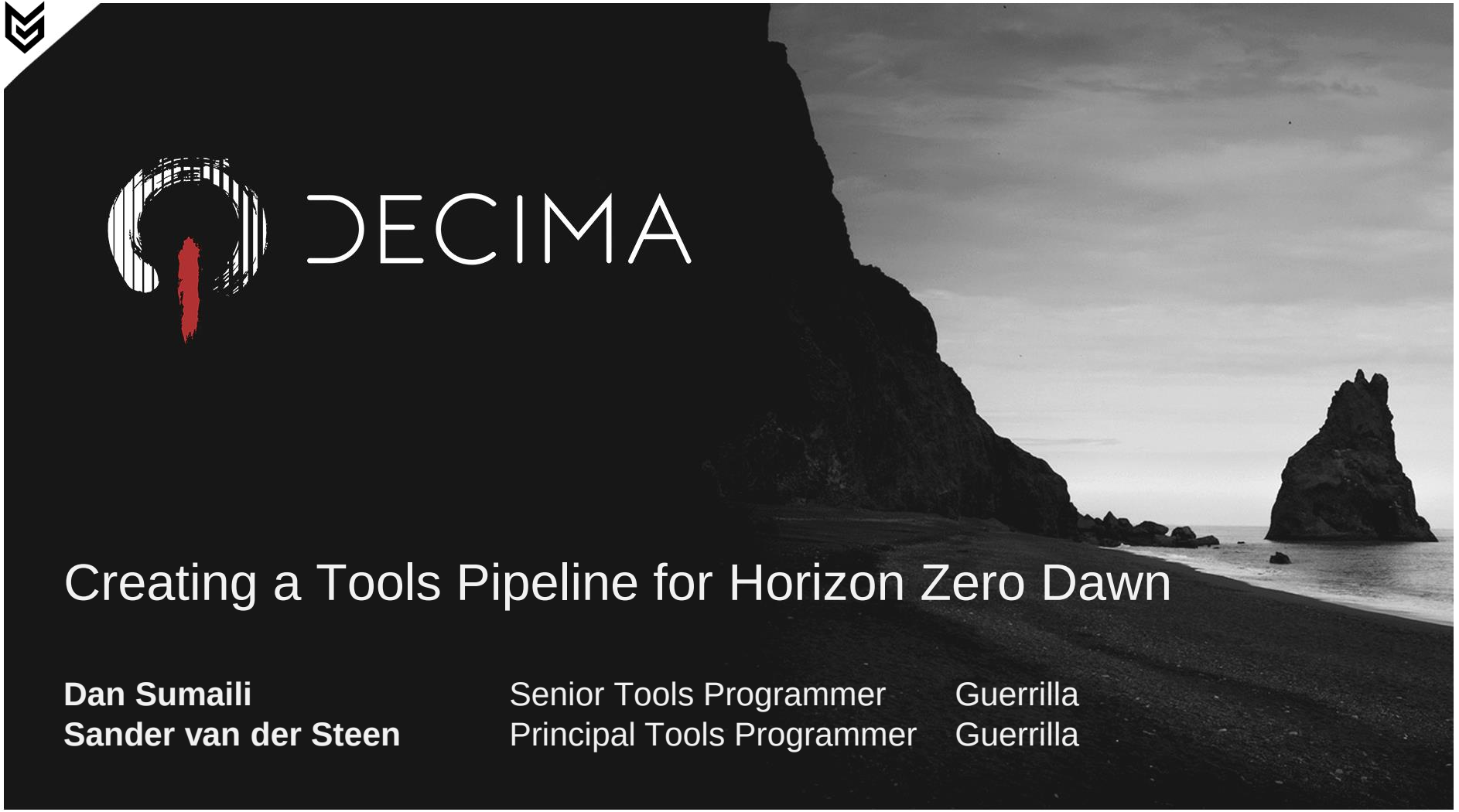
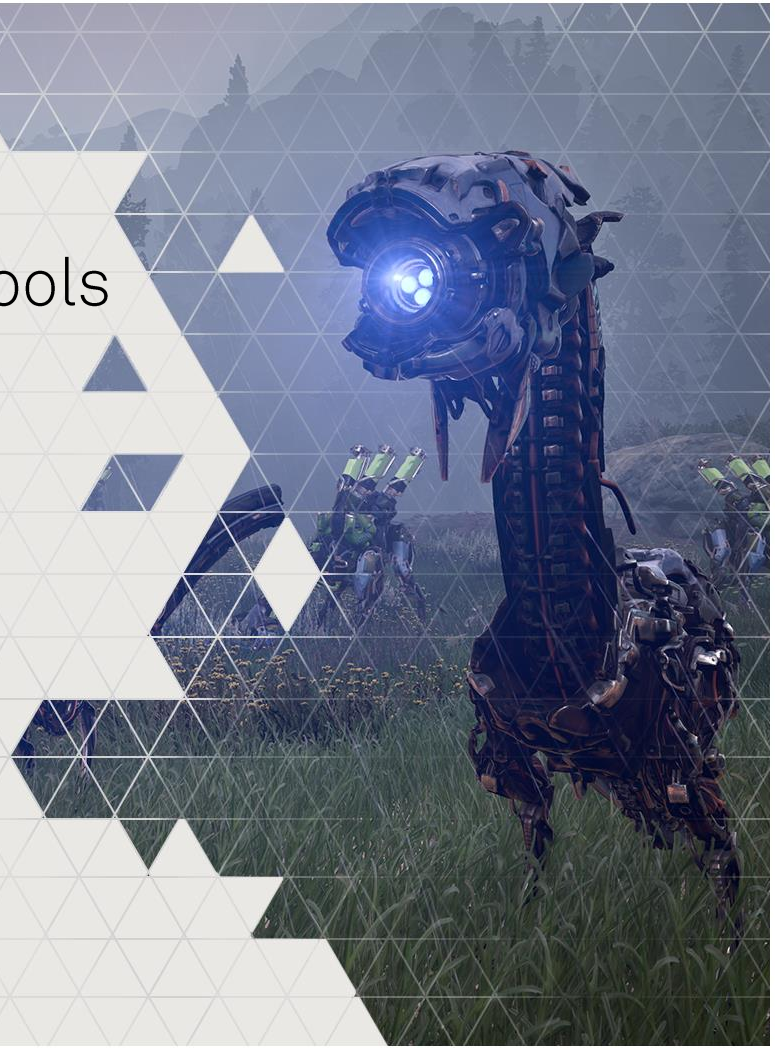



# Creating a Tools Pipeline for Horizon Zero Dawn

**Dan Sumaili** Senior Tools Programmer Guerrilla
**Sander van der Steen** Principal Tools Programmer Guerrilla

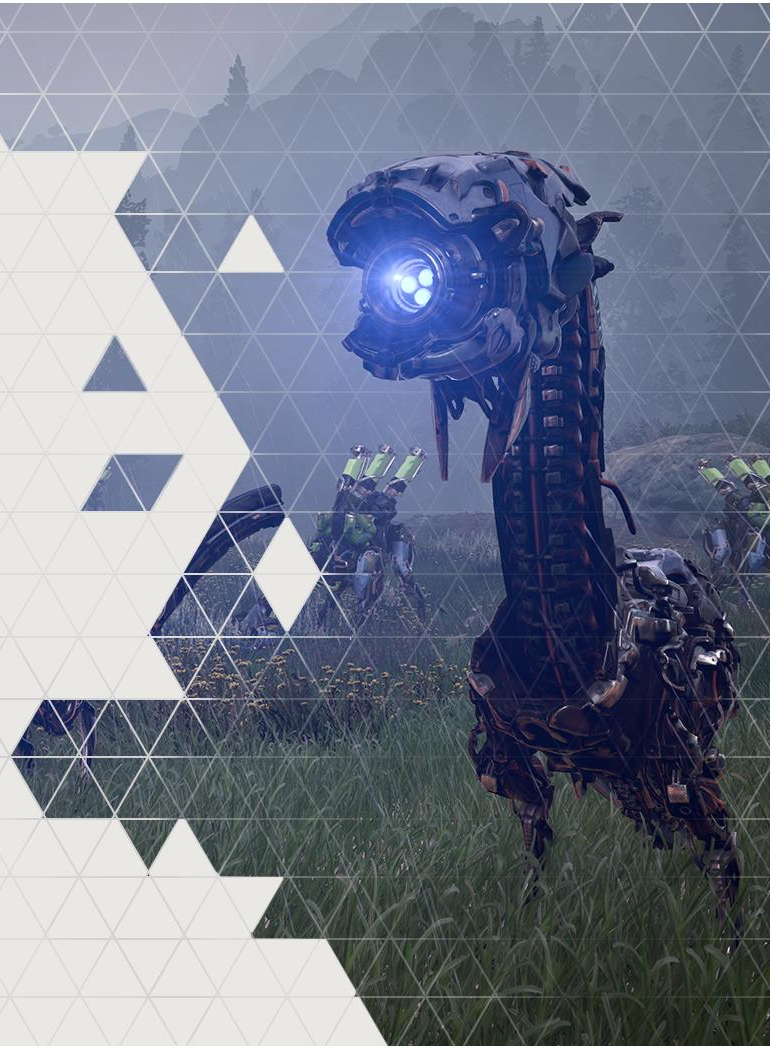
# Takeaways

- Insight into why we rebuilt our tools
- How we did it during production
- How it paid off



# Overview

- Our legacy tools
- Rebuilding the tools
- Building a 3D Editor
- Editing data
- Other tools
- Where we are today



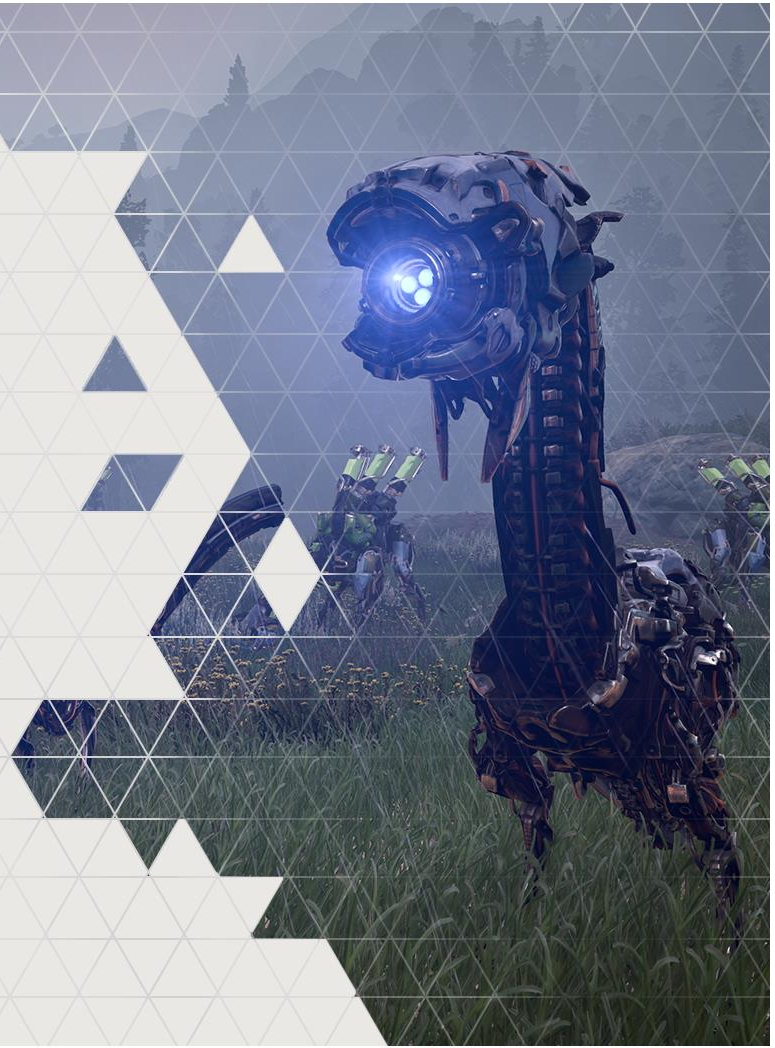
# In the beginning... (Nov 2013)

- We had fragmented tools
  - Mix of languages & frameworks
- Long iteration times (20-30 mins.)
- Dissatisfied Users
  - Far behind our competition

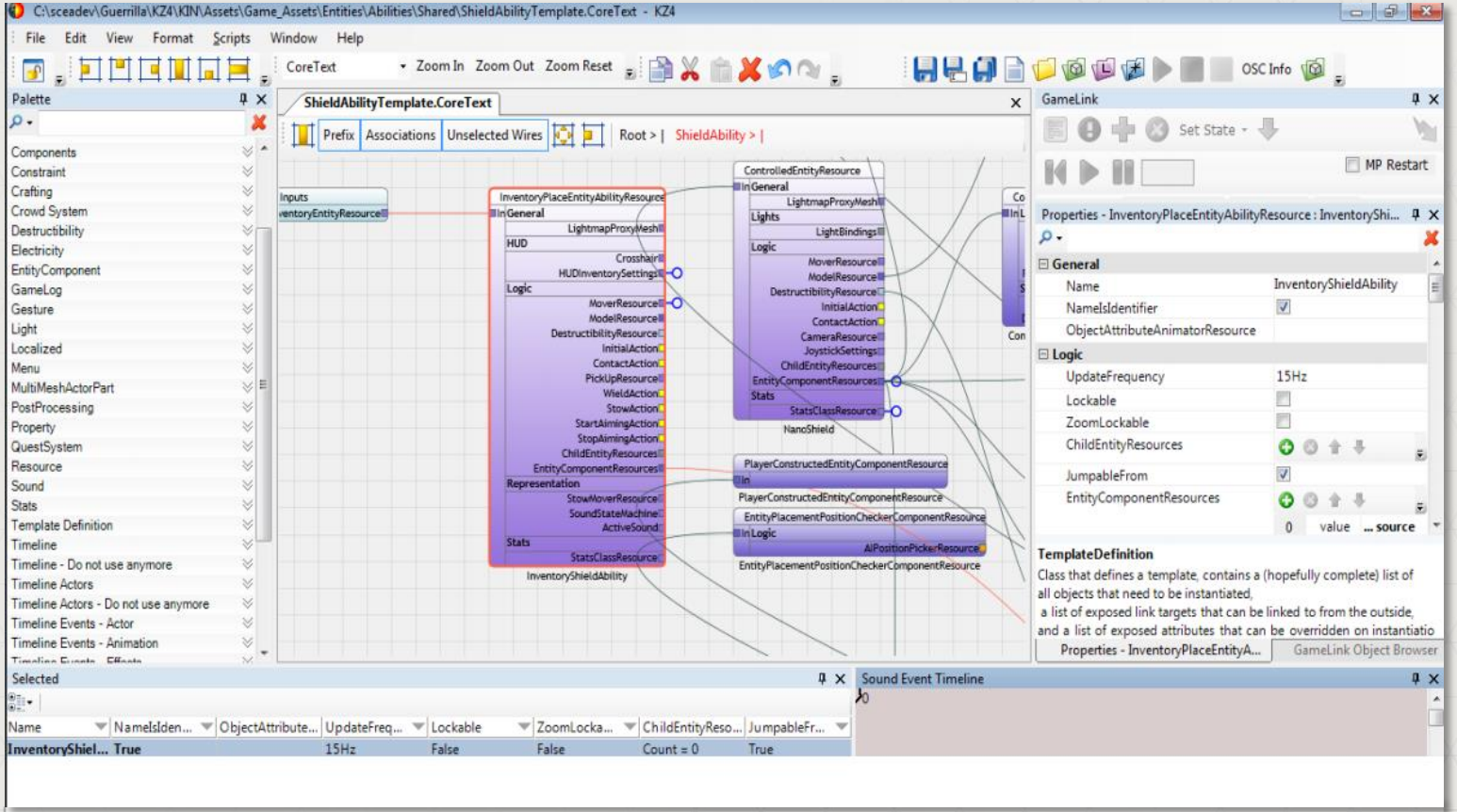


# The CoreText Format

- Intermediate plaintext format
- Used for most data
  - Except Textures and Audio
- **Converted** to binary for game

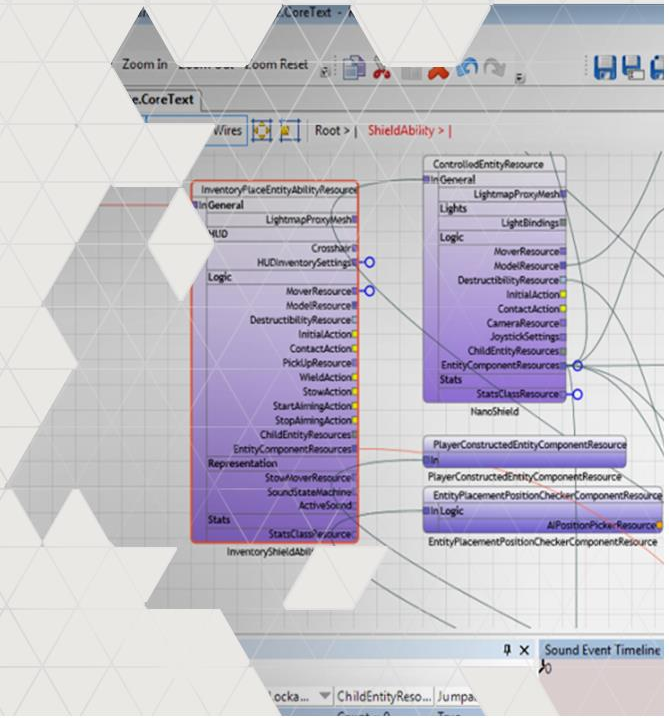


# CoreText Editor

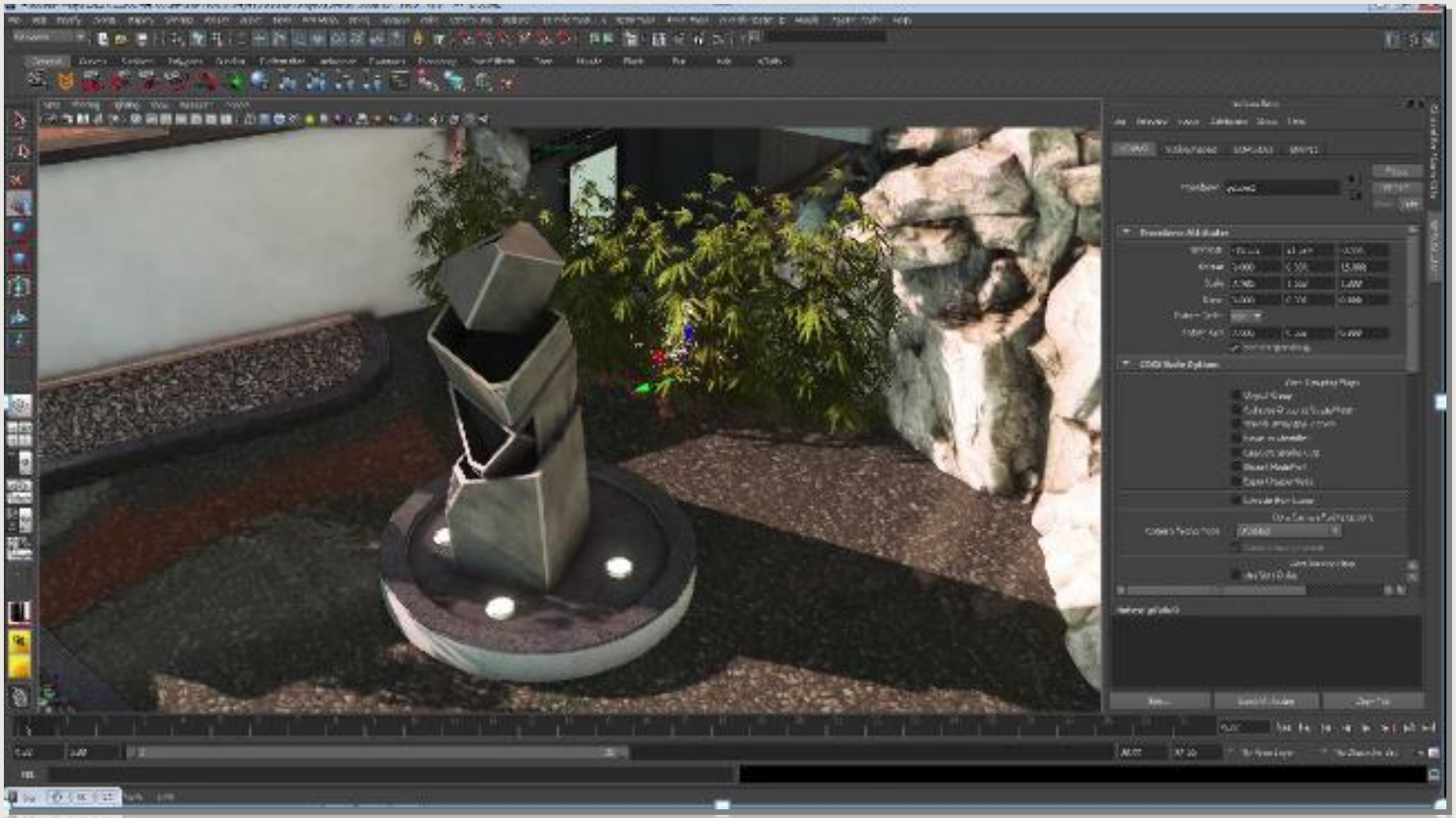


# CoreText Editor

- Used to create & configure game data
- Written in C#, WinForms, ATF
- Absurdly complex code



# Maya Plugin

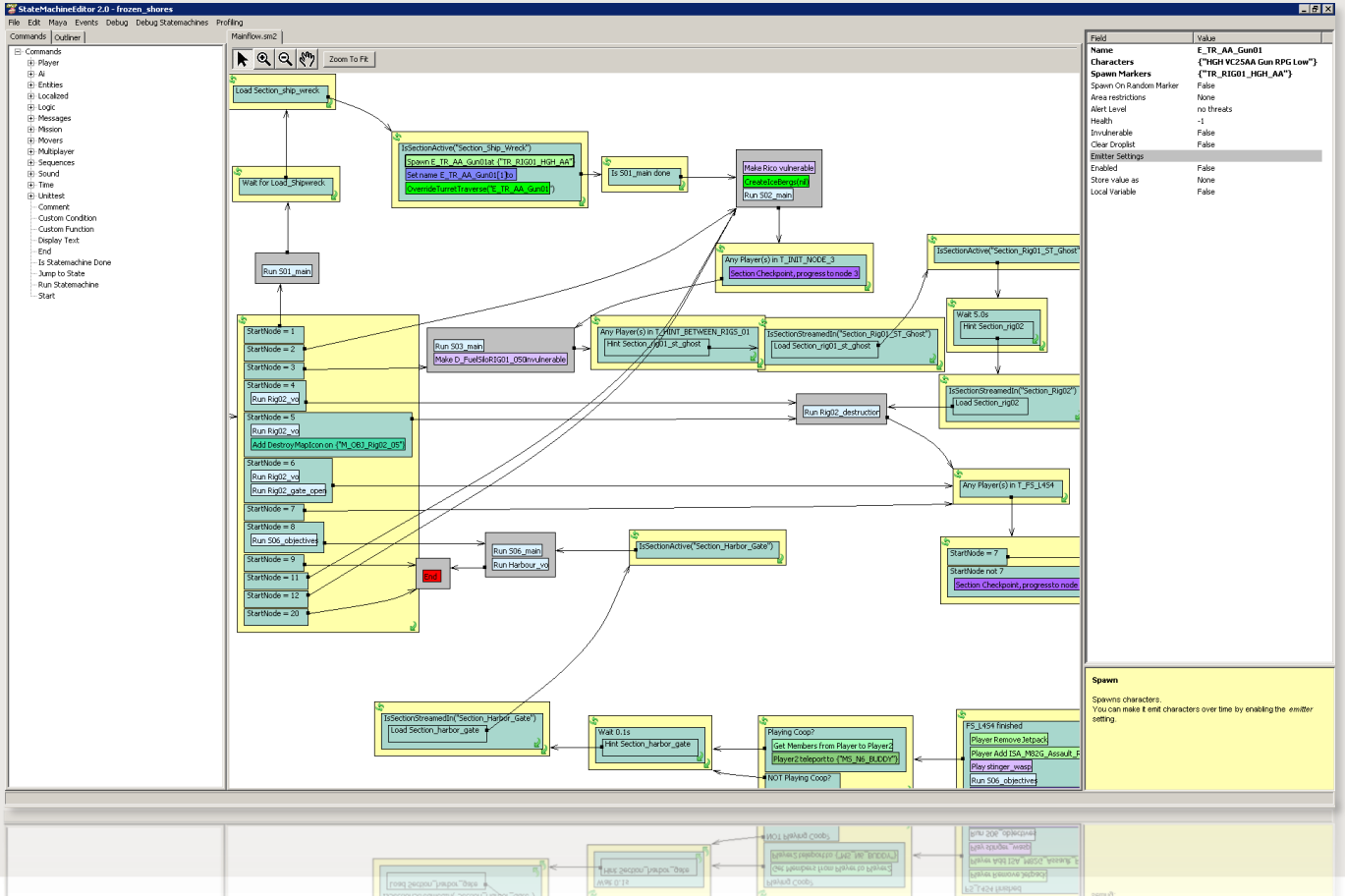


# Maya Plugin

- Exports Maya data to .CoreText
- Used Decima's Renderer
- C++, Python, Maya API
- Killzone Shadow Fall Art Tools
  - http://www.gdcvault.com/play/1020711

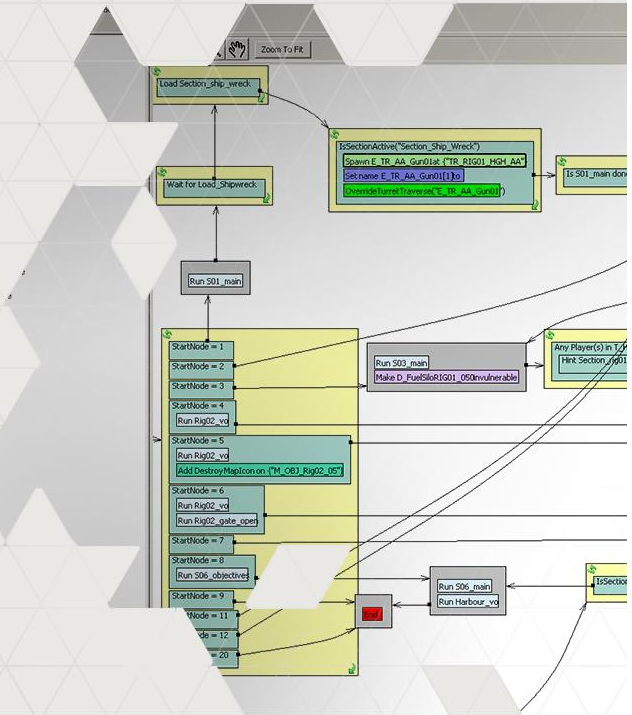


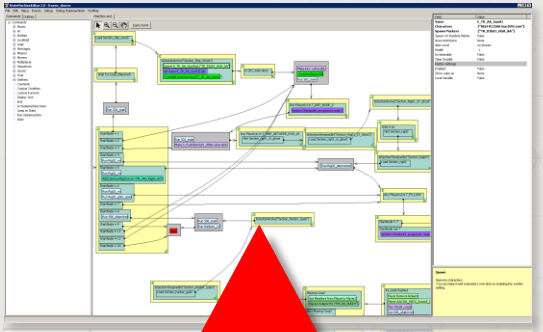
# State Machine Editor – ‘SMED’

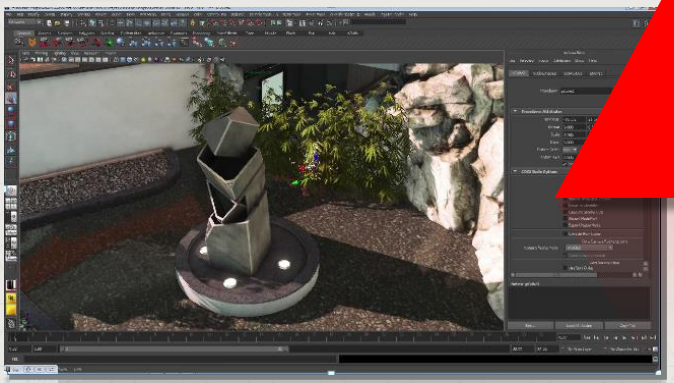


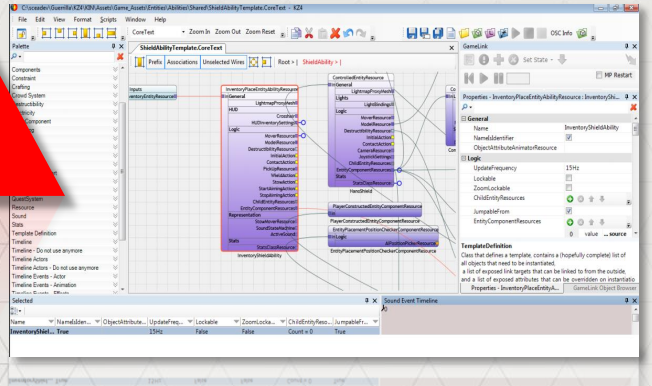
# State Machine Editor – ‘SMED’

- Node-based State Machine editor
- Generates LUA script
- Maintained by a Technical Designer
- Written in Python

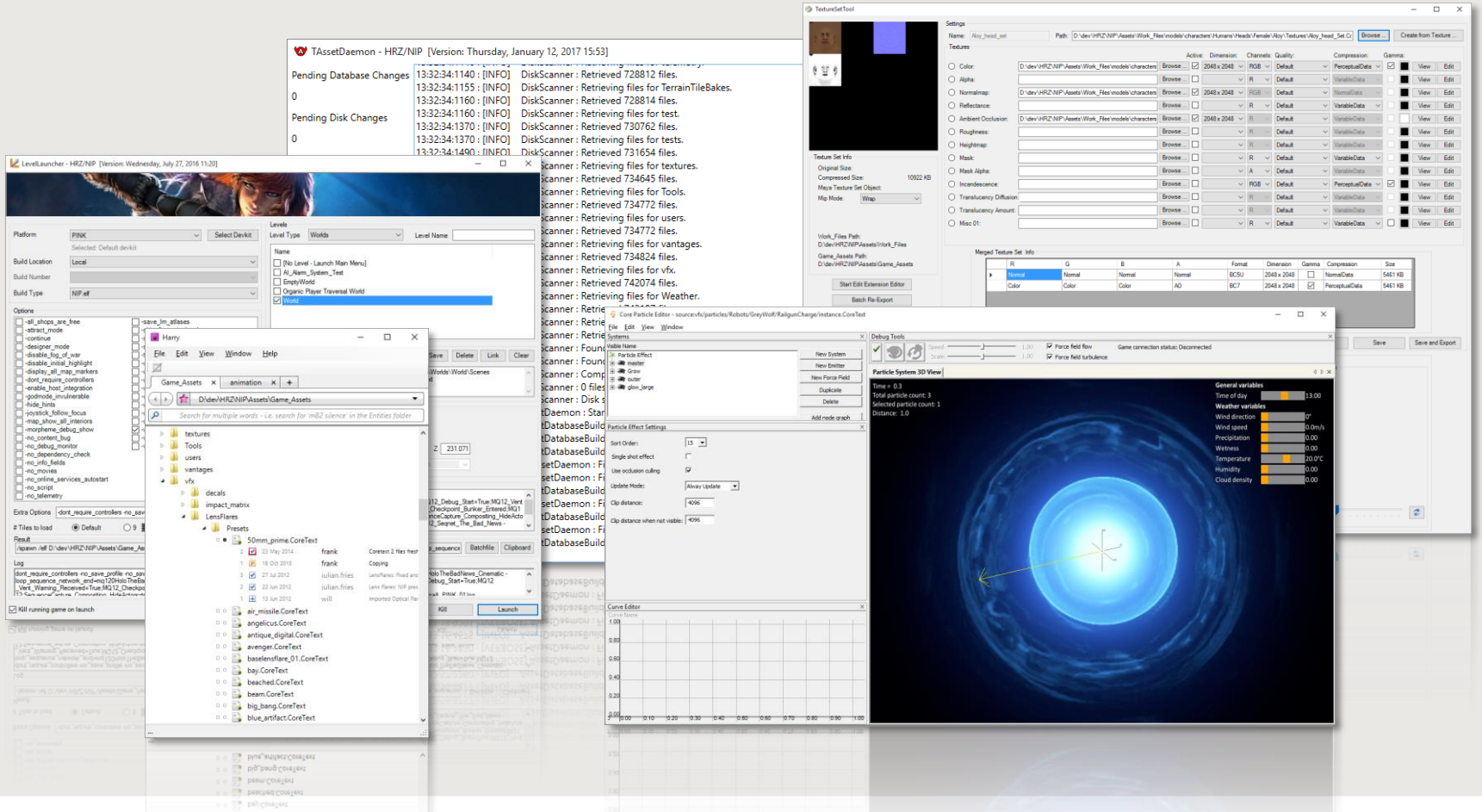






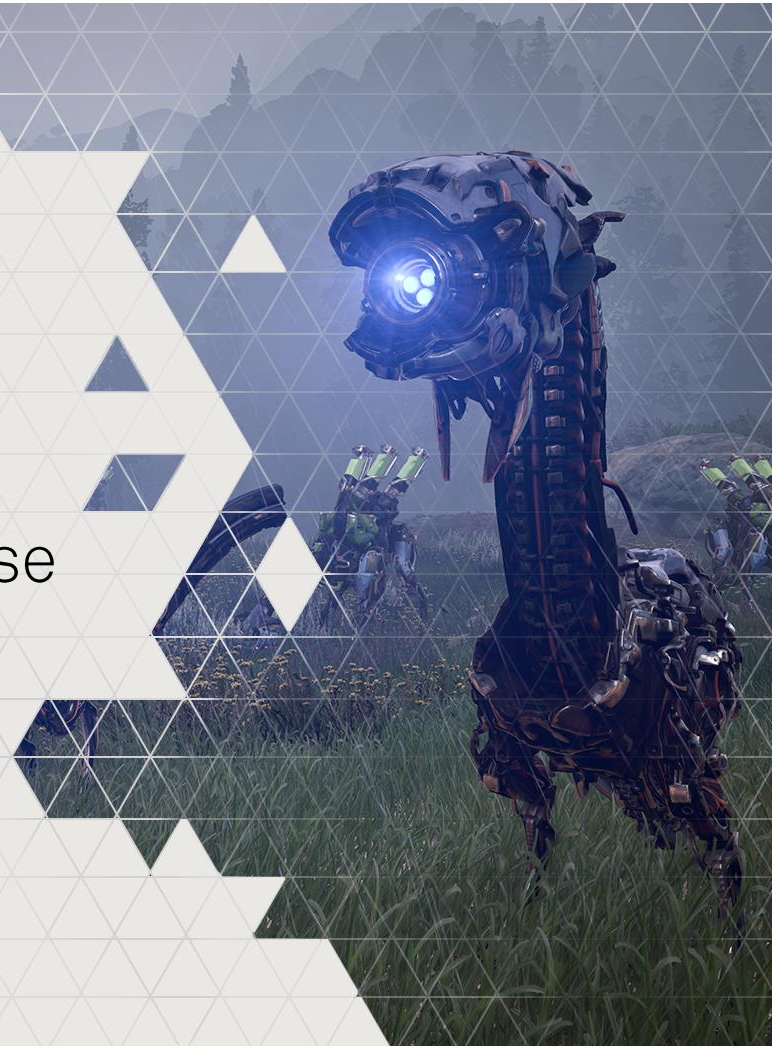


# And many more...



# Debug UI

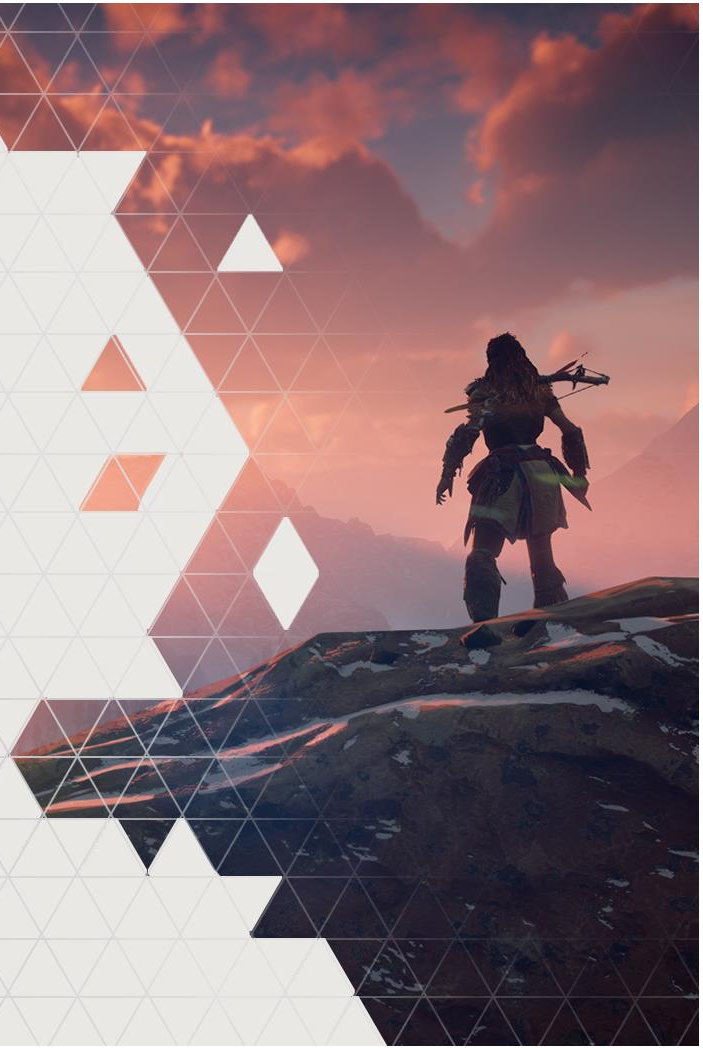
- In-game UI framework
  - Created during **Shadow Fall**
- Programmers found it easy to use
  - Led to many in-game debug tools




Debug UI
Time 353 µs
Pixel count 3,732,315
Selection
Child Packets 1
Time in / outside children 352 µs / 0 µs

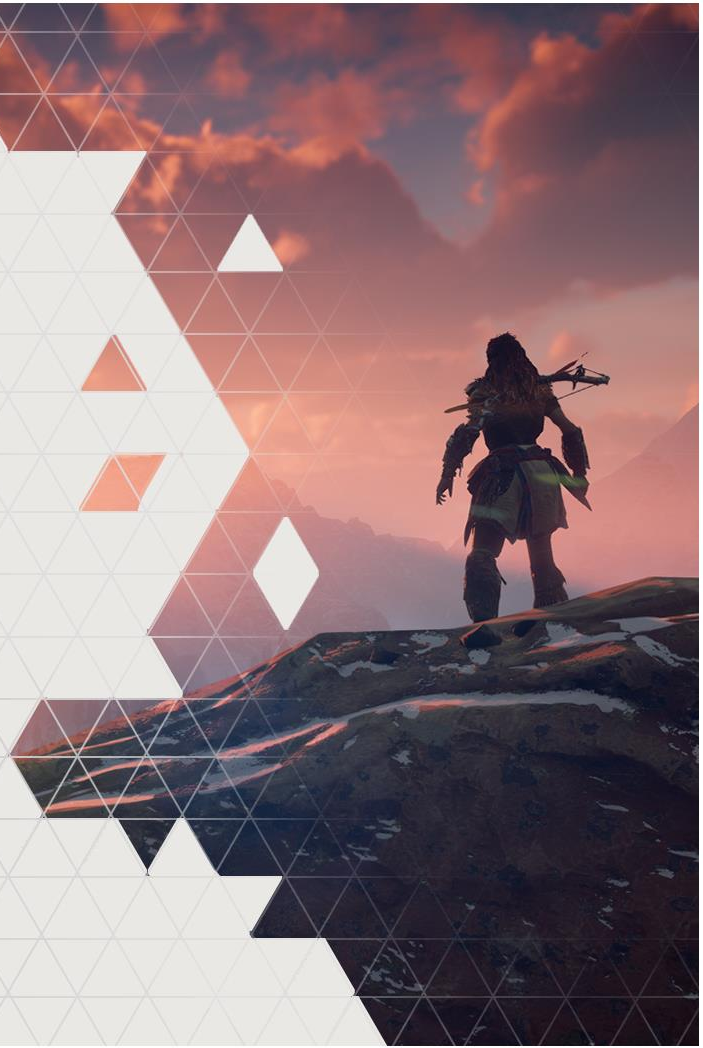
# Horizon Zero Dawn

- Horizon was new for us
- New types of content
- Tools team was small
  - Had to use time and resources efficiently



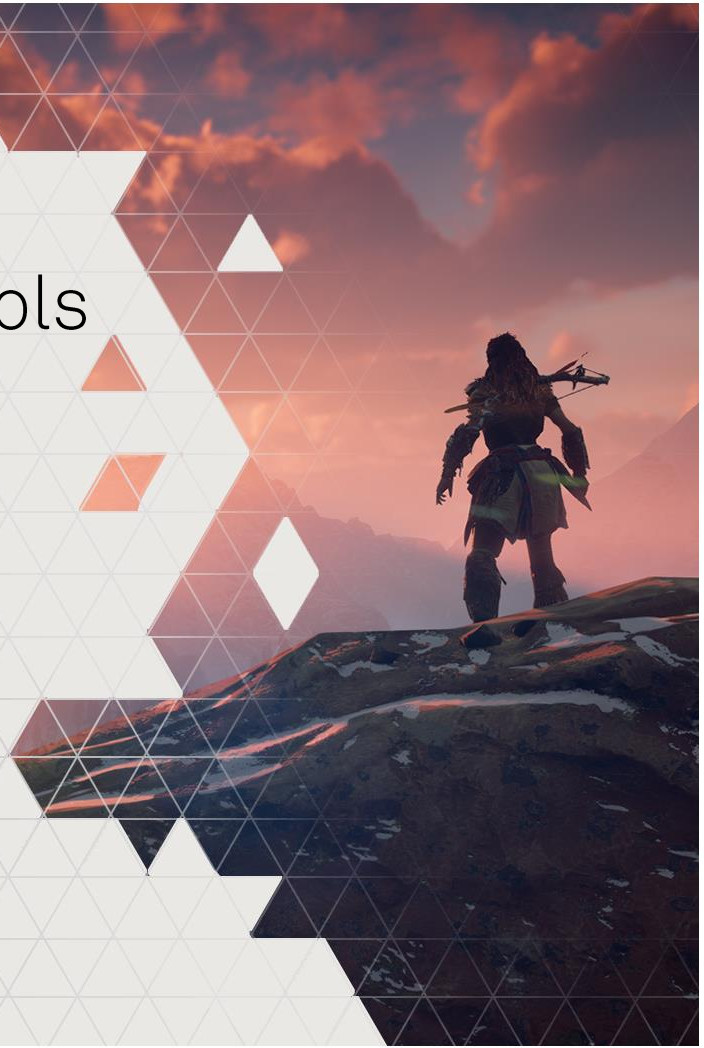
# Horizon Zero Dawn

- The old ways would not scale
- We must begin again…



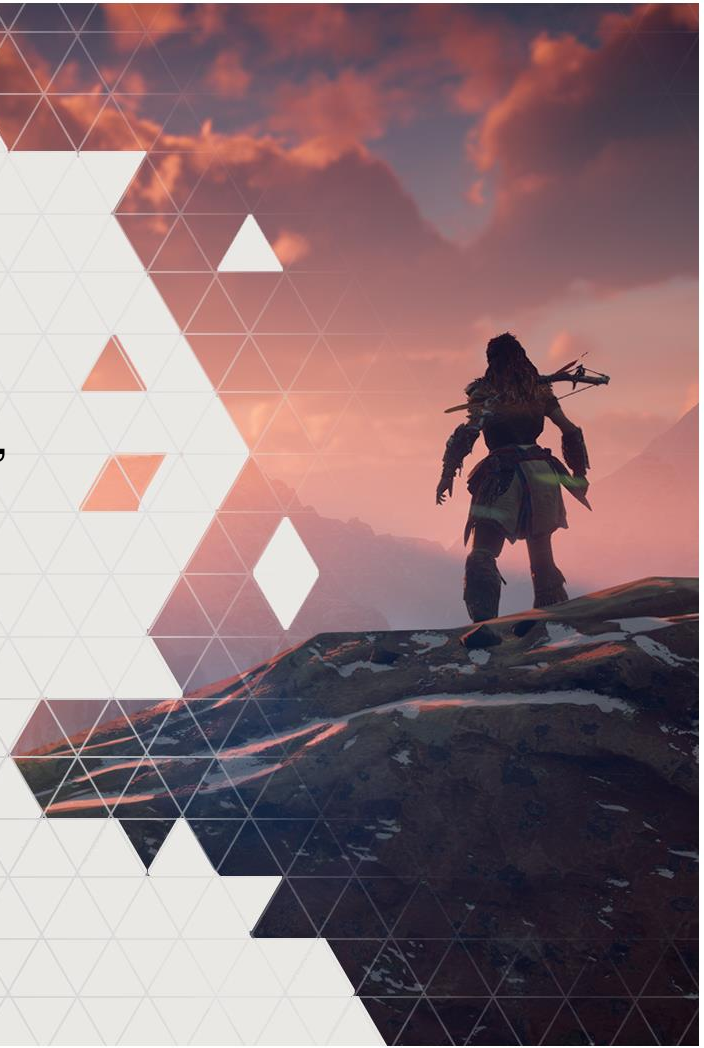
# Tools Goals: Code

- Create a Framework to build Tools
- Re-use **Decima** engine libs
- Making tools should be easy



# Tools Goals: Editor

- Users need tools that fit their work
  - Discipline-specific ‘Editing Contexts’
  - Built from reusable components
- User should be able to stay in **one** Program
  - Decima Editor









DECIMA

# TOOLS FRAMEWORK

## GAME CODE

## CORE

## PIGS

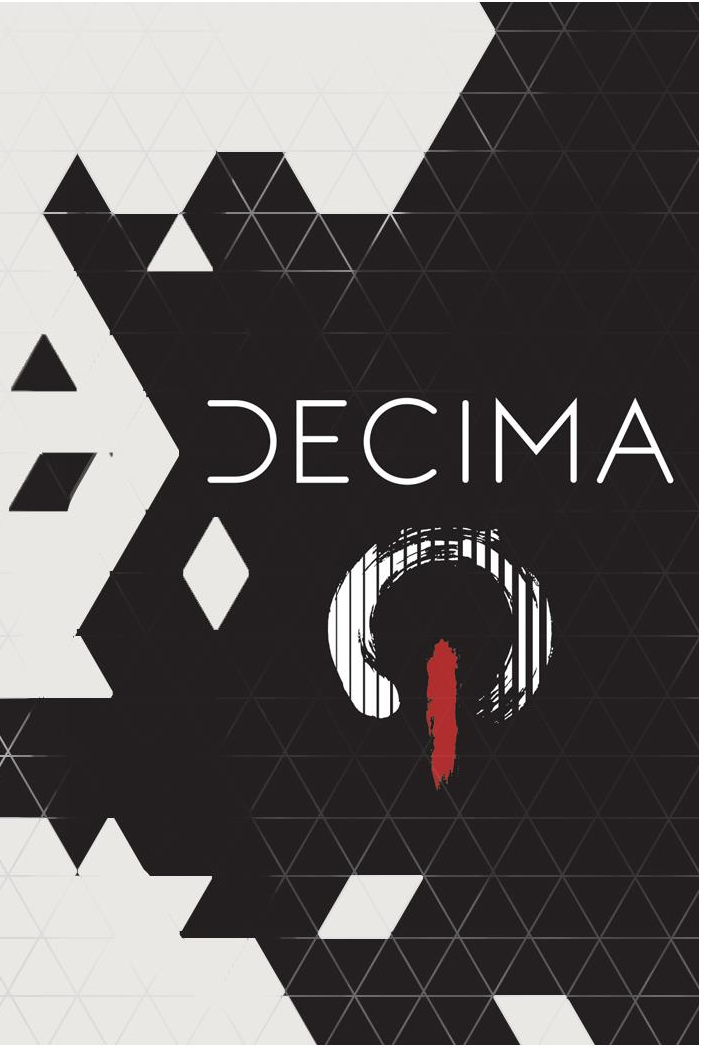
## OS



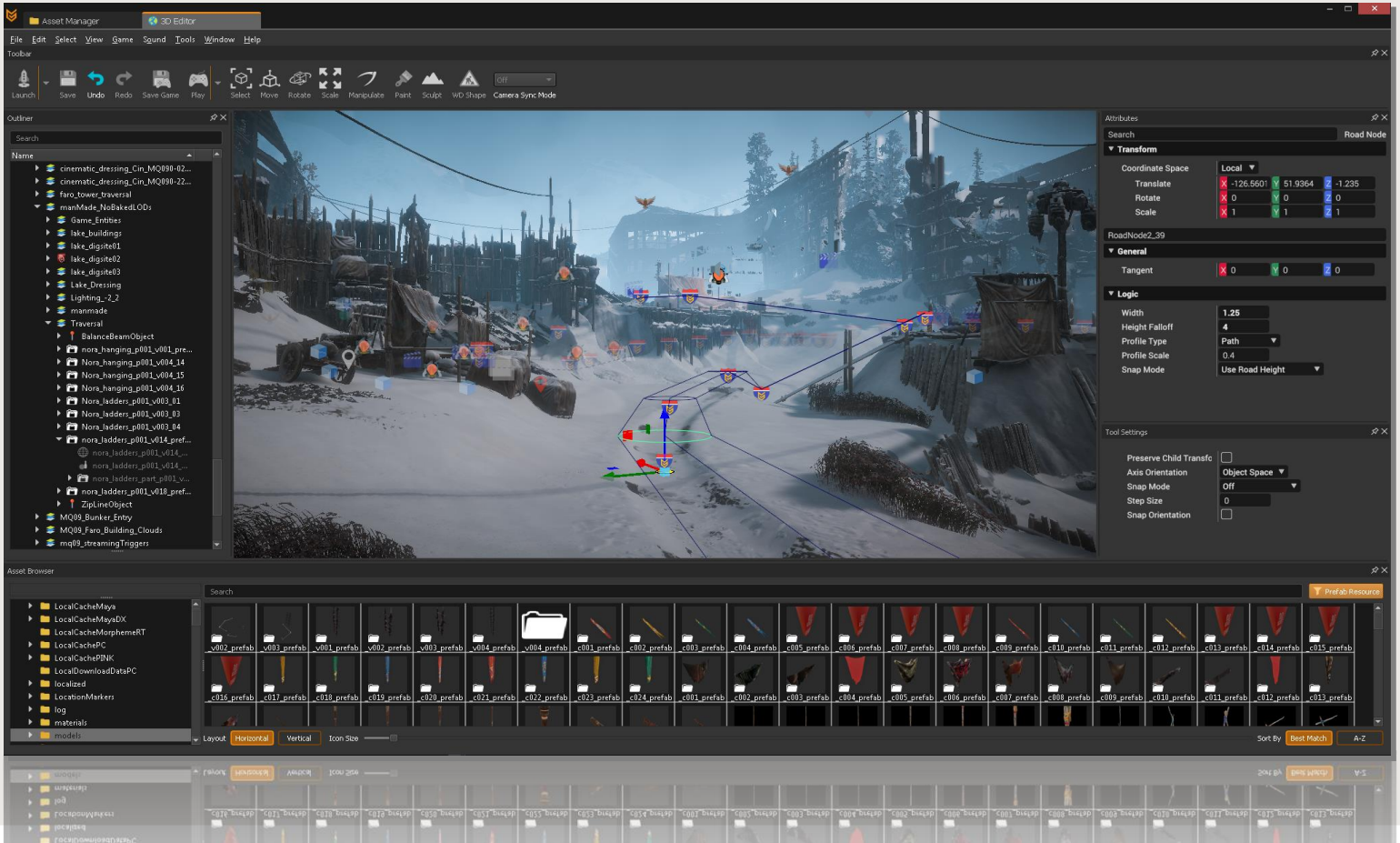



# Decima Engine

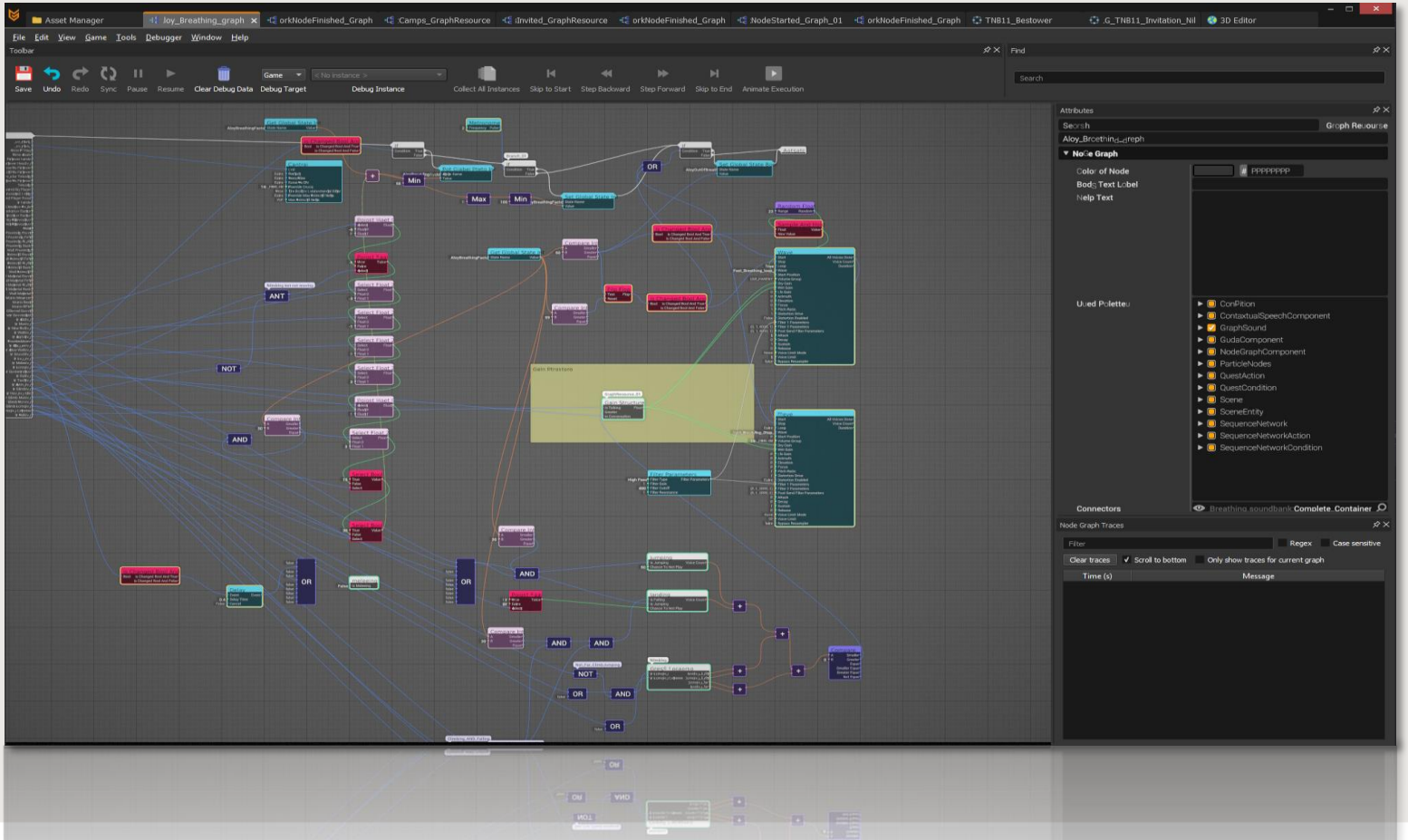
- Data Serialization & Deserialization
- Runtime Type Introspection (RTTI)
- File I/O, Networking, Threads, Jobs
- Image Readers & Writers
- Compression, Hashing, Algorithms
- Unit Testing Framework



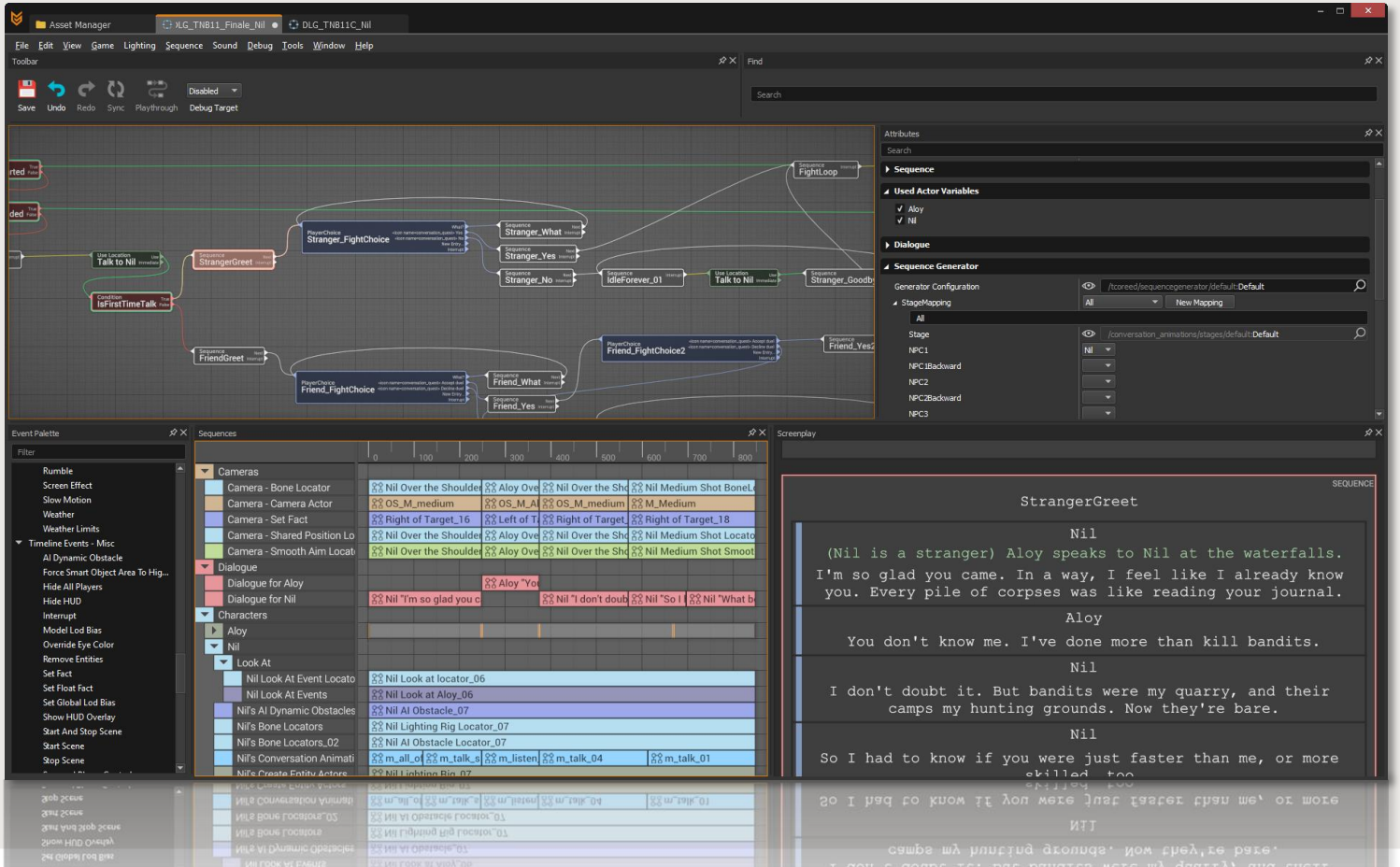
# Level Editor



# Visual Script Editor



# Conversation Editor

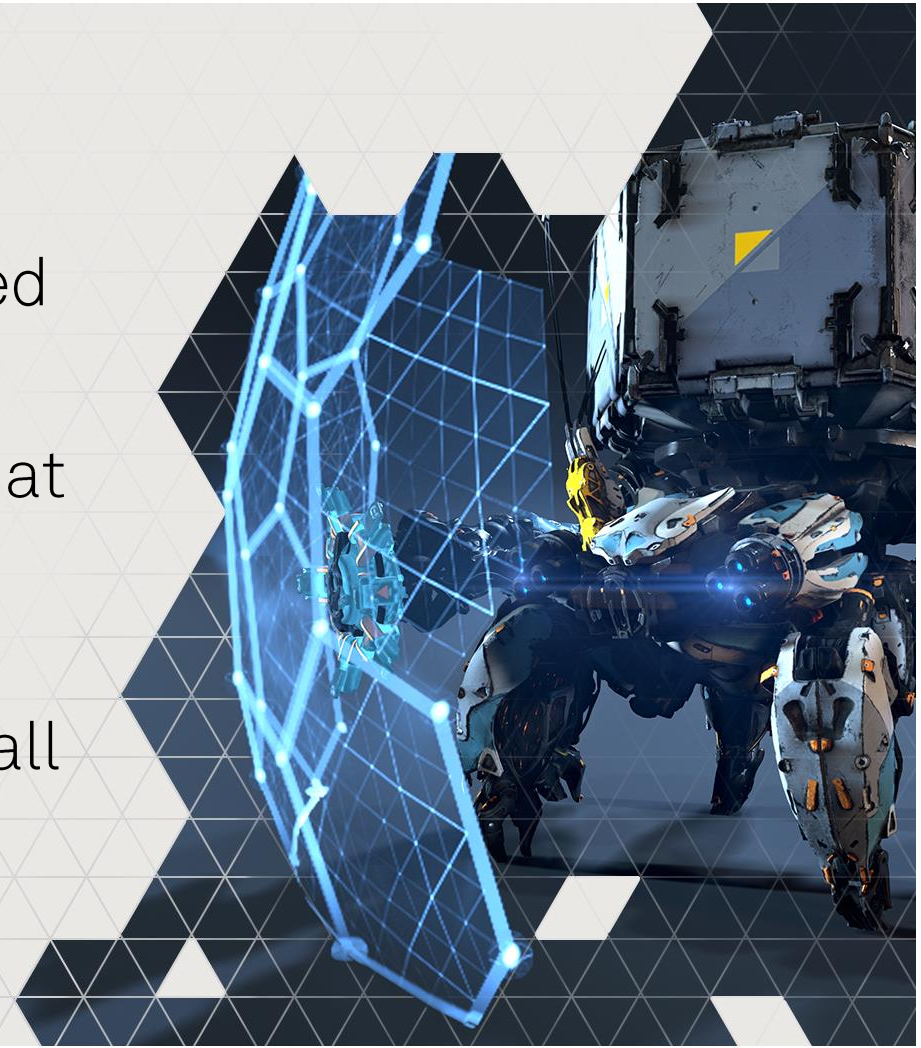




# The first steps to a  new editor

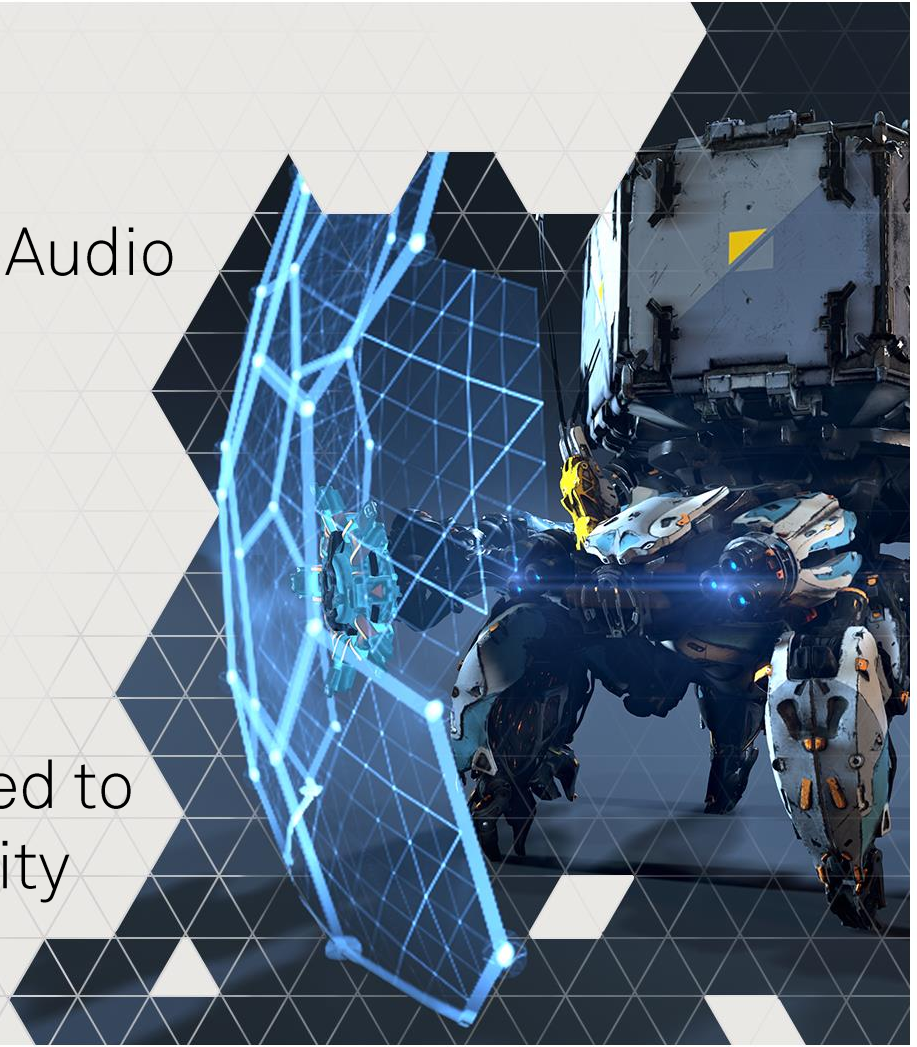
# Where do we begin?

- Production is getting started
- Users already have tools that they’re familiar with
- We would need to start small and and minimize risk



# Where do we begin?

- We decided to focus on the Audio Team
- A small group of users with specific needs
- Their workflow could be used to test the Framework's viability

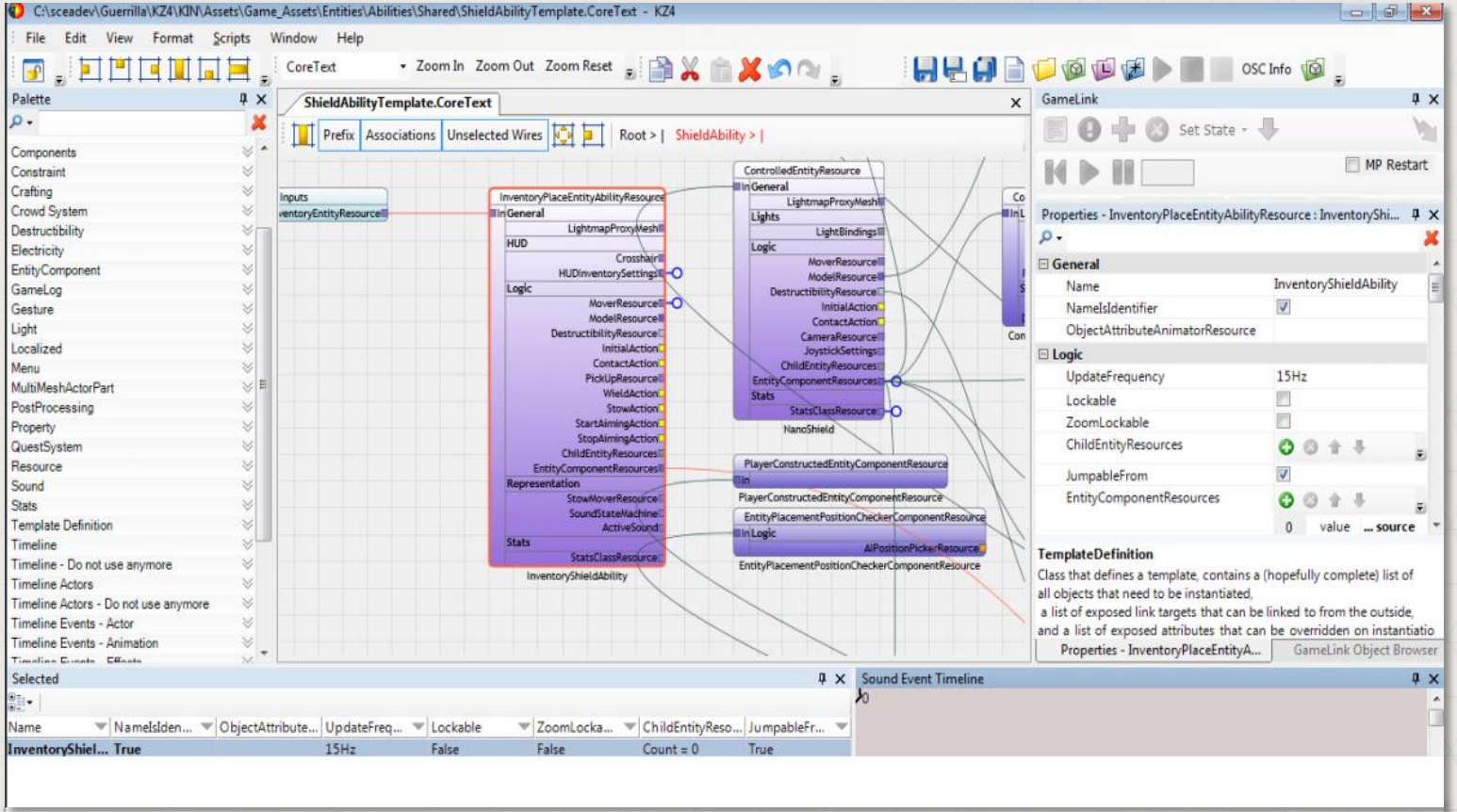




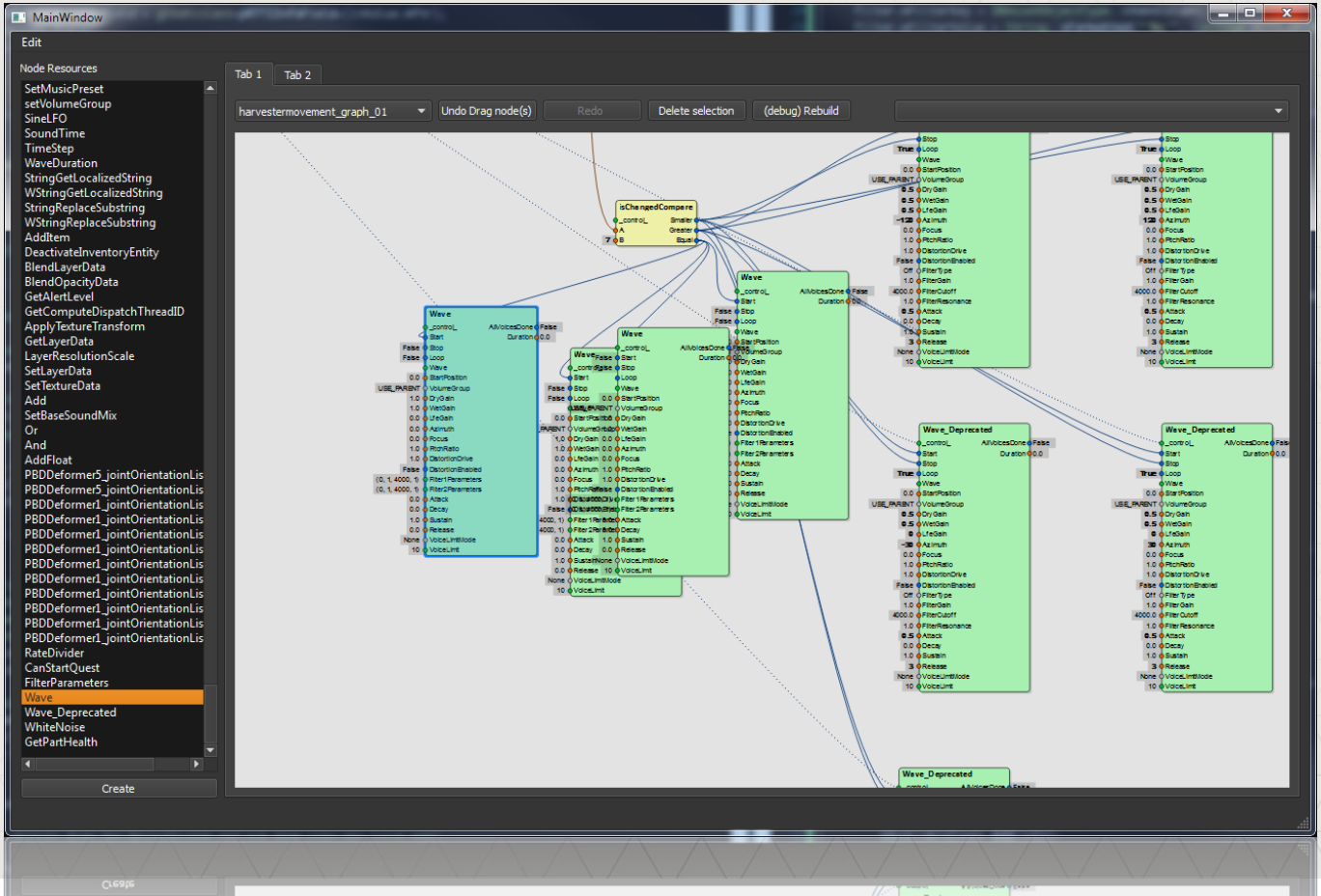
# Where do we begin?

- The Audio Team primarily worked with NodeGraphs
- NodeGraph is our High Performance Visual Scripting language
  - Generates native C++ code, that runs on target
- Proven tech, used for audio in Killzone: Shadow Fall
  - http://www.gdcvault.com/play/1020559 [Varga, Woldhek 2014]

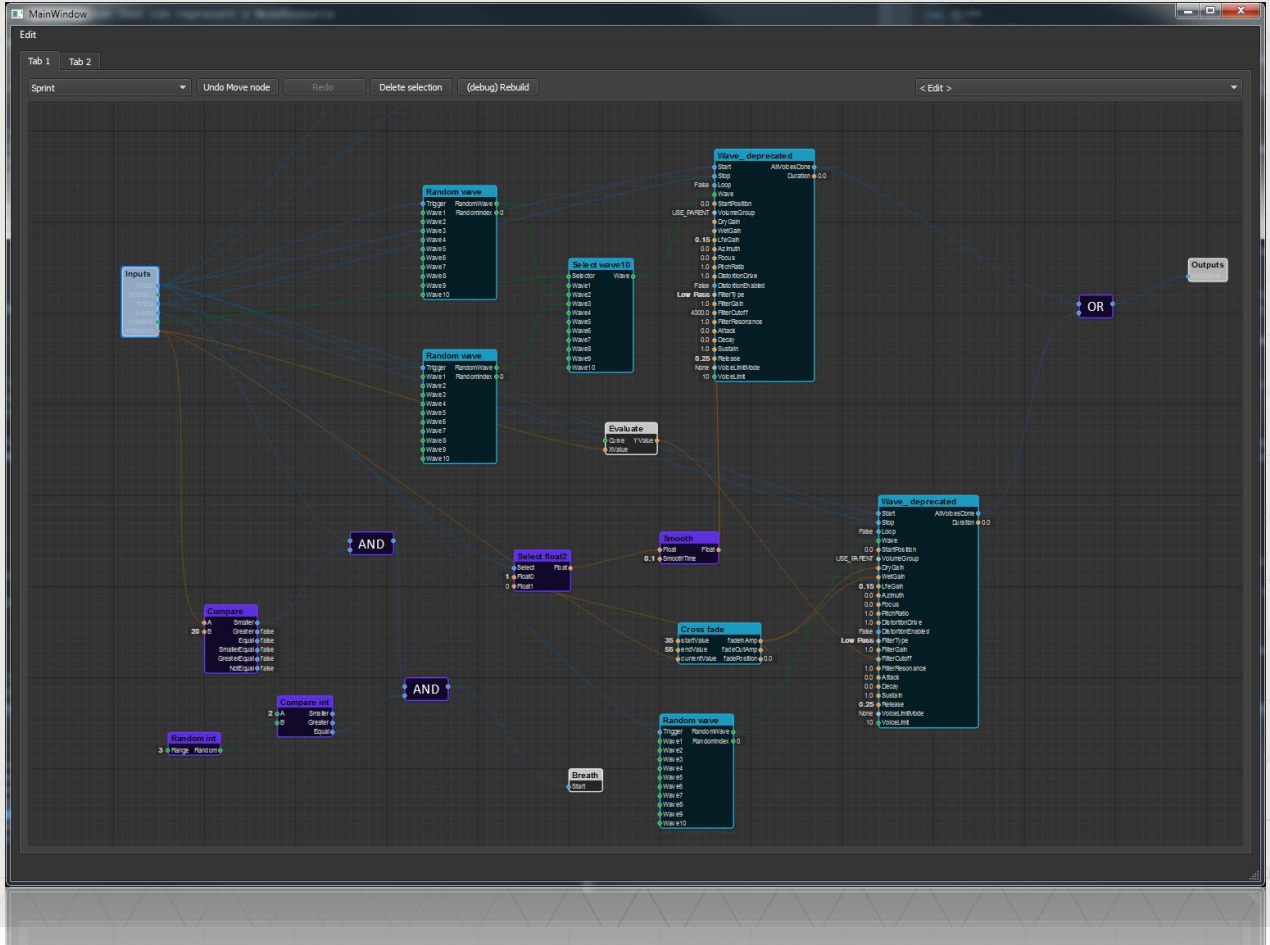
# CoreText Editor



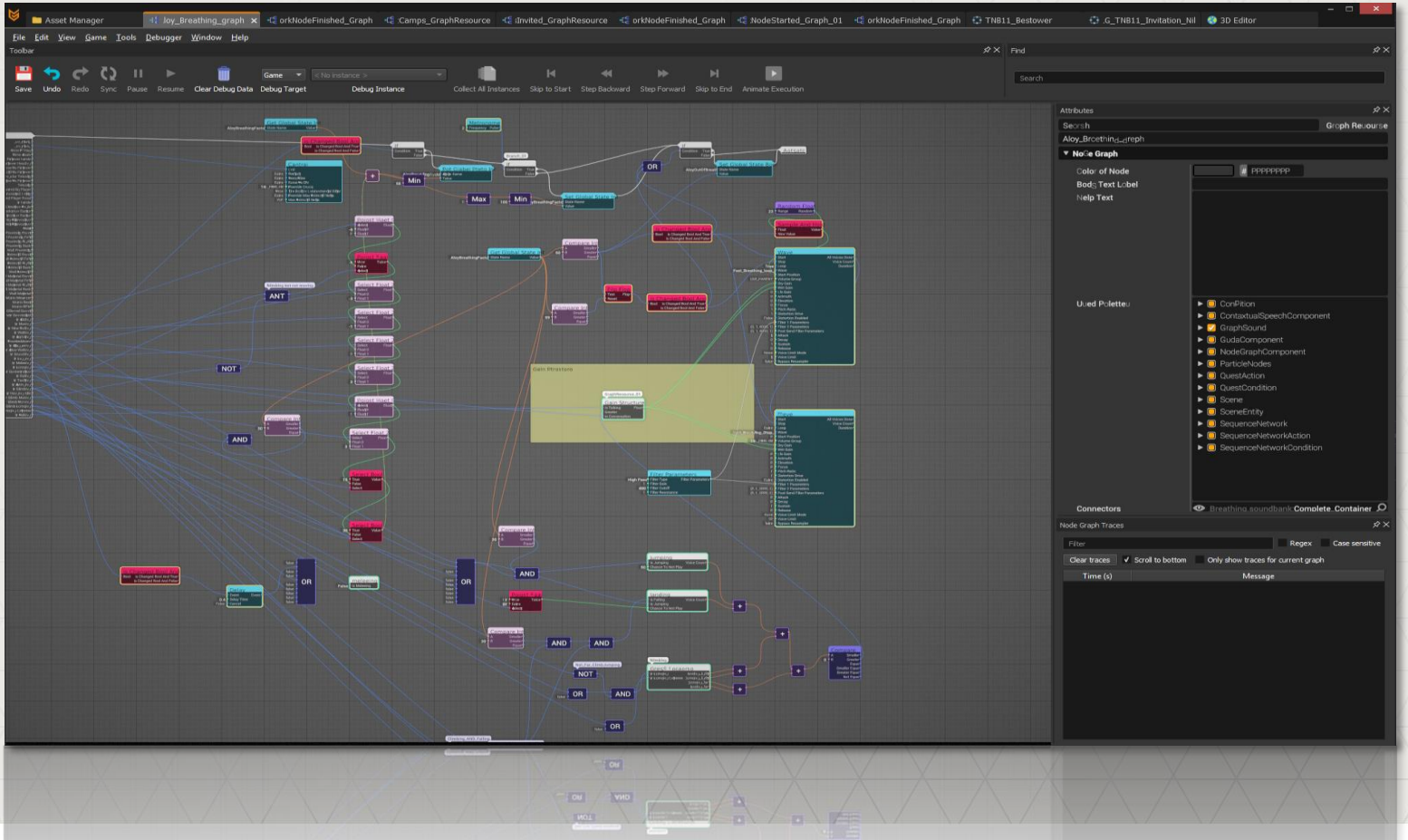
# NodeGraph: Prototype #1



# NodeGraph: Prototype #2

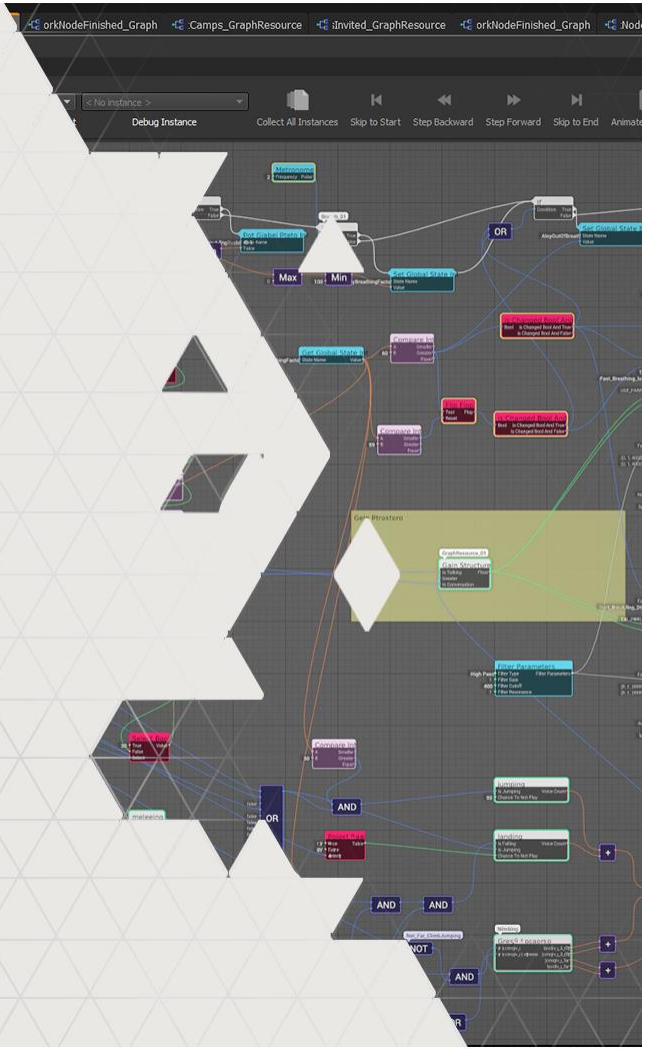


# Node Graph Editor Context



# Node Graph Editor

- We reached feature parity within a few months
- New features could be implemented easily
- Programmers from other teams could also contribute

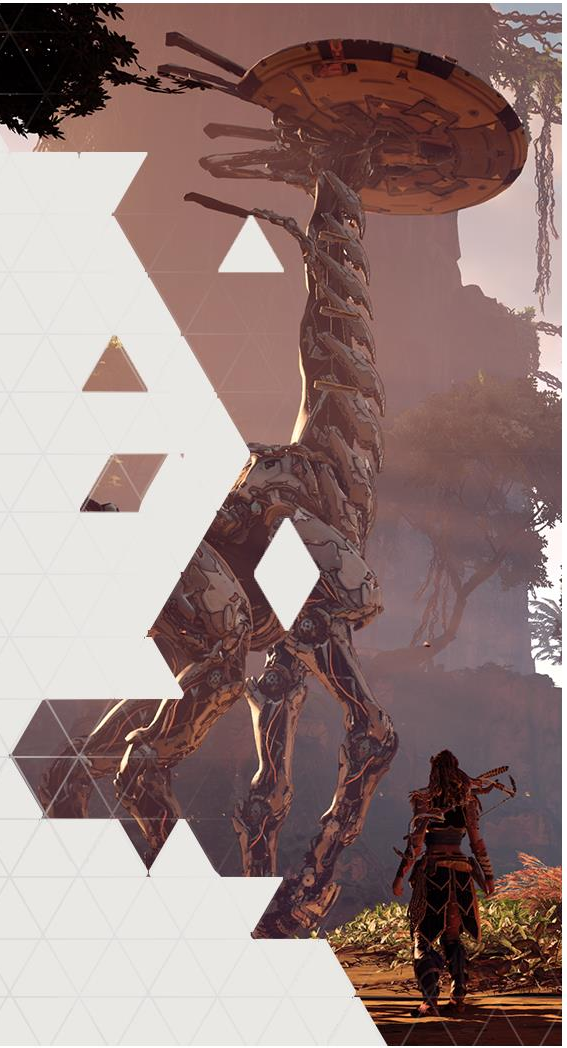


# A new workflow for Game Design

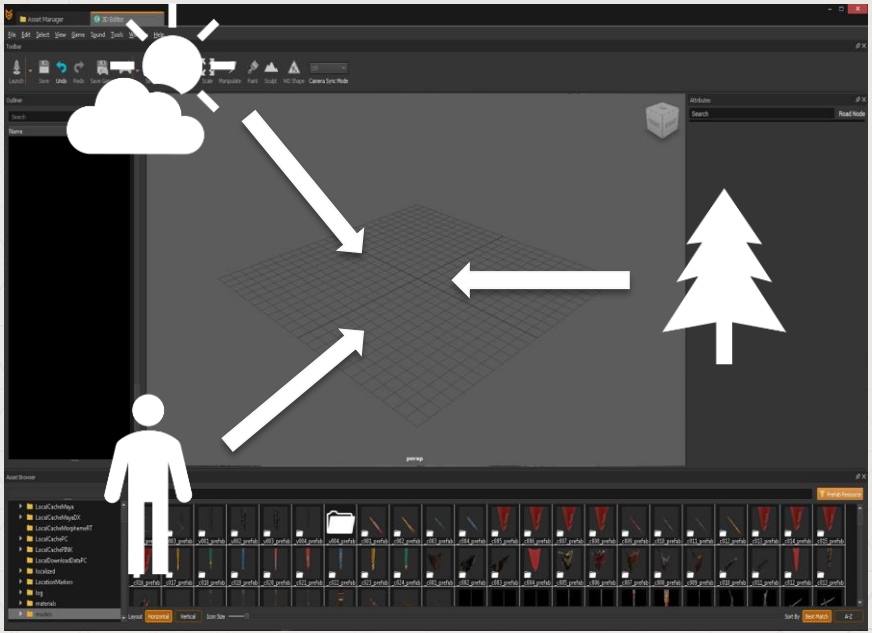
Creating a 3D Editor

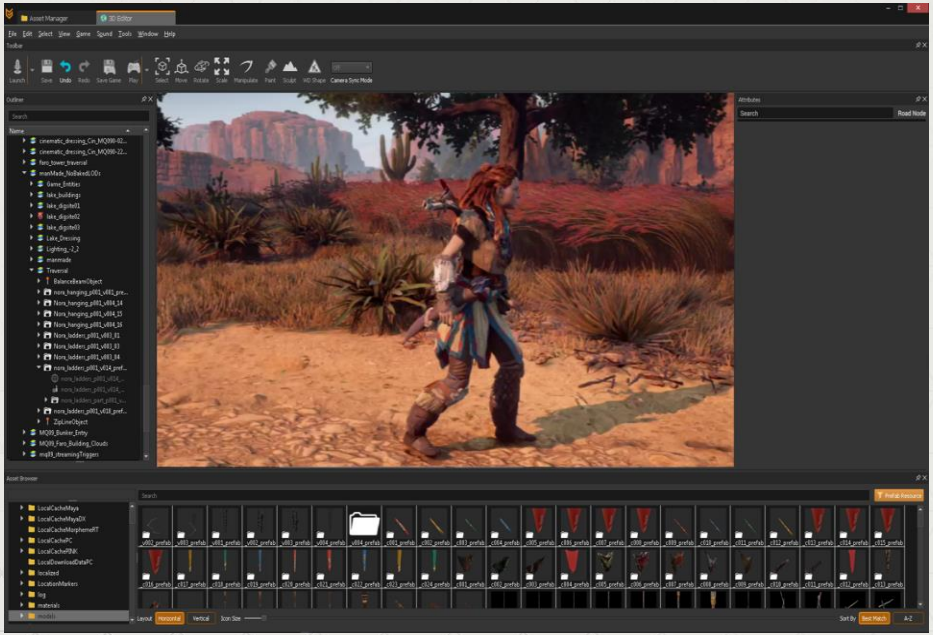
# Why a 3D Editor?

- *Maya* not suitable for quest content
- *3D Editor* was urgently needed
- Focus on Object placement & creation



# Integrate sub-systems or full game?





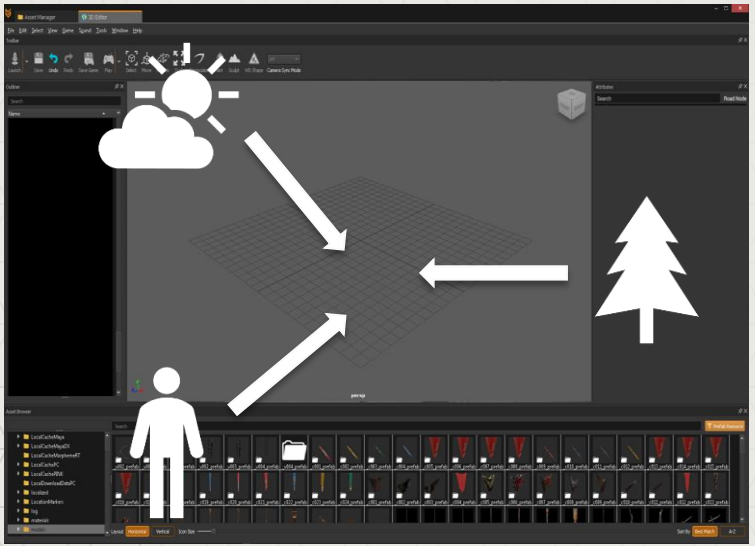
# Integrate only sub-systems into Editor?

+ Purpose build, controllable
+ Fast

- New code-path

  Longer development

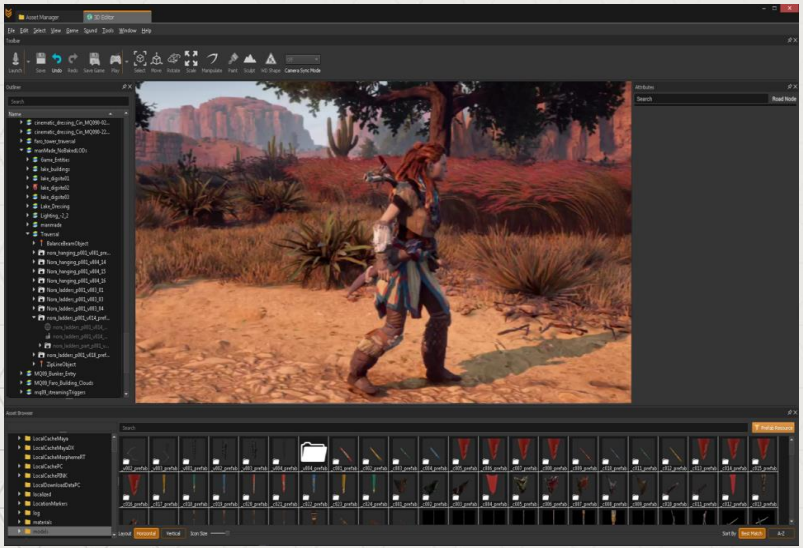
- Visual/behavioral disparity



# Integrate full game into editor?

+ All game systems available
+ *DebugUI* (tools) for free
+ Single code path (play/edit)

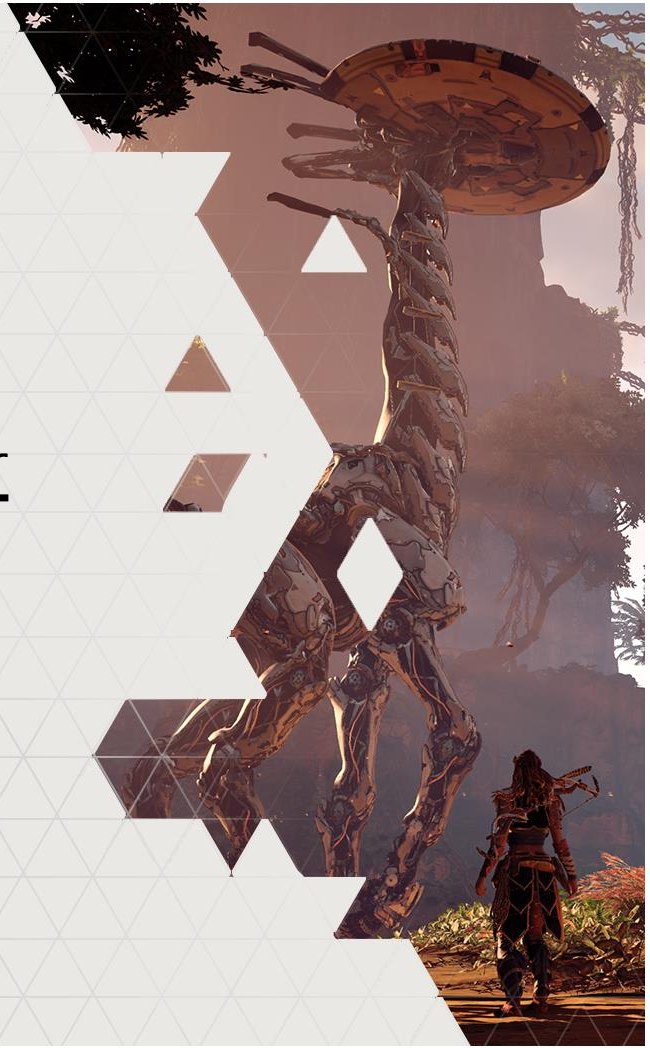
- Overhead of running game



# 3D Editor – Architecture

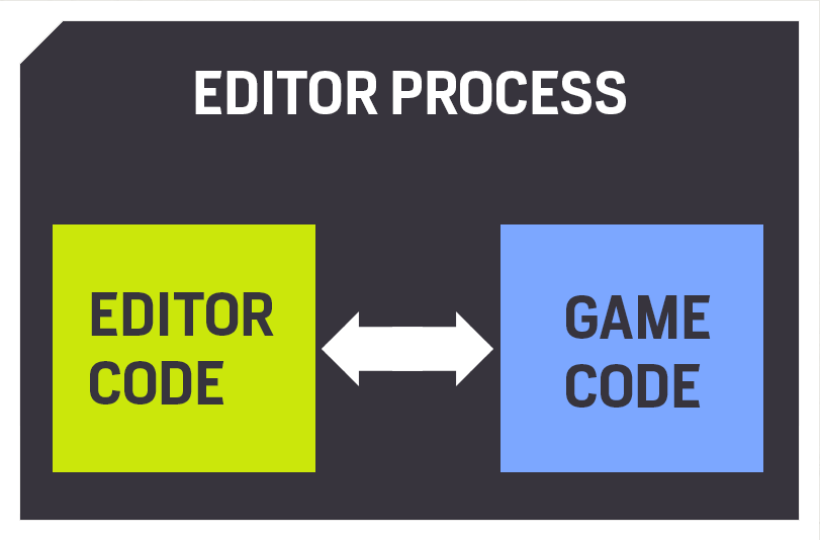
Due to time constraints:

Integrate full game into Editor

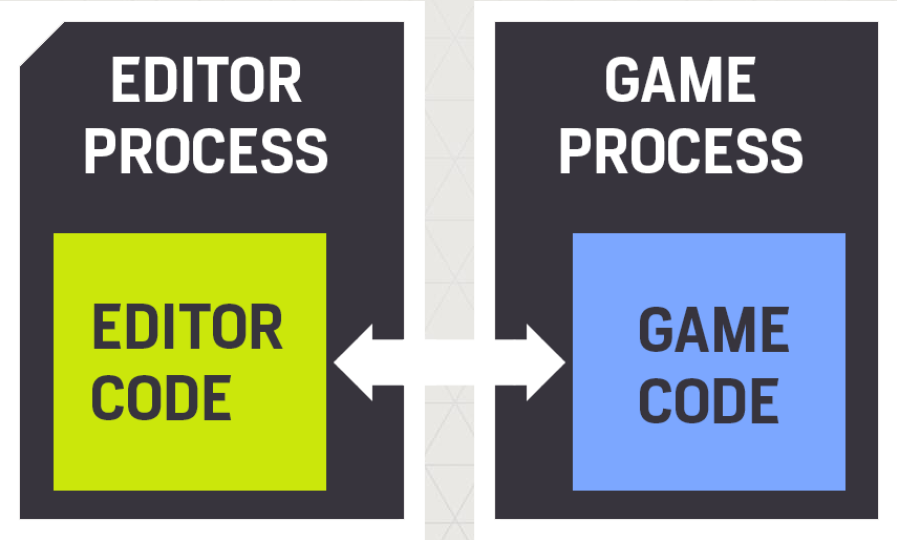


# How to integrate a full game into Editor?

*In* process?




*Out* of process?

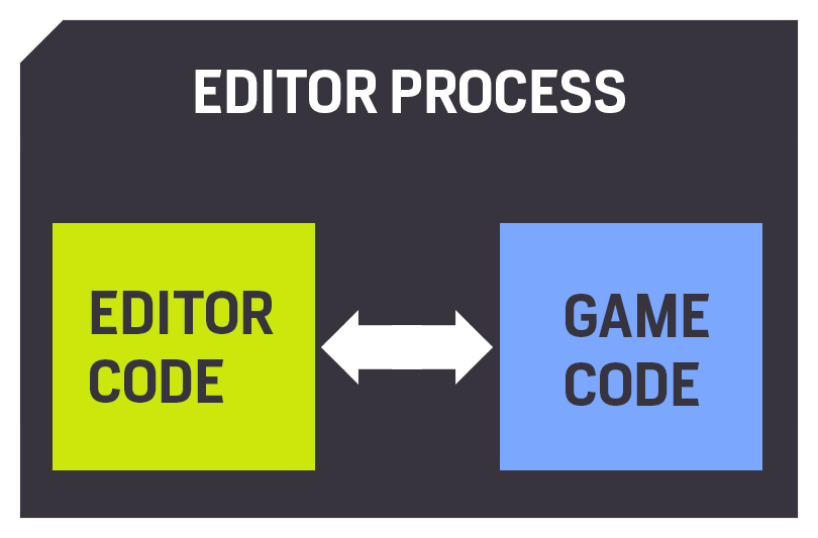


# In-process integration

+ Fast to setup

+ Direct synchronous data changes

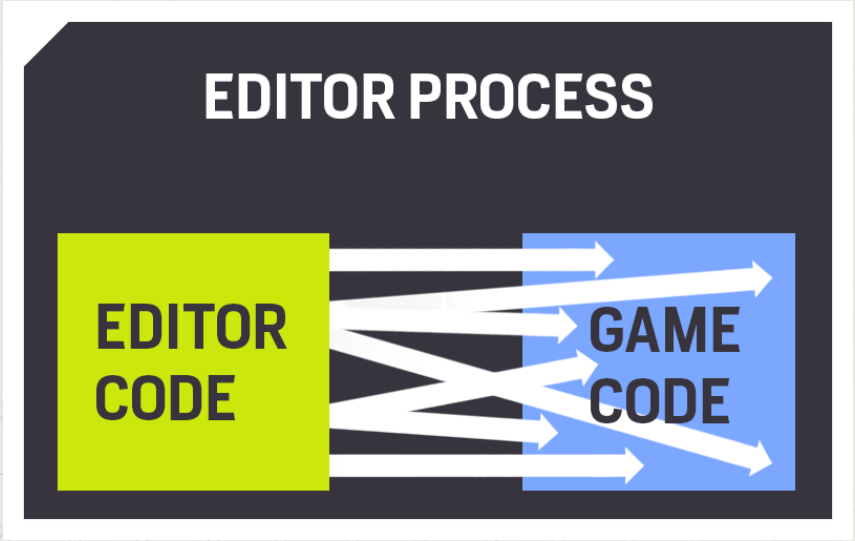
- Crash in game = crash in editor



# In-process integration

+ Fast to setup
+ Direct synchronous data changes

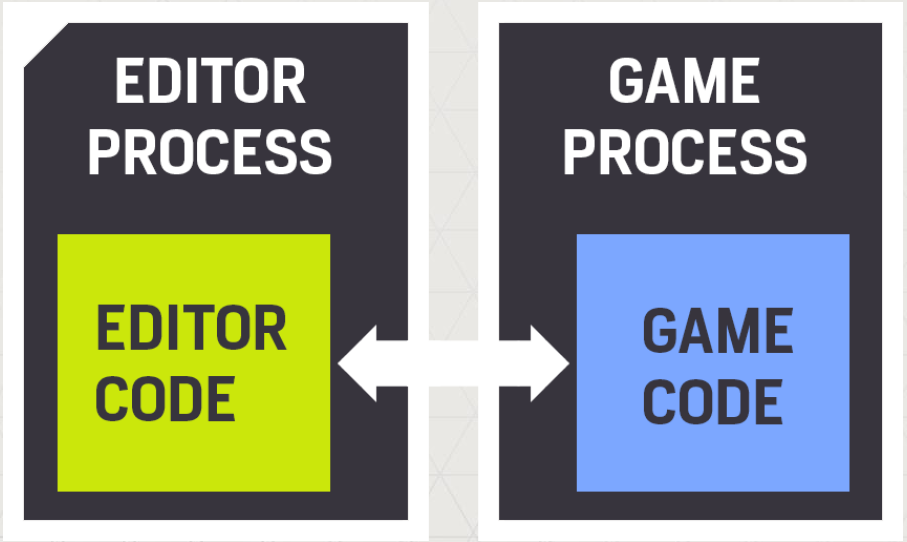
- Instability
- Enforcing code separation



# Out of process integration

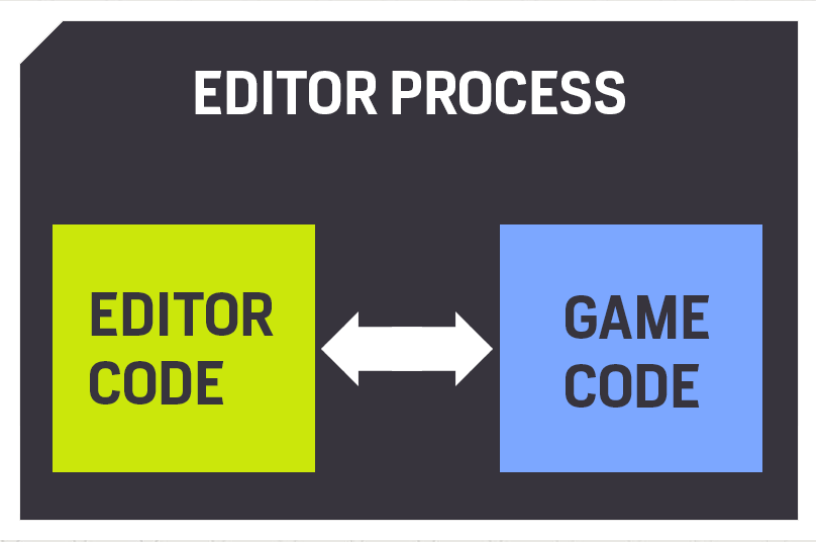
+ Stability
+ Cleaner code
+ Responsive UI

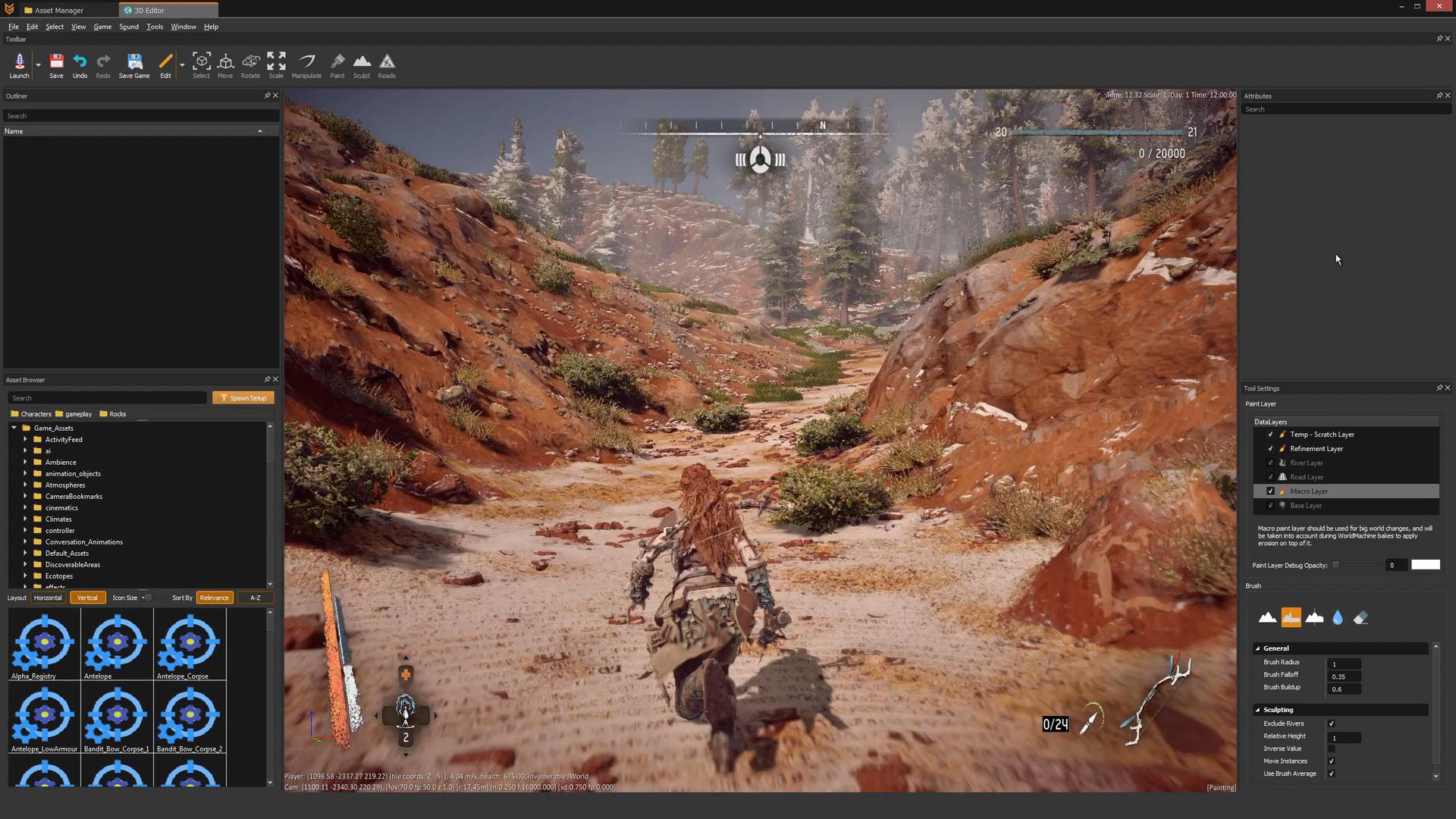
- Longer initial development
- Harder to debug 2 asynchronous processes



# 3D Editor: Initial design

- Develop **in-process** with a full game
- **Goal**: Out of process should remain possible




Asset Manager
3D Editor
File Edit Select View Game Sound Tools Window Help
Toolbar
Launch Save Undo Redo Save Game Edit Select Move Rotate Scale Manipulate Paint Sculpt Roads
Outliner
Search
Name
Asset Browser
Search
Spawn Setup
Characters gameplay Rocks
Game_Assets
ActivityFeed
ai
Ambience
animation_objects
Atmospheres
CameraBookmarks
cinematics
Climates
controller
Conversation_Animations
Default_Assets
DiscoverableAreas
Ecotopes
Layout Horizontal Vertical Icon Size Sort By Relevance A-Z
Alpha_Registry
Antelope
Antelope_Corpse
Antelope_LowArmour
Bandit_Bow_Corpse_1
Bandit_Bow_Corpse_2
Time: 12.32 Scale: 1 Day: 1 Time: 12.00:00
N
20
21
0 / 20000
0/24
Player: (1098.58 -2337.27 219.22) (tile coords: 2, -5 ) 4.04 m/s, health: 675.00, invulnerable, World
Cam: (1100.11 -2340.30 220.29), (fov:70.0 fp:50.0 z:1.0) [r:17.45m] [n:0.250 f:16000.000] [sd:0.750 fp:0.000]
[Painting]
Attributes
Search
Tool Settings
Paint Layer
DataLayers
Temp - Scratch Layer
Refinement Layer
River Layer
Road Layer
Macro Layer
Base Layer
Macro paint layer should be used for big world changes, and will be taken into account during WorldMachine bakes to apply erosion on top of it.
Paint Layer Debug Opacity:
Brush
General
Brush Radius
Brush Falloff
Brush Buildup
Sculpting
Exclude Rivers
Relative Height
Inverse Value
Move Instances
Use Brush Average



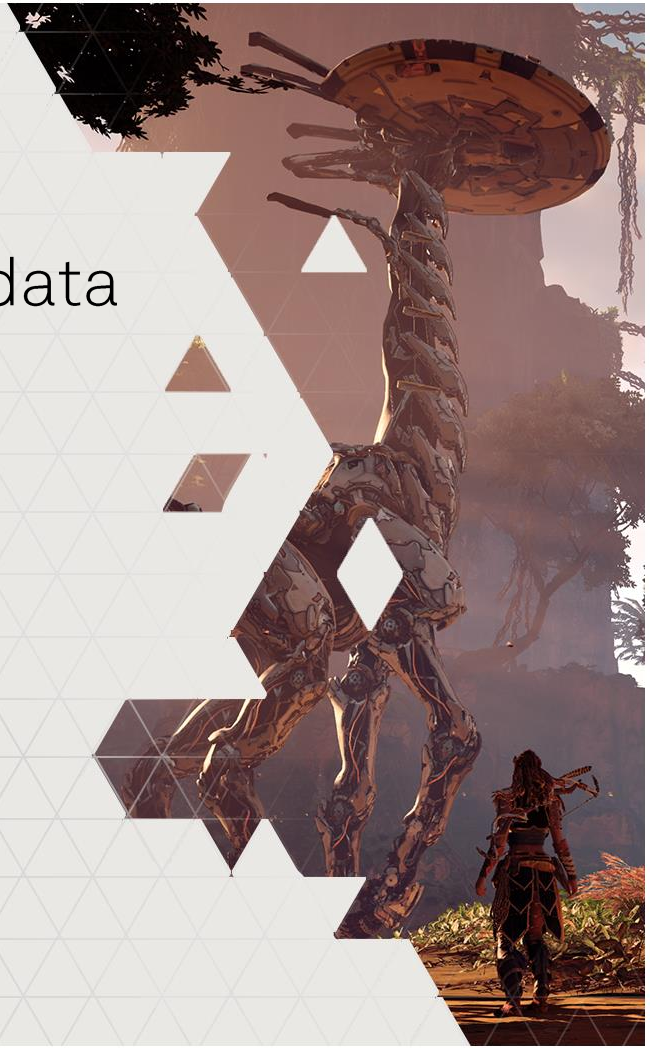
# Editing content in a running game

Required sub-systems for Level Editor

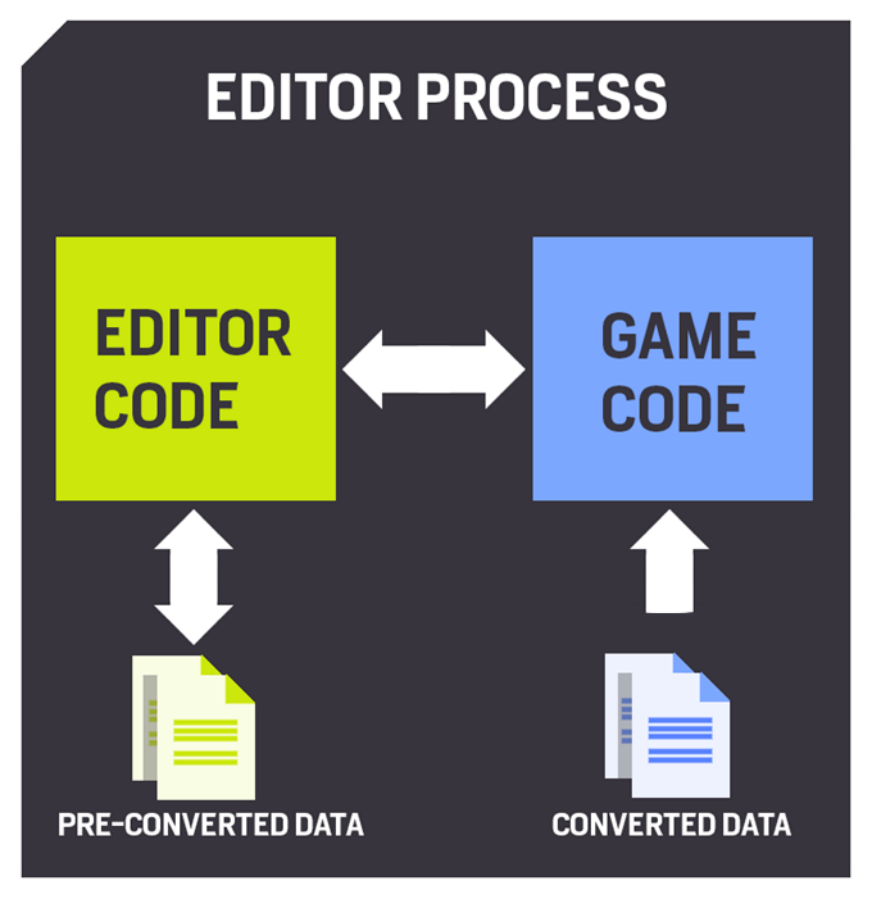
# Game data

- **Problem**: Game reads binary game data

- Creation is *destructive*
  - Shader compilation
  - Meshes optimization
  - Data packing

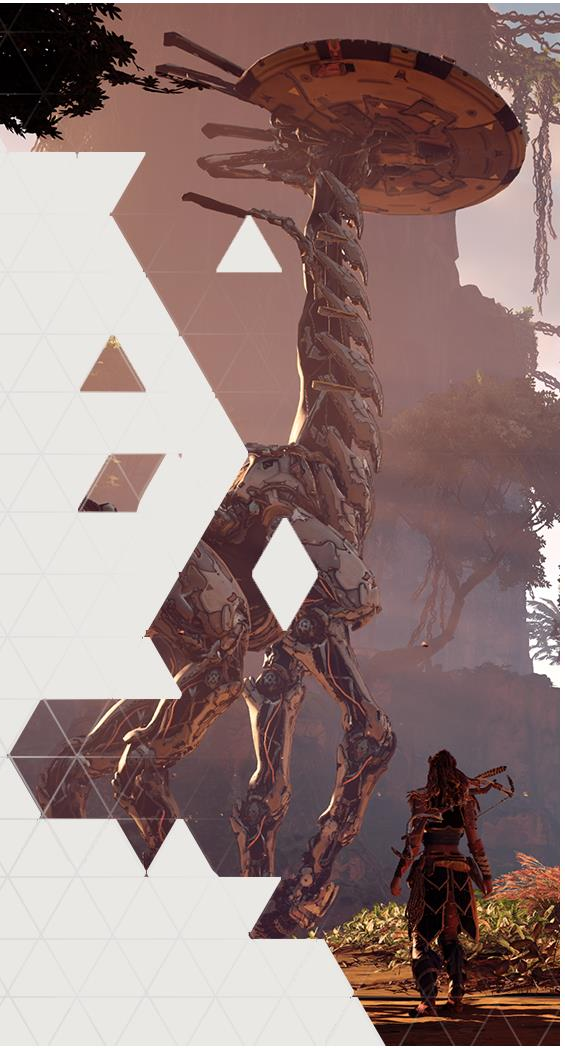


# 3D Editor: Initial design



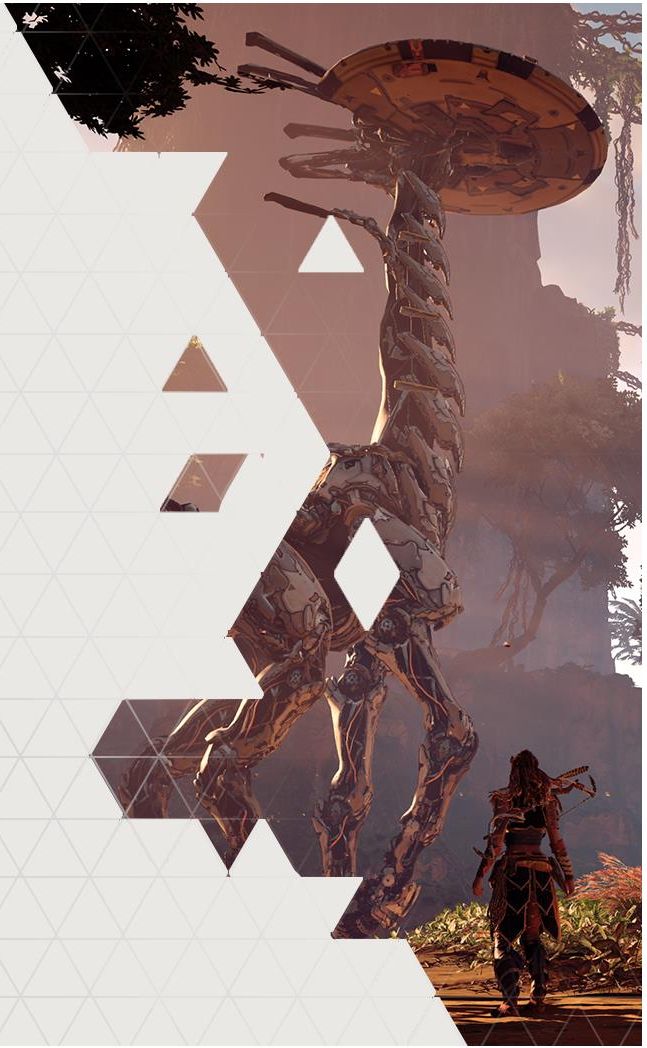
# EditorNode

- Provides **all** editing functionality for an object in game
- Game object does **not** know of Editor Node
- Keeps code clean, no editor specific code interwoven in game code



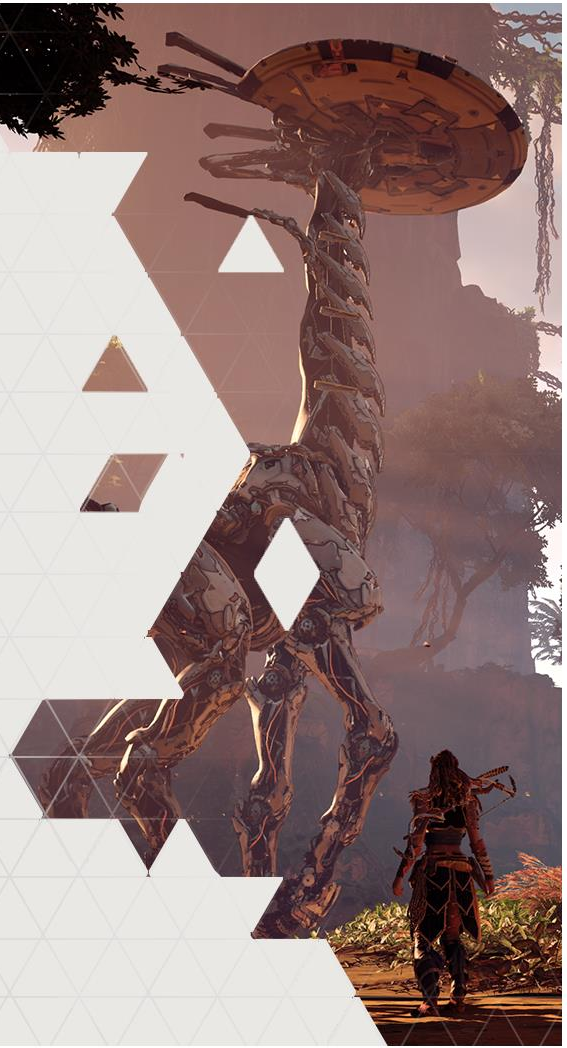
# Only WorldSpace in-game

- Problem: Less hierarchy in game
- Flat list of objects
  - Fast and efficient
  - Transforms in World Space

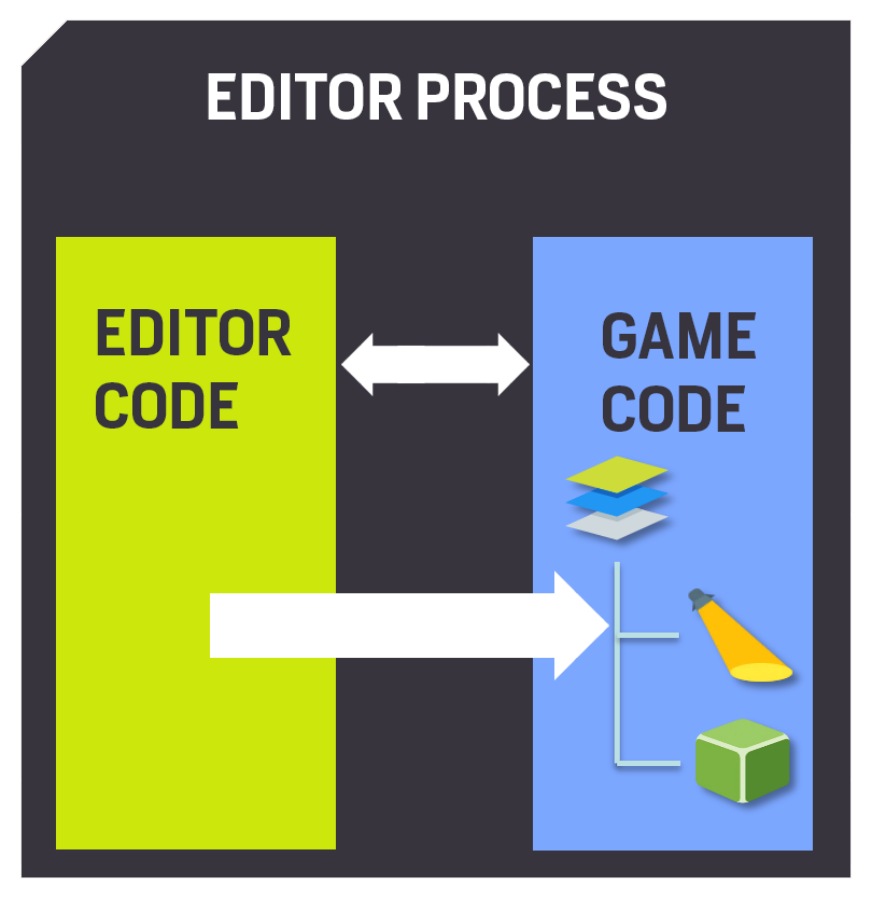


# EditorNodeTree

- Hierarchical structure of in-game data
- Constructed from game data, reflects hierarchy of objects in-game.
  - When entering “*edit mode*”
  - Top down (starting at root)



# Accessing the EditorNodeTree



# 2 Editor Node Trees

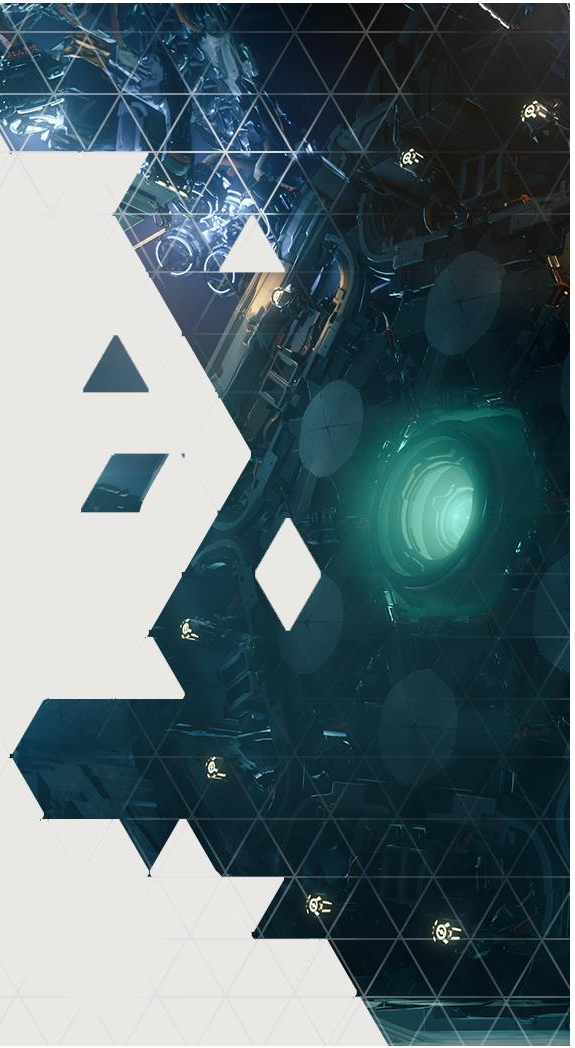




# Making Changes to Objects

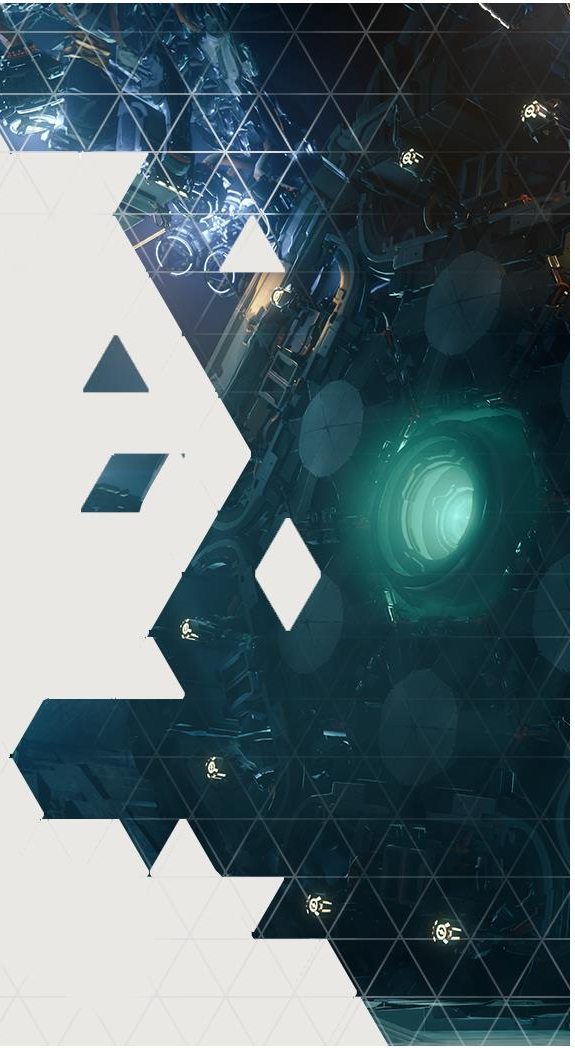
# What we didn't want

- Proxy representations
- String-based accessors
- Editor-specific boilerplate
- Code generation steps in the build



# What we wanted

- Work directly with concrete objects
- Directly modify object members
- Re-use existing systems and utilities to create and transform data

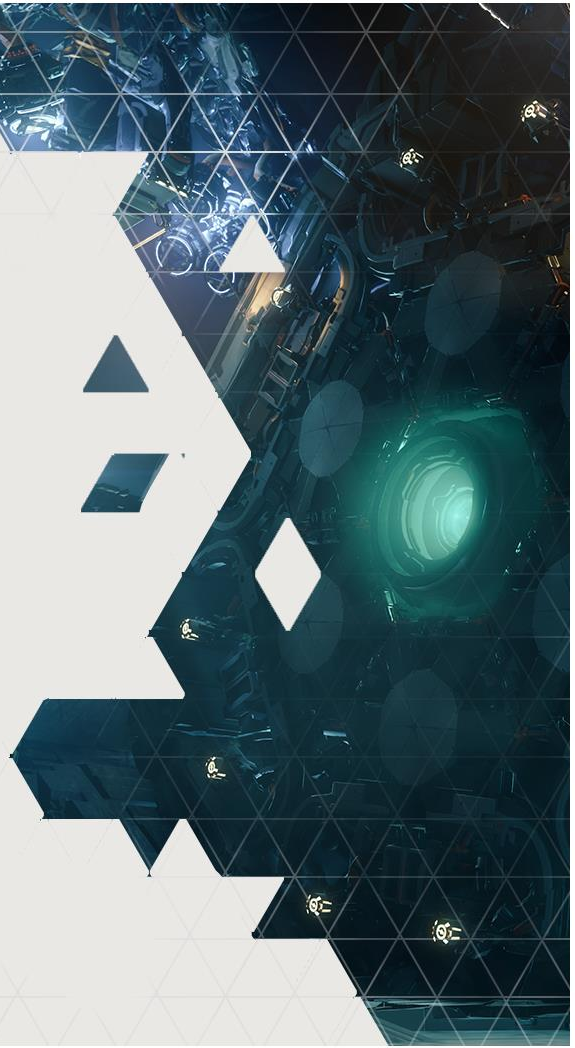


# Making changes to objects

- Being able to make changes is good, but...

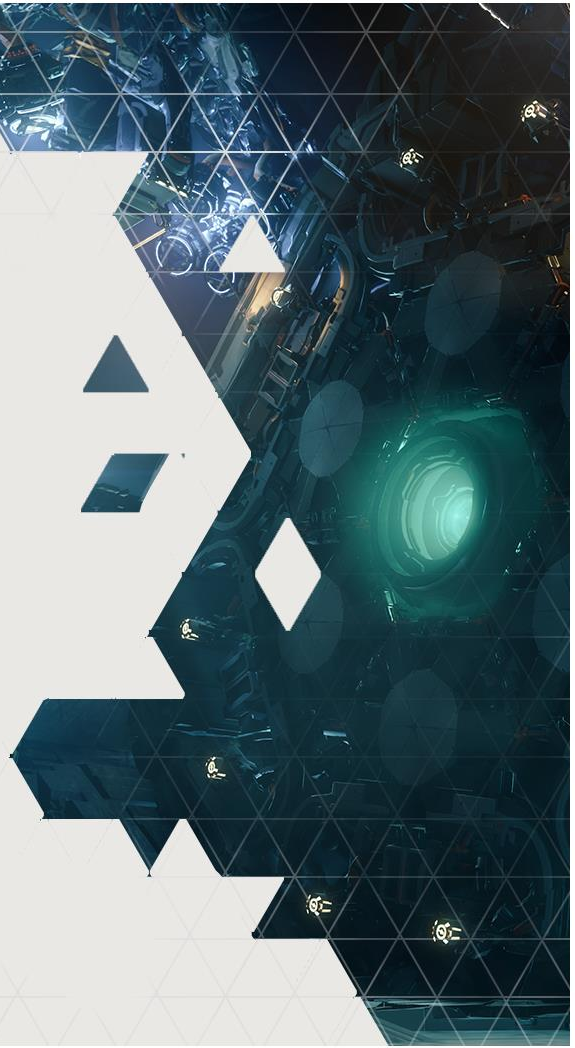
"What is best in life?"

- An **Undo** system is essential for users to remain in control and have the freedom to experiment



# Undo System – Goals

- Simplicity & Reliability
- No virtual **‘Do’** and **‘Undo’** functions
- No new undo code for new features
  - Prevent transformation asymmetry





# Undo System

- The ideal scenario:
  - An undo system that automatically detects object changes, and knows how to revert them
- The compromise, in our case:
  - Beforehand, code indicates which objects will be modified, and when the modifications are completed

# Undo System – Transactions

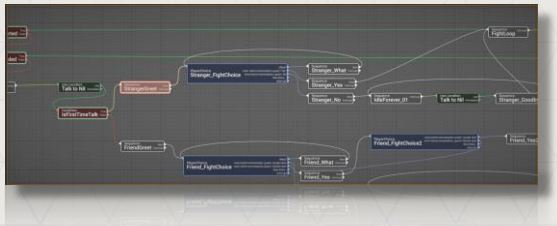
- Describes all modifications in a change
  - Commands represent individual changes
- Built on the RTTI system
- Subscribers are notified of all transactions



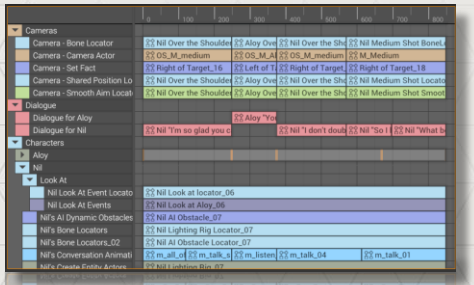
# Transaction Notifications





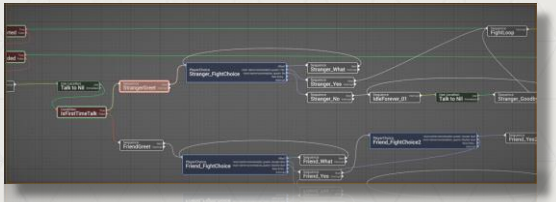
ADD OBJ
ADD OBJ
SET VALUE
SET VALUE
DEL OBJ



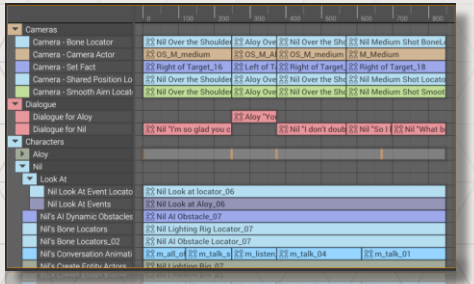
# Transaction Notifications





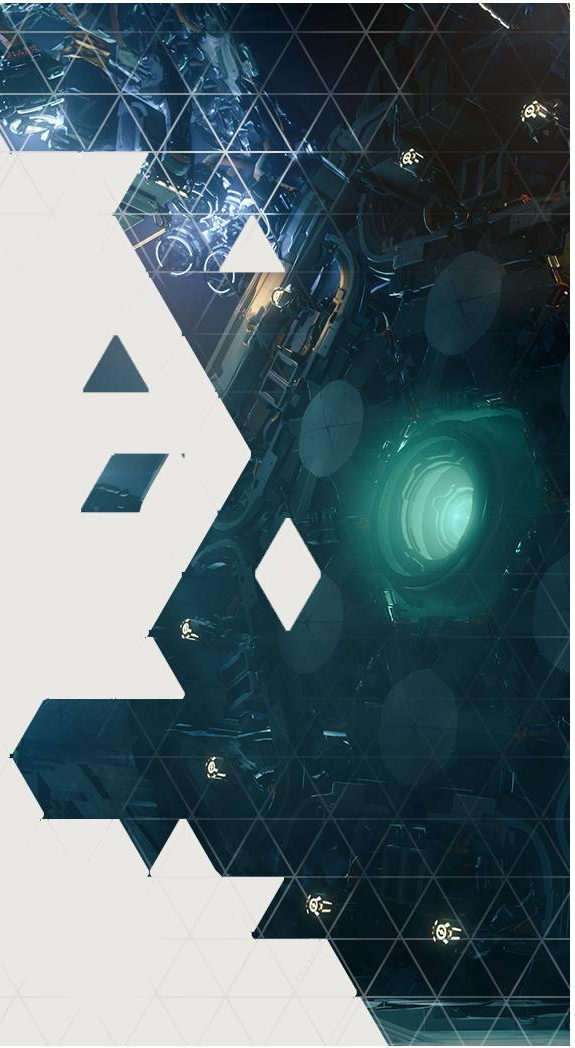



- Code Separation
- Changes can be made from anywhere
- No knowledge of do and undo in editor code



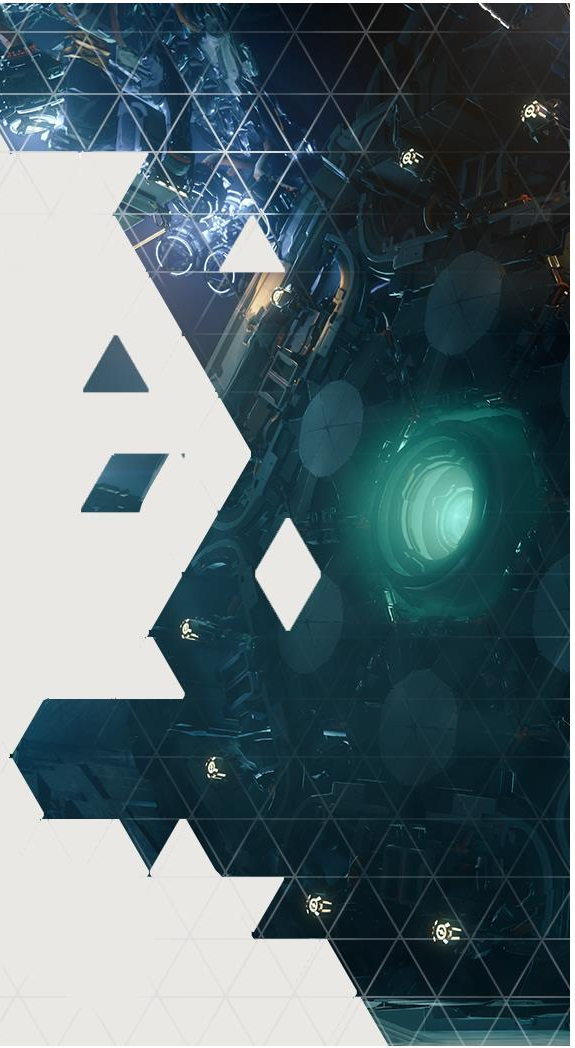
# Undo System – DiffUtil

- Building transactions is hard
- DiffUtil builds them
  - Produces transactions by looking at changes that have already happened



# Undo System – DiffUtil

- Before changes are made:
  - DiffUtil takes **snapshot** of objects' state
- After changes:
  - Analyzes what has changed between the snapshot and the current state



# DiffUtil – Code Example

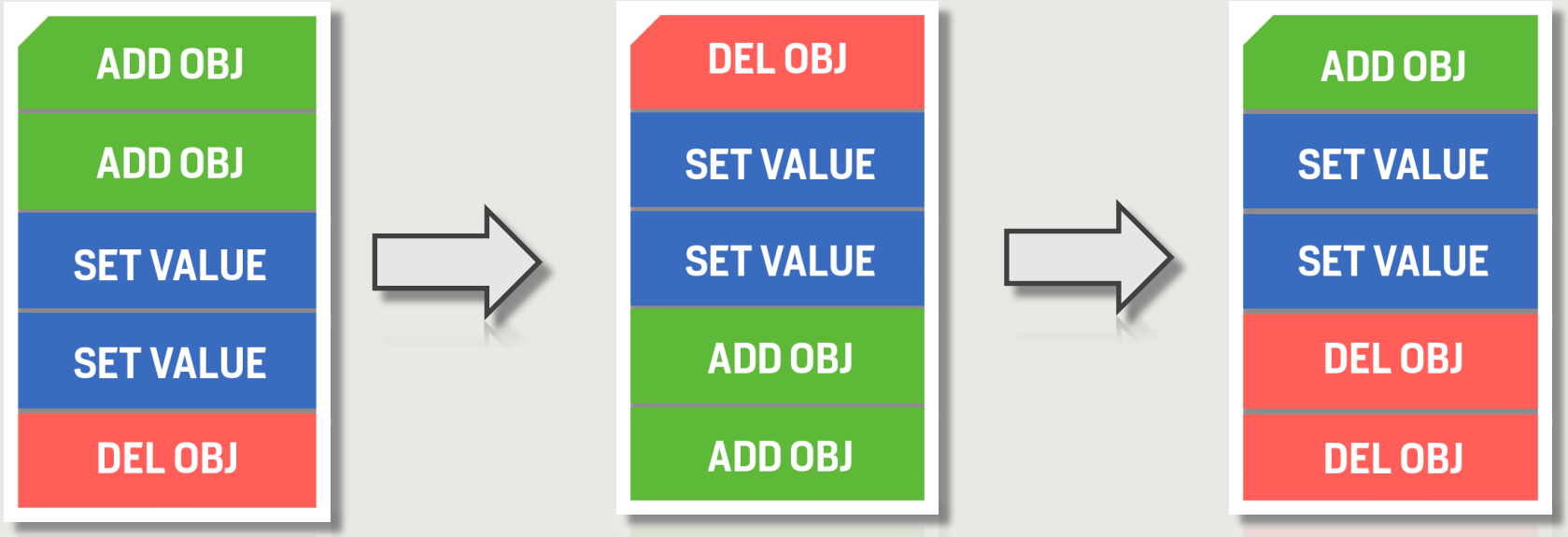
```
// Get the spotlight to edit...
pSpotLightResource spotlight = GetSpotlightToEdit();

// Take a snapshot of the spotlight's current state
DiffUtil diff;
diff.TakeSnapshot(spotlight);

// Update the cone angle
spotlight->SetConeAngle(30.0f);

// Indicate that we have finished making changes
diff.CommitChanges("Set Cone Angle to 30");
```

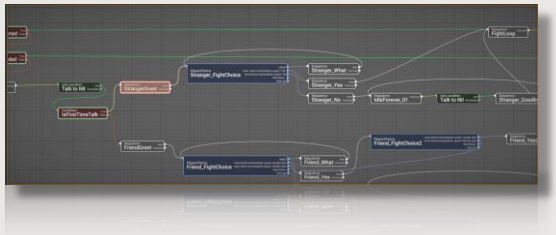
# Undo System – Reverting Changes



# Applying the Undo Transaction





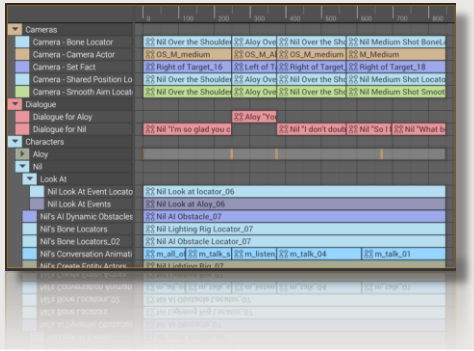
ADD OBJ

SET VALUE

SET VALUE

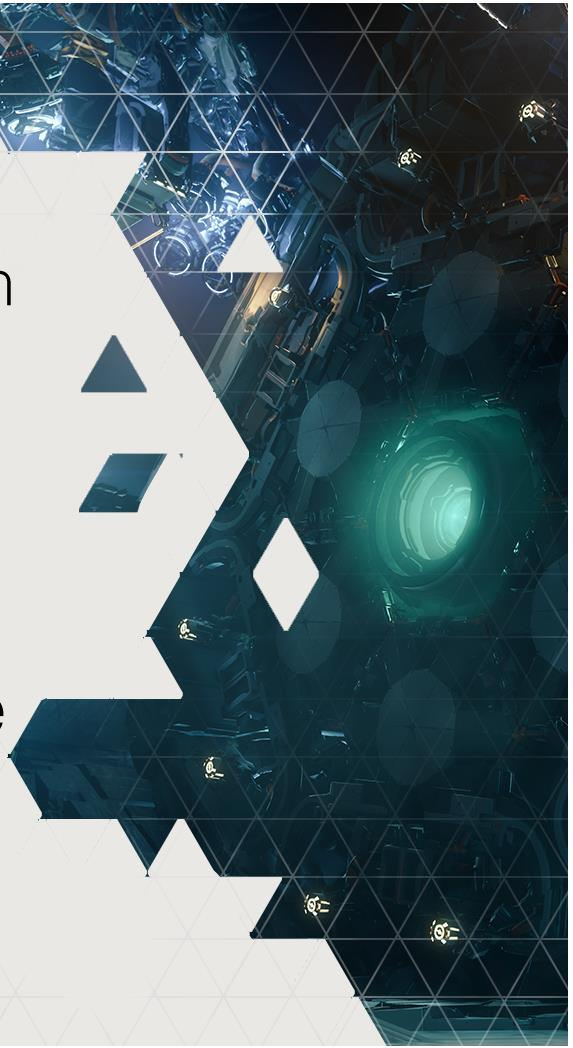
DEL OBJ

DEL OBJ



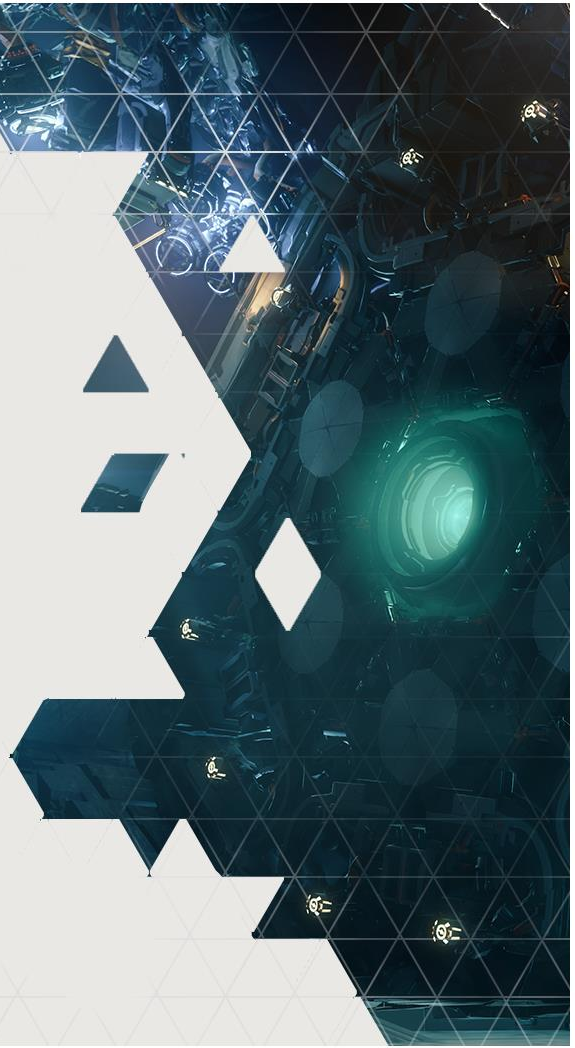
# What worked?

- A reliable, transparent system that can represent arbitrarily complex changes
- No distinction between **do** and **undo**
- Stable code, that changed very little in the time since it was written



# What didn’t work?

- Easy for programmers to forget to use a DiffUtil
- Transactions validated in Debug
  - But this doesn’t catch everything
- Doesn’t support filesystem operations



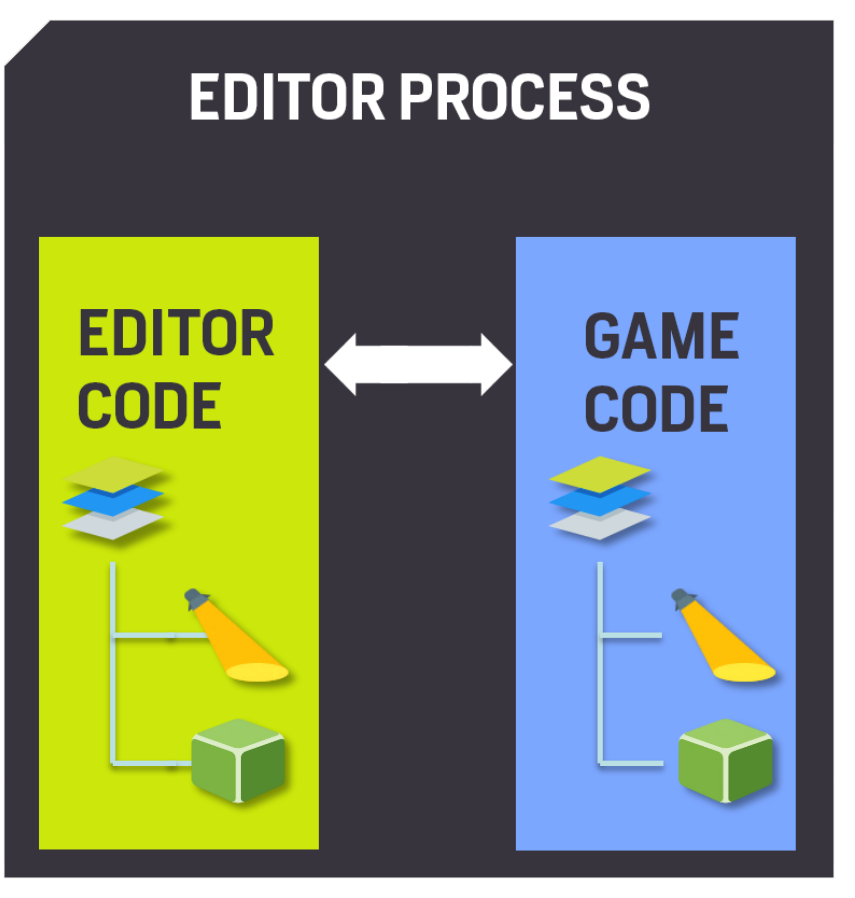
# Transactions in the 3D Editor

# Changes in the 3D Editor

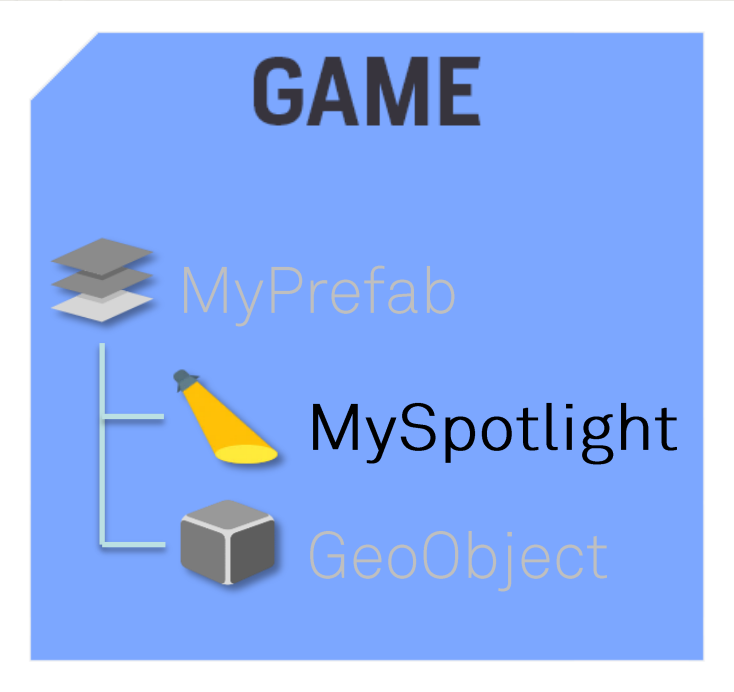
- Transactions to the Game



# EditorNodeTree in space conversion

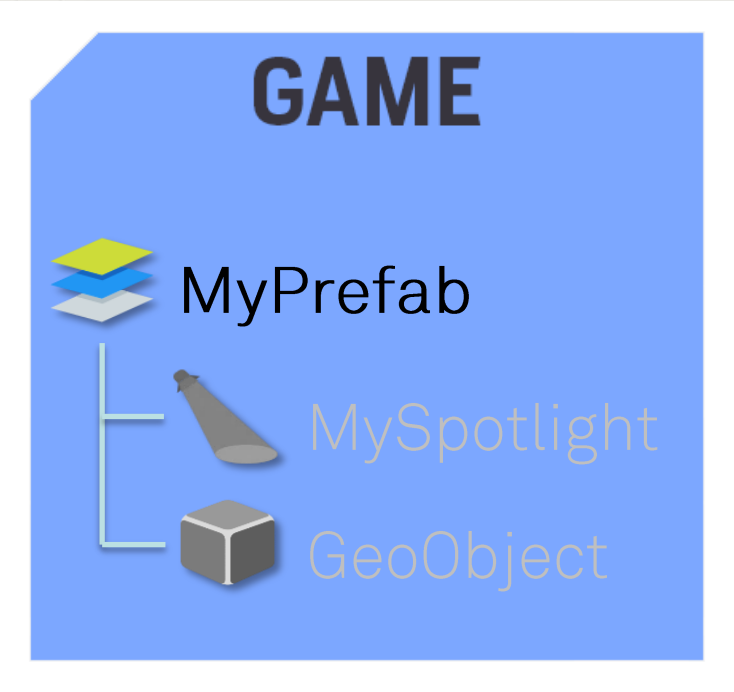


# 3D Editor – Applying a change in the tree



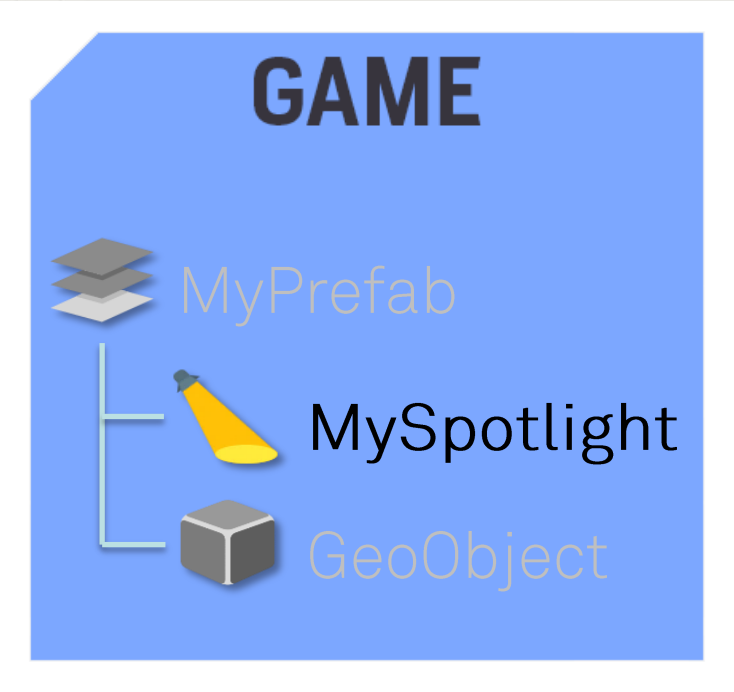

- Modify local space of *MySpotlight*

# 3D Editor – Applying a change in the tree



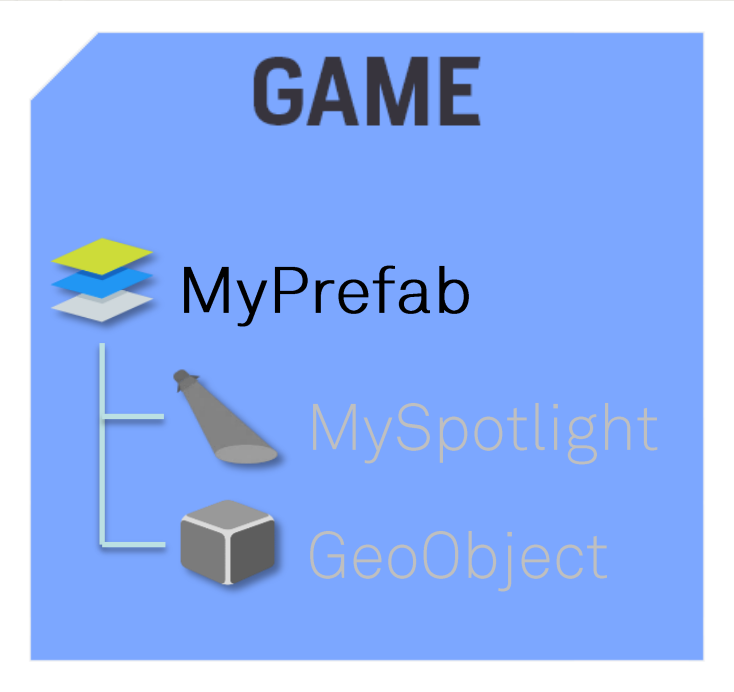

- *MyPrefab* multiplies transform change with it's local matrix
- Modified transaction passed to *MySpotLight*

# 3D Editor – Applying a change in the tree




- *MySpotlight* consumes the change
- The modified transaction is in WorldSpace

# 3D Editor – Applying a change in the tree

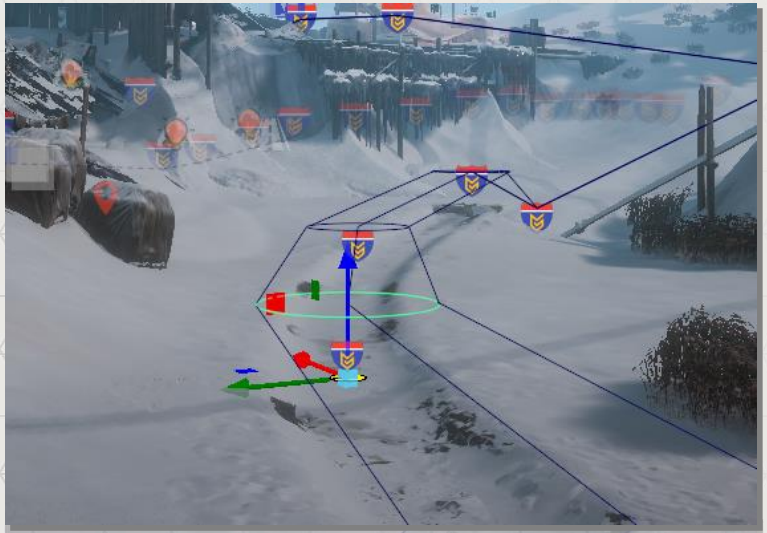



- *[Optional]: MyPrefab* invalidates cached data or updates dependencies


Time: 0.13 Scale: 1 Day: 1 Time: 12:00:06 (Paus
Edit
layer: (2816.02 -768.01 164.06) (tile coords: 5, -2 cam tile coords: 5, -2 ), 0.05 m/s, health: 210.00, invulnerable, World | Total play time: 00:14:30
am: (2896.22 -751.29 177.28), fwd: (0.23 0.71 -0.67). [fov:70.0 fp:70.0 z:1.0] [r:41.05m] [n:0.200 f:16000.000] [sd:0.750 fp:0.000] [gfxmode:None] [hdr:Off] [dm:0.83] 1.0] f:0.8] [name: FlyoverCamera] [TVcap: 1920x1080 HDR: No]
Build: Unknown 0/0 Thu Feb 2 15:56:05 2017 - ReleaseSymRtDl
[Editor, Paintin

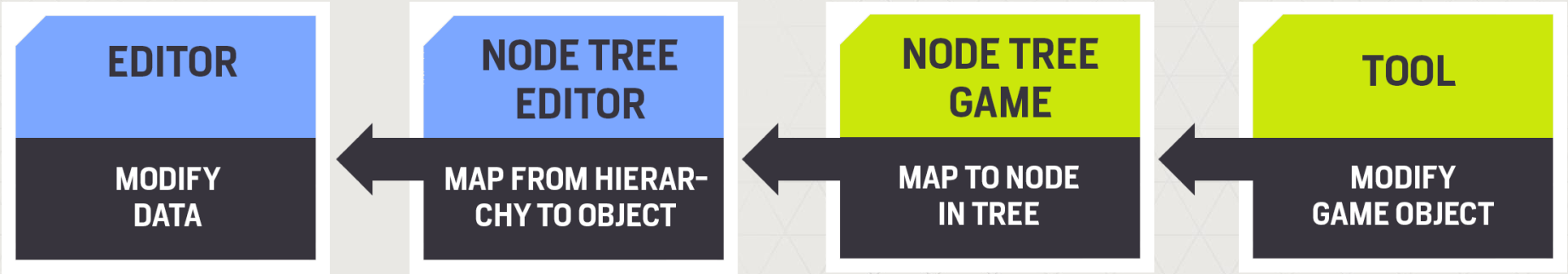
# Interactive changes

- **Problem**: Interactive changes start *in-game*
  - Orientate an object in viewport
  - Painting vegetation
  - Terrain sculpting
  - Etc.



# Naïve implementation of tool change

- (Tool) Transactions from the Game ?



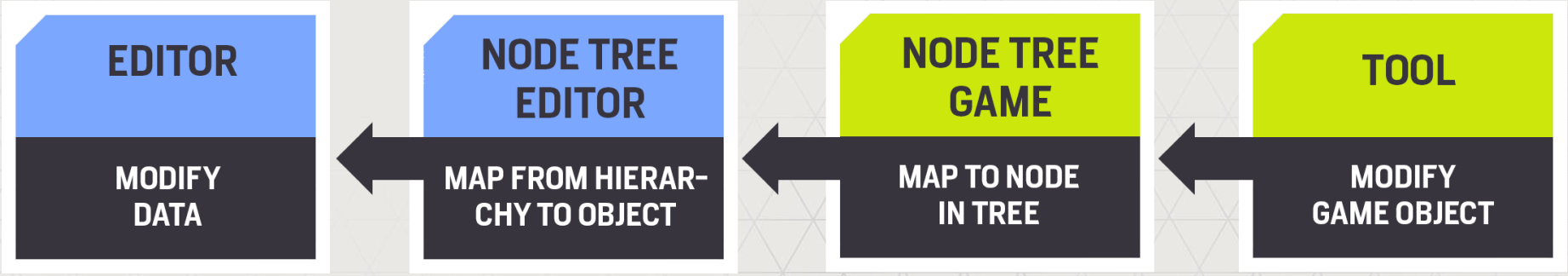
# Naïve implementation of tool change

- (Tool) Transactions from the Game ?



# Naïve implementation of tool change

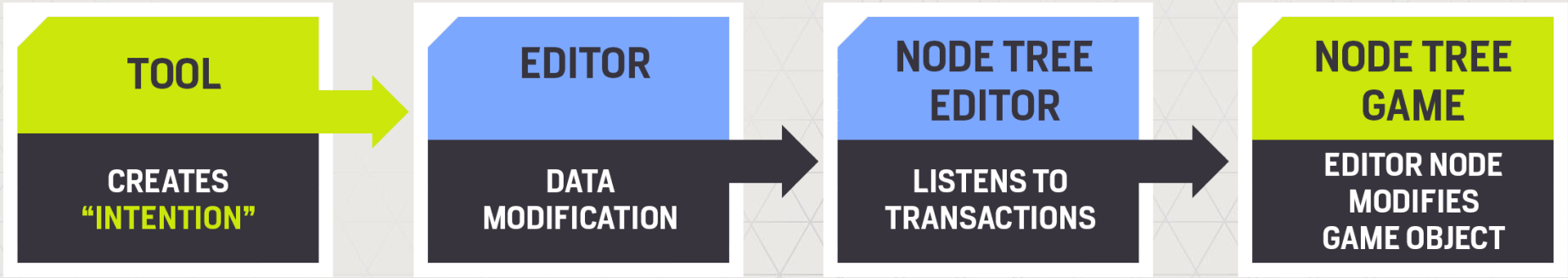
- Who has authority over data?
- Impossible due to dataloss in conversion
- Creates a cycle going back to game



# Tool changes

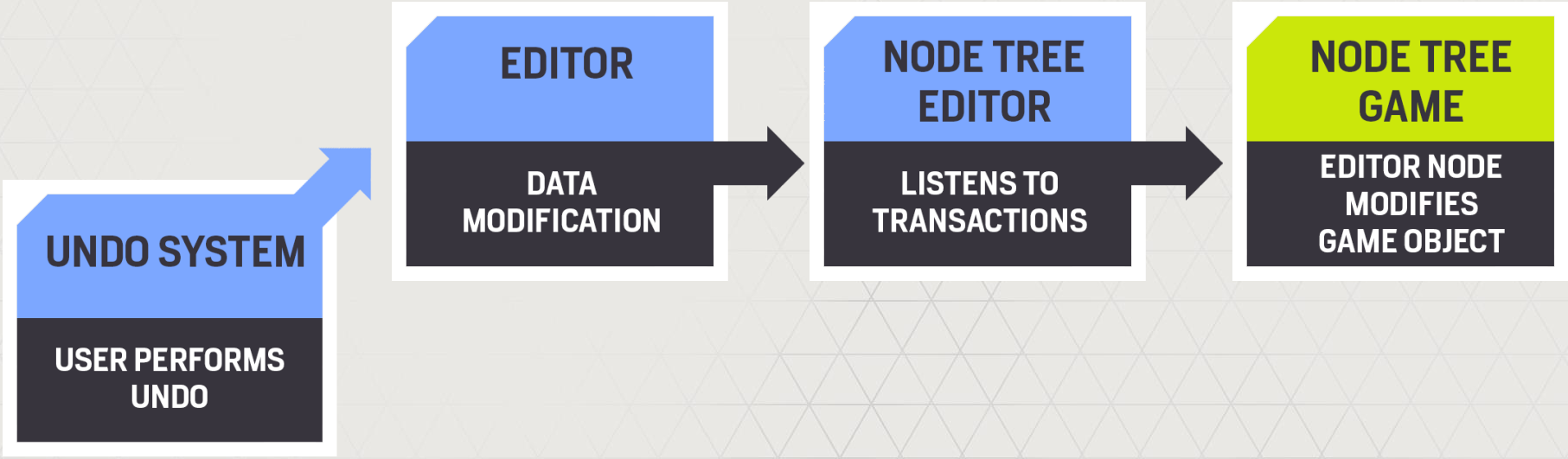
Solution:

- Tool does not modify data
- Editor keeps authority over data



# Undoing tool changes

Undo works the same way!





# GameSync

Updating structural changes in the game

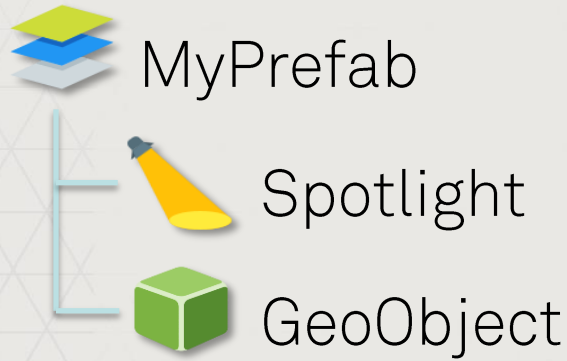
# GameSync

- *Gamesync* handles structural changes to the game
- Pre-dates Tools Framework
- Creates patch file to apply in game



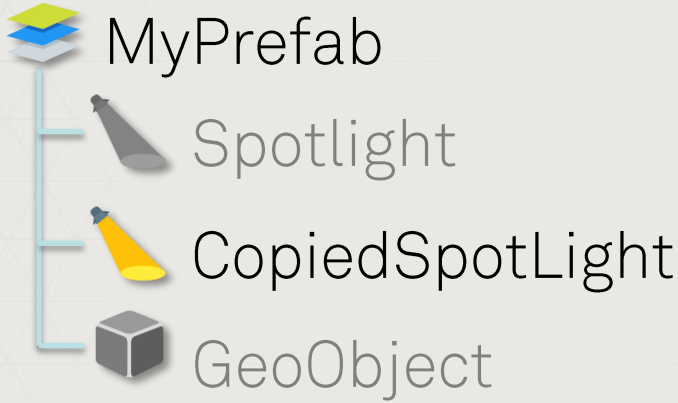
# GameSync

- Create a copy of *Spotlight*






# GameSync – Analyze change






# GameSync – Patch file



MyPrefab (*modified*)

CopiedSpotLight



# GameSync – Conversion







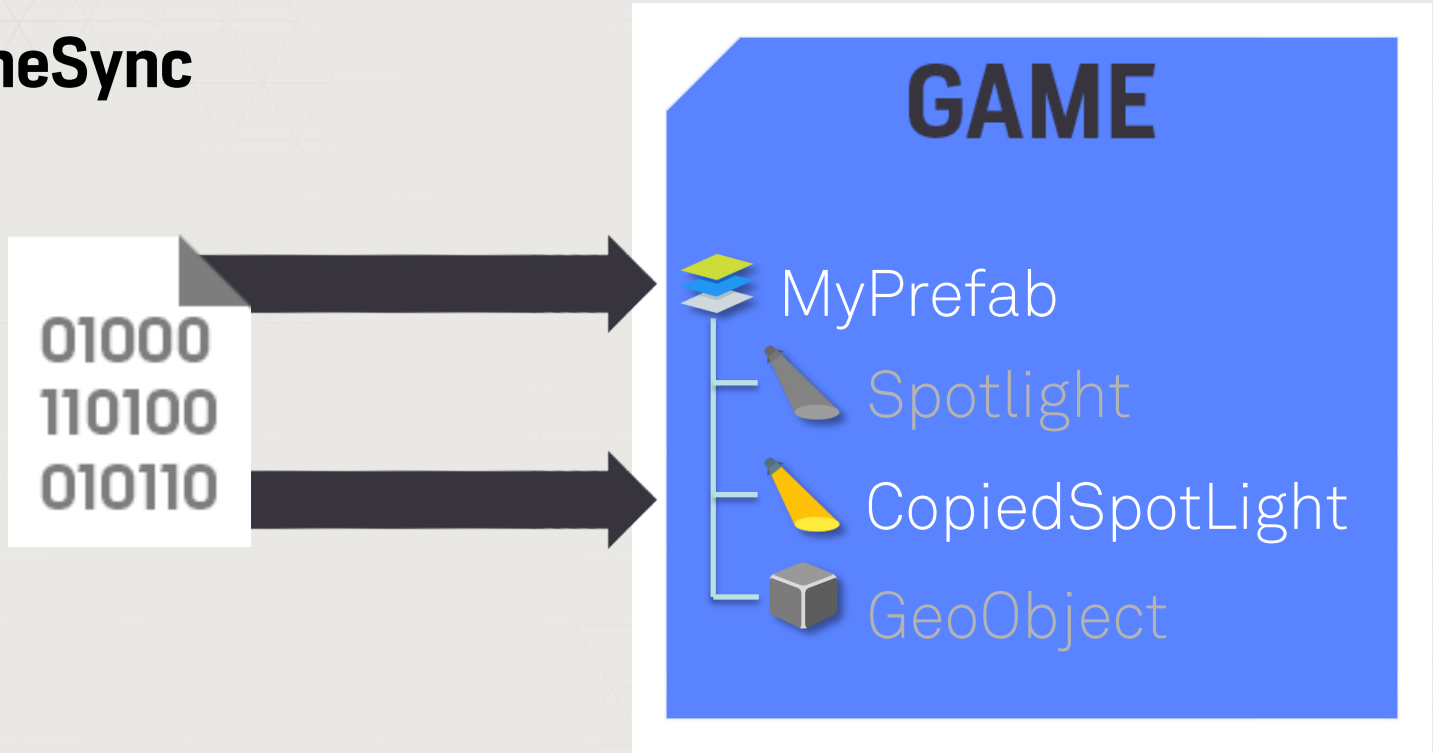


# GameSync



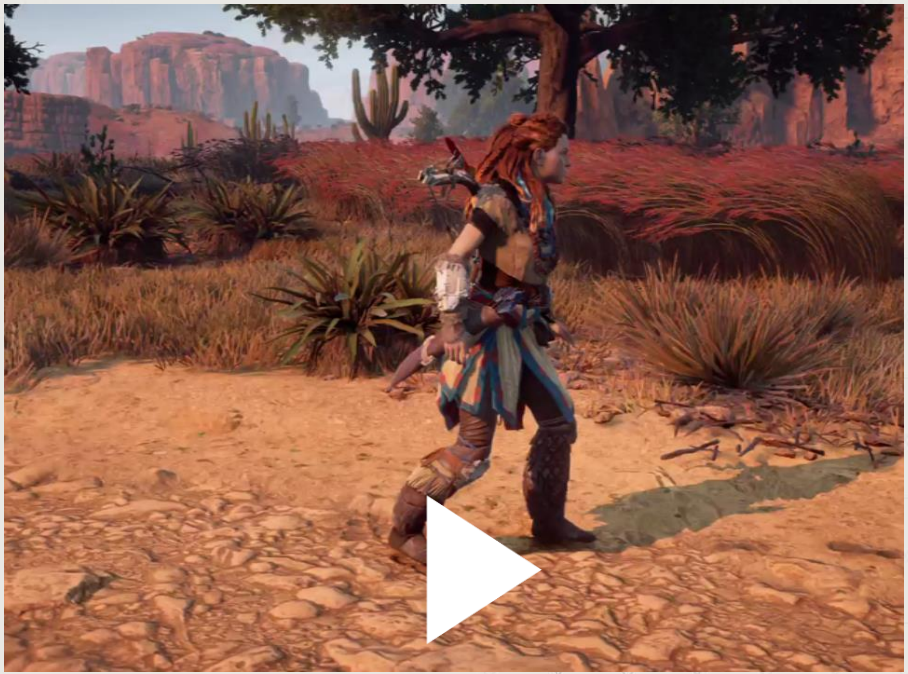
ANALYSE CHANGE → CREATE PATCH FILE → CONVERT PATCH FILE → PAUSE GAME SYSTEMS → APPLY PATCH → RESUME GAME

# GameSync




ANALYSE CHANGE → CREATE PATCH FILE → CONVERT PATCH FILE → PAUSE GAME SYSTEMS → APPLY PATCH → RESUME GAME

# GameSync



ANALYSE CHANGE → CREATE PATCH FILE → CONVERT PATCH FILE → PAUSE GAME SYSTEMS → APPLY PATCH → RESUME GAME

# GameSync – The Good

- Works on all changes throughout Editor
- Can sync to target as well (PS4)
- No game restart required



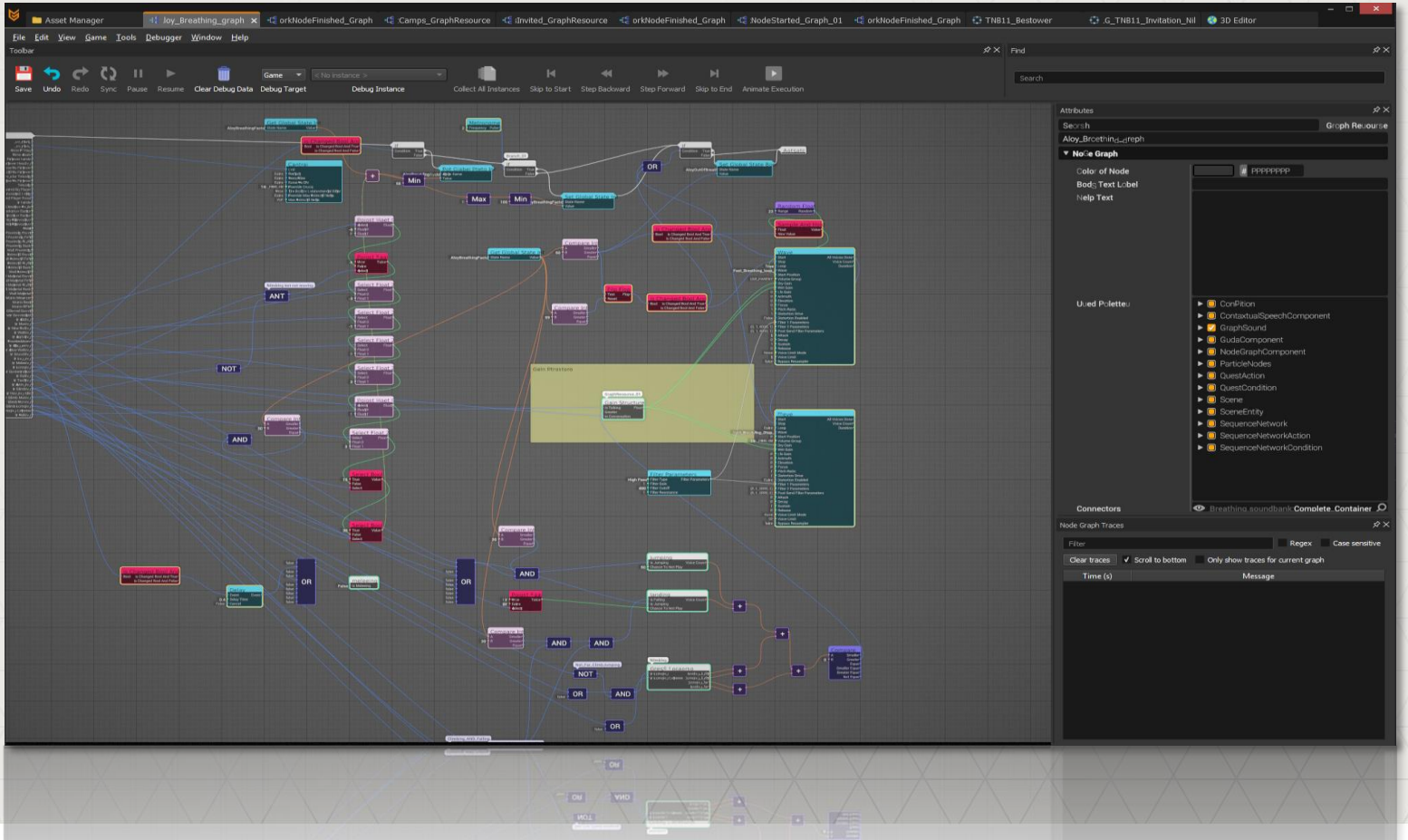
# GameSync – The Bad

- Systems were not designed to work with GameSync
- The change analysis code became very complex
- Performance issues

- Inconsistent behavior
  - Crashes, incomplete updates
- Low level of confidence from users

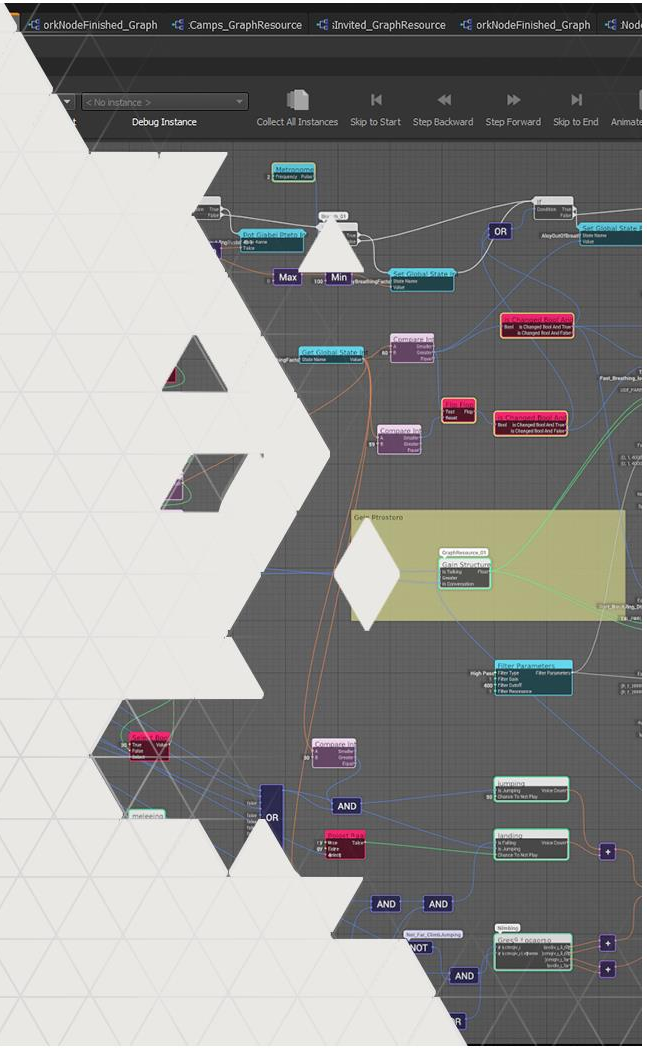
# Other Tools

# NodeGraph Debugger



# NodeGraph Debugger

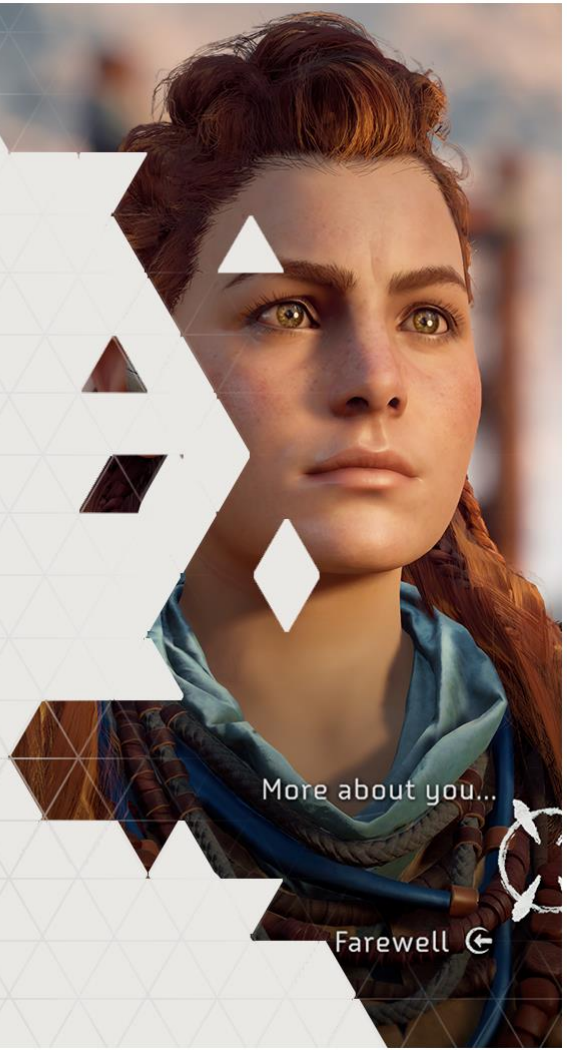
- Powerful debugging features
  - Variable inspection
  - Execution visualization
  - Historical debugging
- Minimal involvement from the tools team
  - Almost all of the work was done by game tech



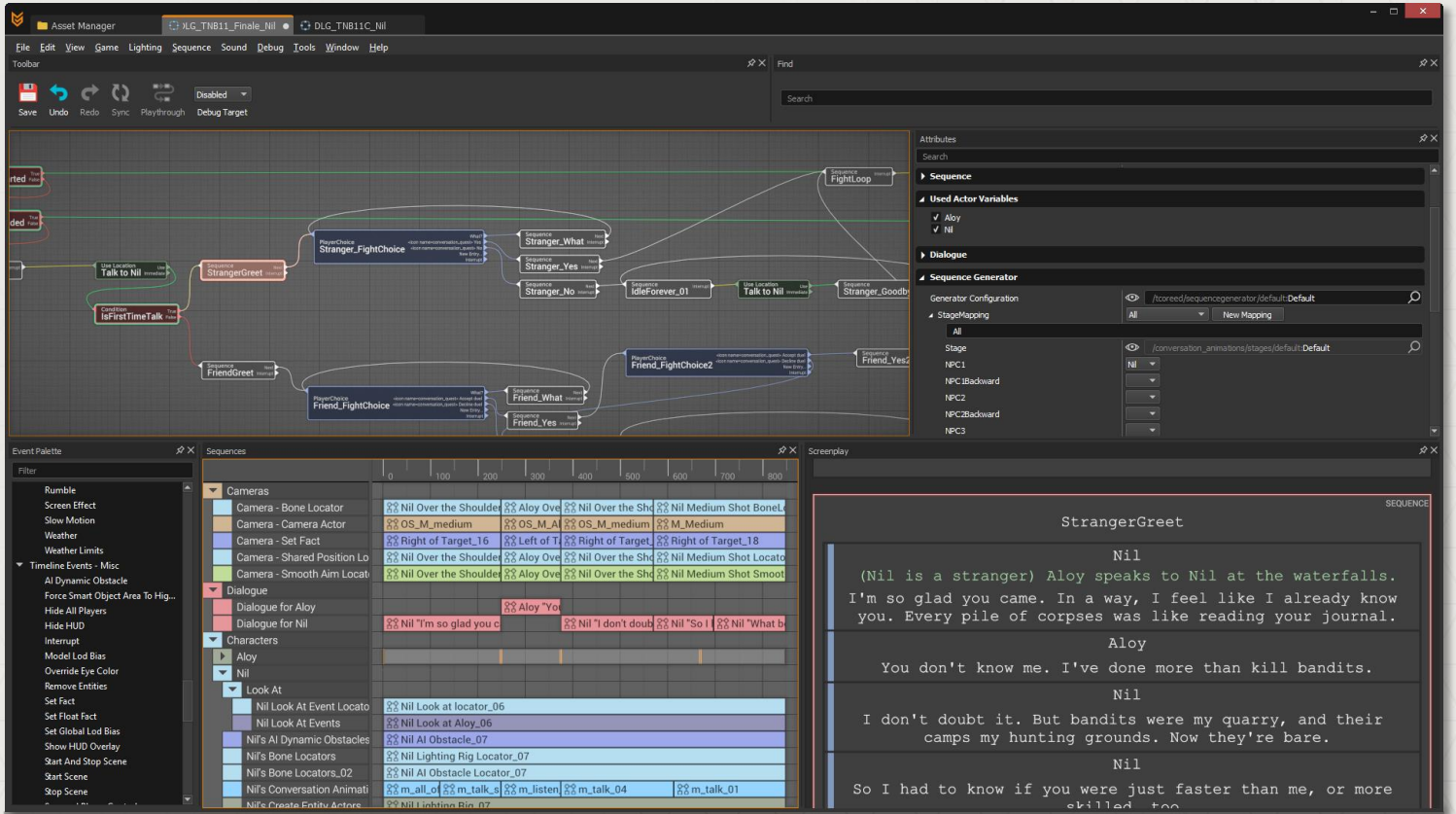


# Interactive Conversations

- A new feature for Horizon
- Free to build a workflow from scratch
- Built specifically for:
  - Writers
  - Quest designers
  - Cinematic artists

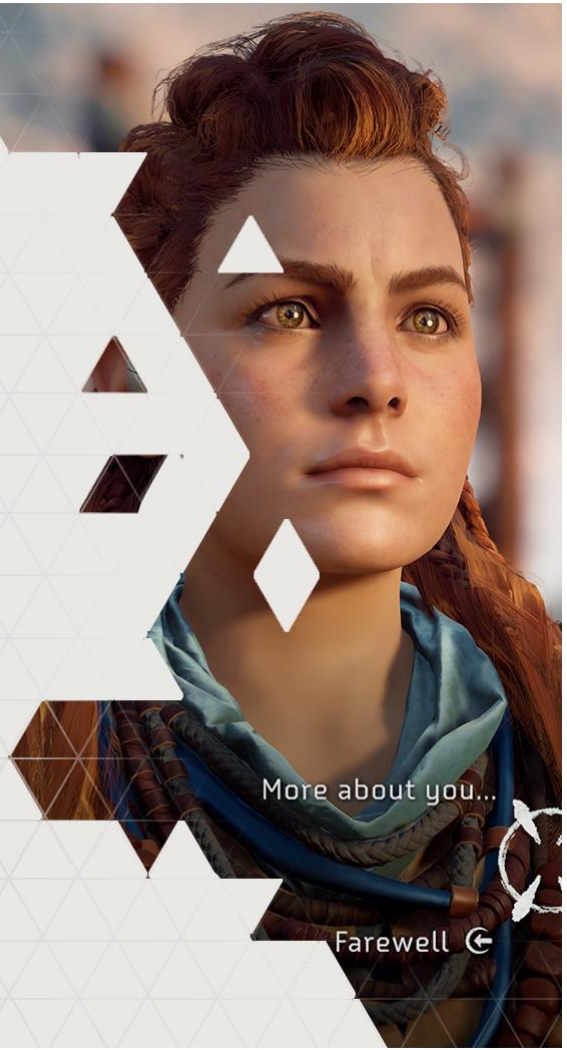


# Conversation Editor Context



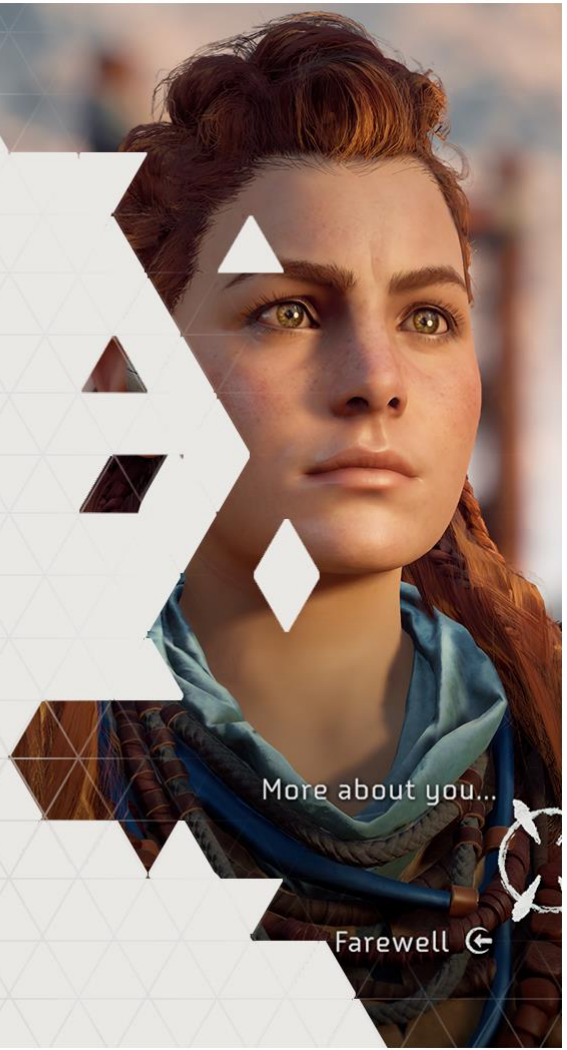
# Conversation Generator

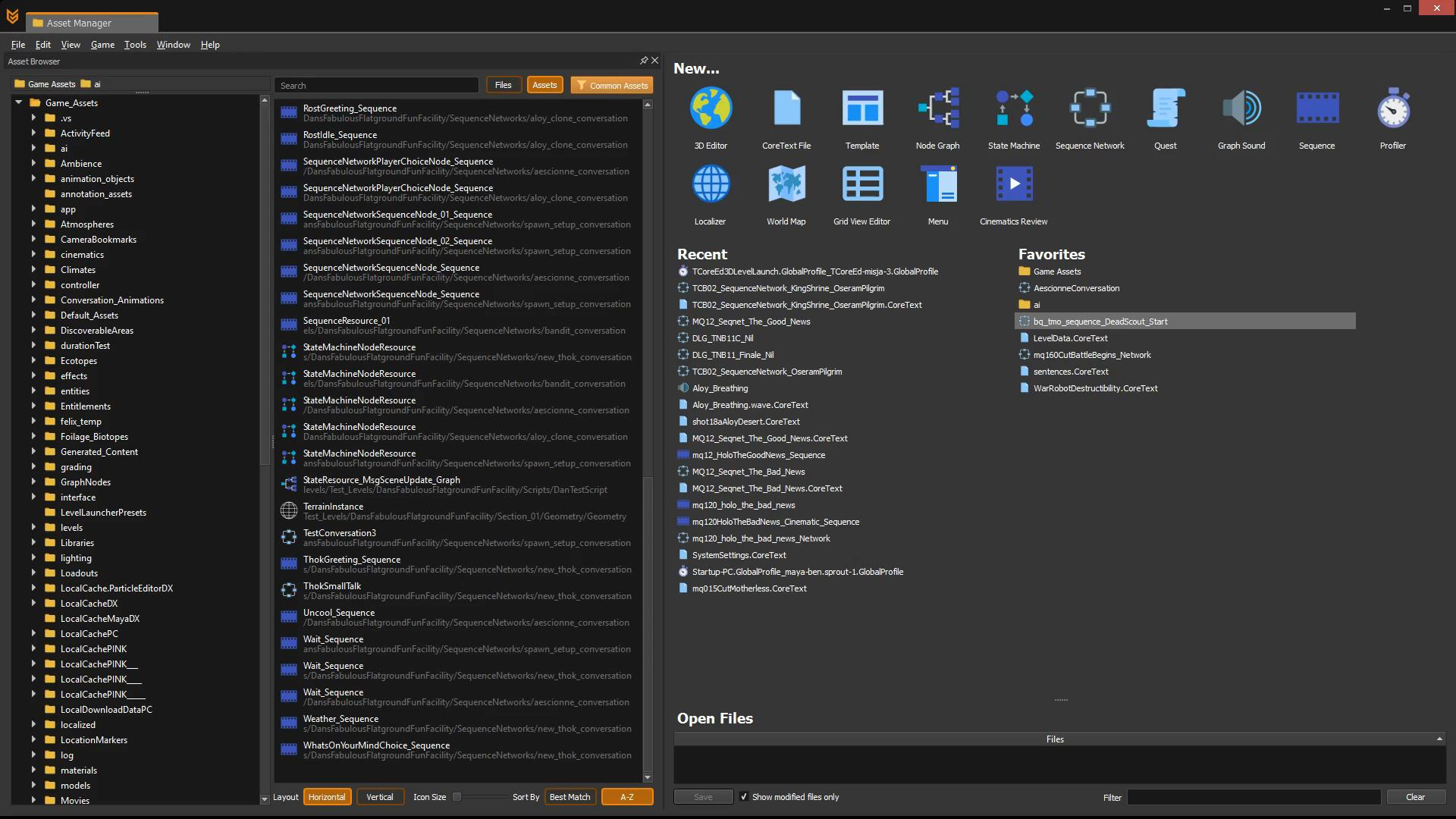
- This conversation was auto-generated
- Camera cuts, animation, gestures selected by rule-based system
- RoboVoice and lip-sync animations created for mock-up content



# Conversation Generator

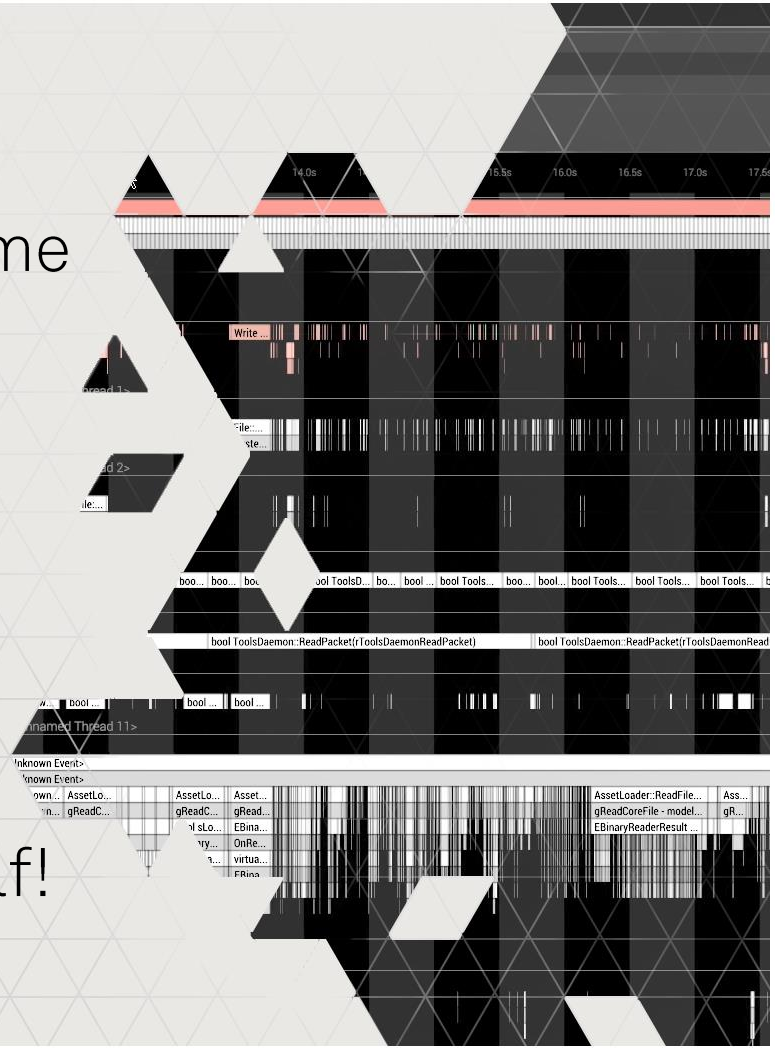
- Inspired by the work of **CD Projekt RED** and **BioWare**
- Was possible because it was easy to build in the new framework
- Behind the Scenes of Cinematic Dialogues in ‘The Witcher 3: Wild Hunt’
  - http://www.gdcvault.com/play/1023285/



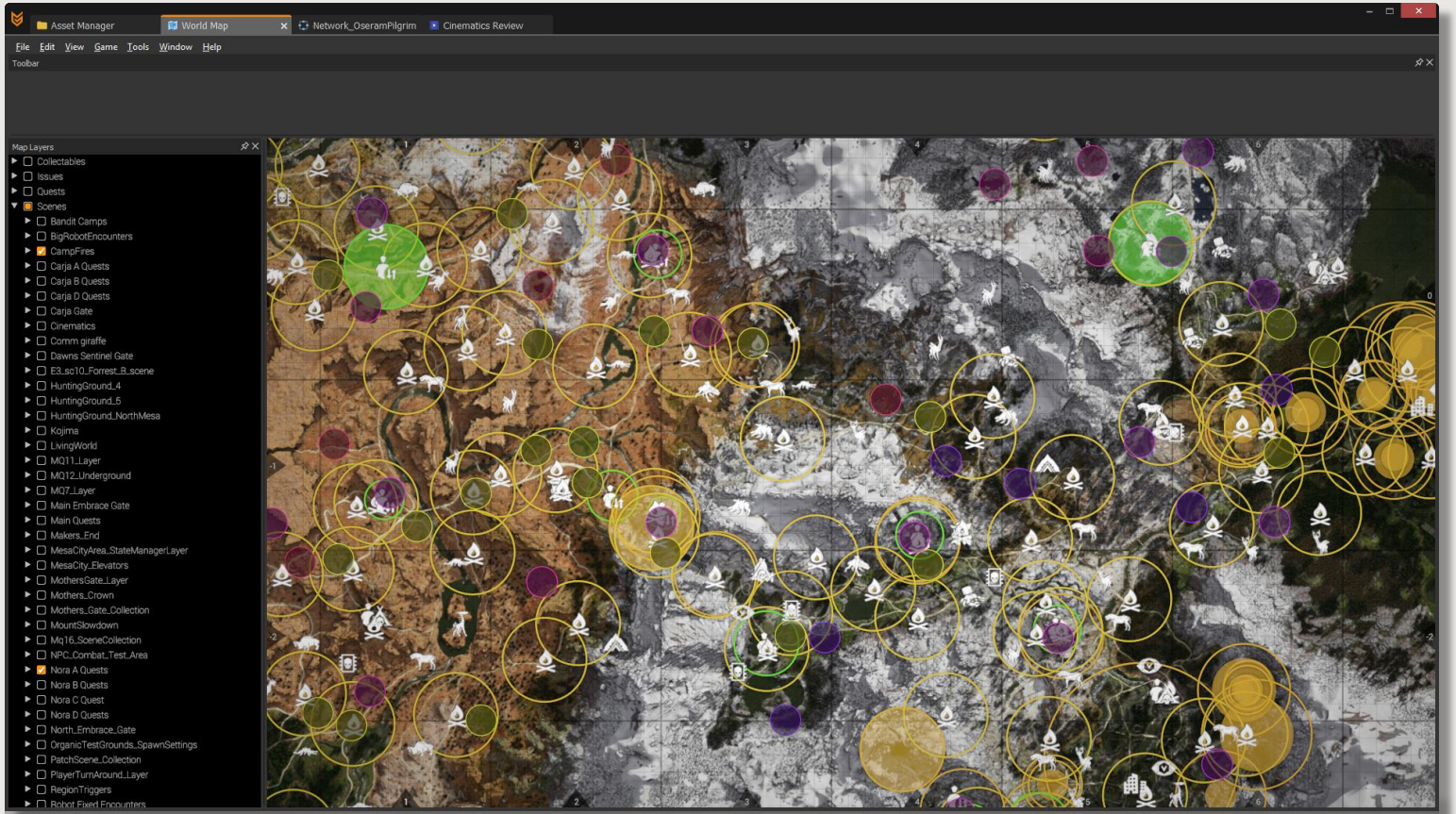

Asset Manager
File Edit View Game Tools Window Help
Asset Browser
Game Assets ai
Search
Files
Assets
Common Assets
Game_Assets
.vs
ActivityFeed
ai
Ambience
animation_objects
annotation_assets
app
Atmospheres
CameraBookmarks
cinematics
Climates
controller
Conversation_Animations
Default_Assets
DiscoverableAreas
durationTest
Ecotopes
effects
entities
Entitlements
felix_temp
Foilage_Biotopes
Generated_Content
grading
GraphNodes
interface
LevelLauncherPresets
levels
Libraries
lighting
Loadouts
LocalCache.ParticleEditorDX
LocalCacheDX
LocalCacheMayaDX
LocalCachePC
LocalCachePINK
LocalCachePINK__
LocalCachePINK___
LocalCachePINK____
LocalDownloadDataPC
localized
LocationMarkers
log
materials
models
Movies
RostGreeting_Sequence
DansFabulousFlatgroundFunFacility/SequenceNetworks/aloy_clone_conversation
RostIdle_Sequence
DansFabulousFlatgroundFunFacility/SequenceNetworks/aloy_clone_conversation
SequenceNetworkPlayerChoiceNode_Sequence
/DansFabulousFlatgroundFunFacility/SequenceNetworks/aescionne_conversation
SequenceNetworkPlayerChoiceNode_Sequence
DansFabulousFlatgroundFunFacility/SequenceNetworks/aloy_clone_conversation
SequenceNetworkSequenceNode_01_Sequence
ansFabulousFlatgroundFunFacility/SequenceNetworks/spawn_setup_conversation
SequenceNetworkSequenceNode_02_Sequence
ansFabulousFlatgroundFunFacility/SequenceNetworks/spawn_setup_conversation
SequenceNetworkSequenceNode_Sequence
/DansFabulousFlatgroundFunFacility/SequenceNetworks/aescionne_conversation
SequenceNetworkSequenceNode_Sequence
ansFabulousFlatgroundFunFacility/SequenceNetworks/spawn_setup_conversation
SequenceResource_01
els/DansFabulousFlatgroundFunFacility/SequenceNetworks/bandit_conversation
StateMachineNodeResource
s/DansFabulousFlatgroundFunFacility/SequenceNetworks/new_thok_conversation
StateMachineNodeResource
els/DansFabulousFlatgroundFunFacility/SequenceNetworks/bandit_conversation
StateMachineNodeResource
/DansFabulousFlatgroundFunFacility/SequenceNetworks/aescionne_conversation
StateMachineNodeResource
DansFabulousFlatgroundFunFacility/SequenceNetworks/aloy_clone_conversation
StateMachineNodeResource
ansFabulousFlatgroundFunFacility/SequenceNetworks/spawn_setup_conversation
StateResource_MsgSceneUpdate_Graph
levels/Test_Levels/DansFabulousFlatgroundFunFacility/Scripts/DanTestScript
TerrainInstance
Test_Levels/DansFabulousFlatgroundFunFacility/Section_01/Geometry/Geometry
TestConversation3
ansFabulousFlatgroundFunFacility/SequenceNetworks/spawn_setup_conversation
ThokGreeting_Sequence
s/DansFabulousFlatgroundFunFacility/SequenceNetworks/new_thok_conversation
ThokSmallTalk
s/DansFabulousFlatgroundFunFacility/SequenceNetworks/new_thok_conversation
Uncool_Sequence
/DansFabulousFlatgroundFunFacility/SequenceNetworks/aescionne_conversation
Wait_Sequence
ansFabulousFlatgroundFunFacility/SequenceNetworks/spawn_setup_conversation
Wait_Sequence
s/DansFabulousFlatgroundFunFacility/SequenceNetworks/new_thok_conversation
Wait_Sequence
/DansFabulousFlatgroundFunFacility/SequenceNetworks/aescionne_conversation
Weather_Sequence
s/DansFabulousFlatgroundFunFacility/SequenceNetworks/new_thok_conversation
WhatsOnYourMindChoice_Sequence
s/DansFabulousFlatgroundFunFacility/SequenceNetworks/new_thok_conversation
Layout
Horizontal
Vertical
Icon Size
Sort By
Best Match
A-Z
New...
3D Editor
CoreText File
Template
Node Graph
State Machine
Sequence Network
Quest
Graph Sound
Sequence
Profiler
Localizer
World Map
Grid View Editor
Menu
Cinematics Review
Recent
TCoreEd3DLevelLaunch.GlobalProfile_TCoreEd-misja-3.GlobalProfile
TCB02_SequenceNetwork_KingShrine_OseramPilgrim
TCB02_SequenceNetwork_KingShrine_OseramPilgrim.CoreText
MQ12_Seqnet_The_Good_News
DLG_TNB11C_Nil
DLG_TNB11_Finale_Nil
TCB02_SequenceNetwork_OseramPilgrim
Aloy_Breathing
Aloy_Breathing.wave.CoreText
shot18aAloyDesert.CoreText
MQ12_Seqnet_The_Good_News.CoreText
mq12_HoloTheGoodNews_Sequence
MQ12_Seqnet_The_Bad_News
MQ12_Seqnet_The_Bad_News.CoreText
mq120_holo_the_bad_news
mq120HoloTheBadNews_Cinematic_Sequence
mq120_holo_the_bad_news_Network
SystemSettings.CoreText
Startup-PC.GlobalProfile_maya-ben.sprout-1.GlobalProfile
mq015CutMotherless.CoreText
Favorites
Game Assets
AescionneConversation
ai
bq_tmo_sequence_DeadScout_Start
LevelData.CoreText
mq160CutBattleBegins_Network
sentences.CoreText
WarRobotDestructibility.CoreText
Open Files
Files
Save
Show modified files only
Filter
Clear

# Profiler

- Built on top of an existing in-game profiler framework
- Real-time data from any app running the profiler framework
- Editor runs the profiler framework, so it can profile itself!

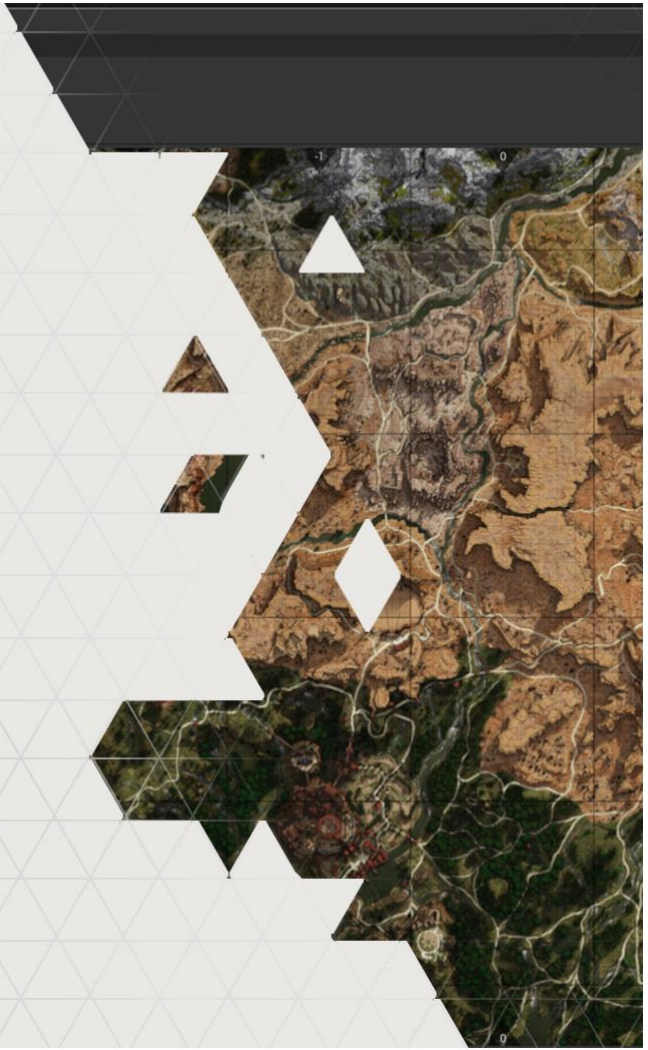


# World Map


Asset Manager
World Map
Network_OseramPilgrim
Cinematics Review
File Edit View Game Tools Window Help
Toolbar
Map Layers
Collectables
Issues
Quests
Scenes
Bandit Camps
BigRobotEncounters
CampFires
Carja A Quests
Carja B Quests
Carja D Quests
Carja Gate
Cinematics
Comm giraffe
Dawns Sentinel Gate
E3_sc10_Forrest_B_scene
HuntingGround_4
HuntingGround_5
HuntingGround_NorthMesa
Kojima
LivingWorld
MQ11_Layer
MQ12_Underground
MQ7_Layer
Main Embrace Gate
Main Quests
Makers_End
MesaCityArea_StateManagerLayer
MesaCity_Elevators
MothersGate_Layer
Mothers_Crown
Mothers_Gate_Collection
MountSlowdown
Mq16_SceneCollection
NPC_Combat_Test_Area
Nora A Quests
Nora B Quests
Nora C Quest
Nora D Quests
North_Embrace_Gate
OrganicTestGrounds_SpawnSettings
PatchScene_Collection
PlayerTurnAround_Layer
RegionTriggers
Robot Fixed Encounters

# World Map

- Visualizes game features on a 2D world map
- Pluggable system for overlays
  - Quests
  - JIRA bugs
- Came online late in the project, and wasn't heavily used



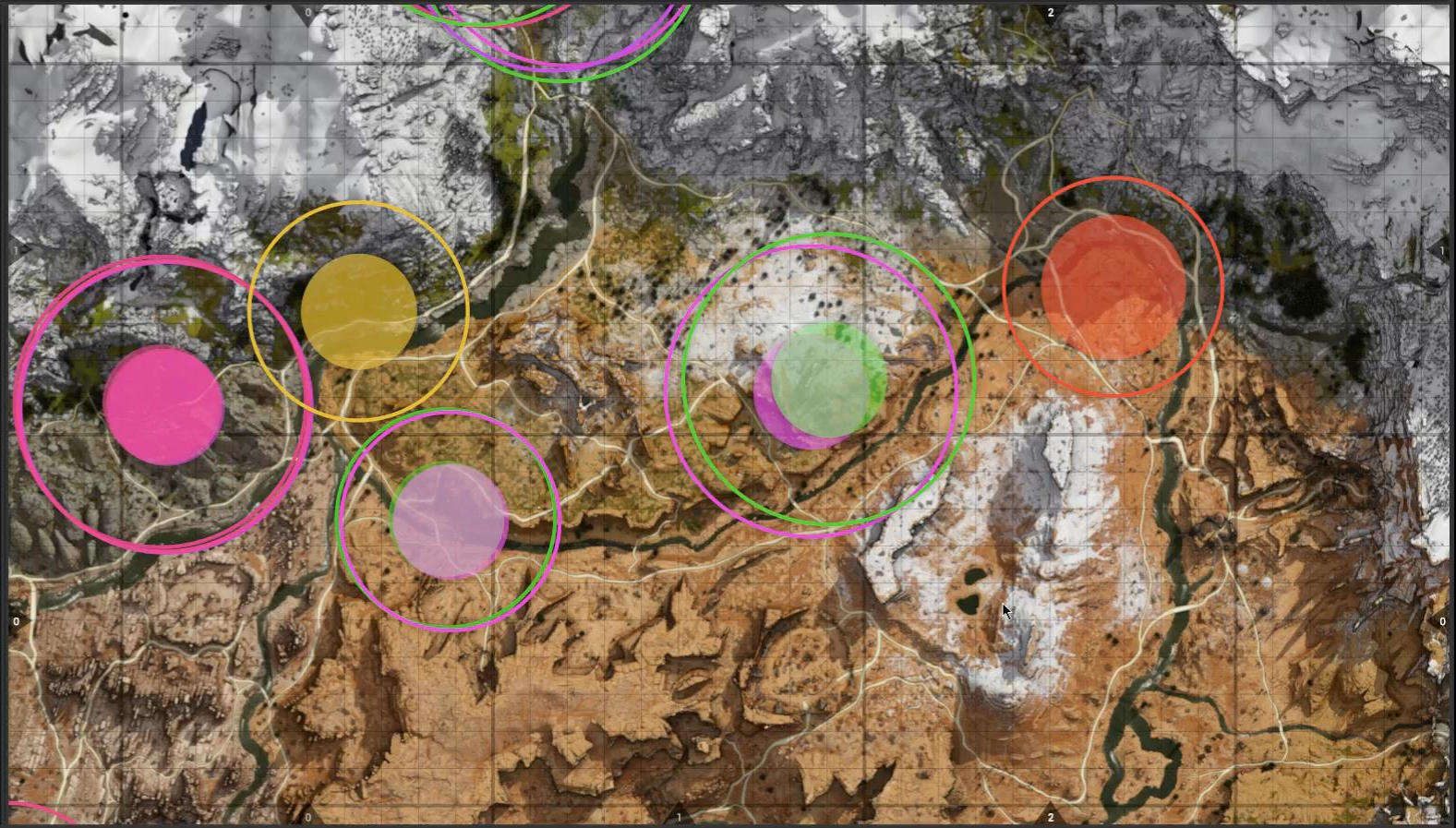
Asset Manager | World Map

File Edit View Game Tools Window Help

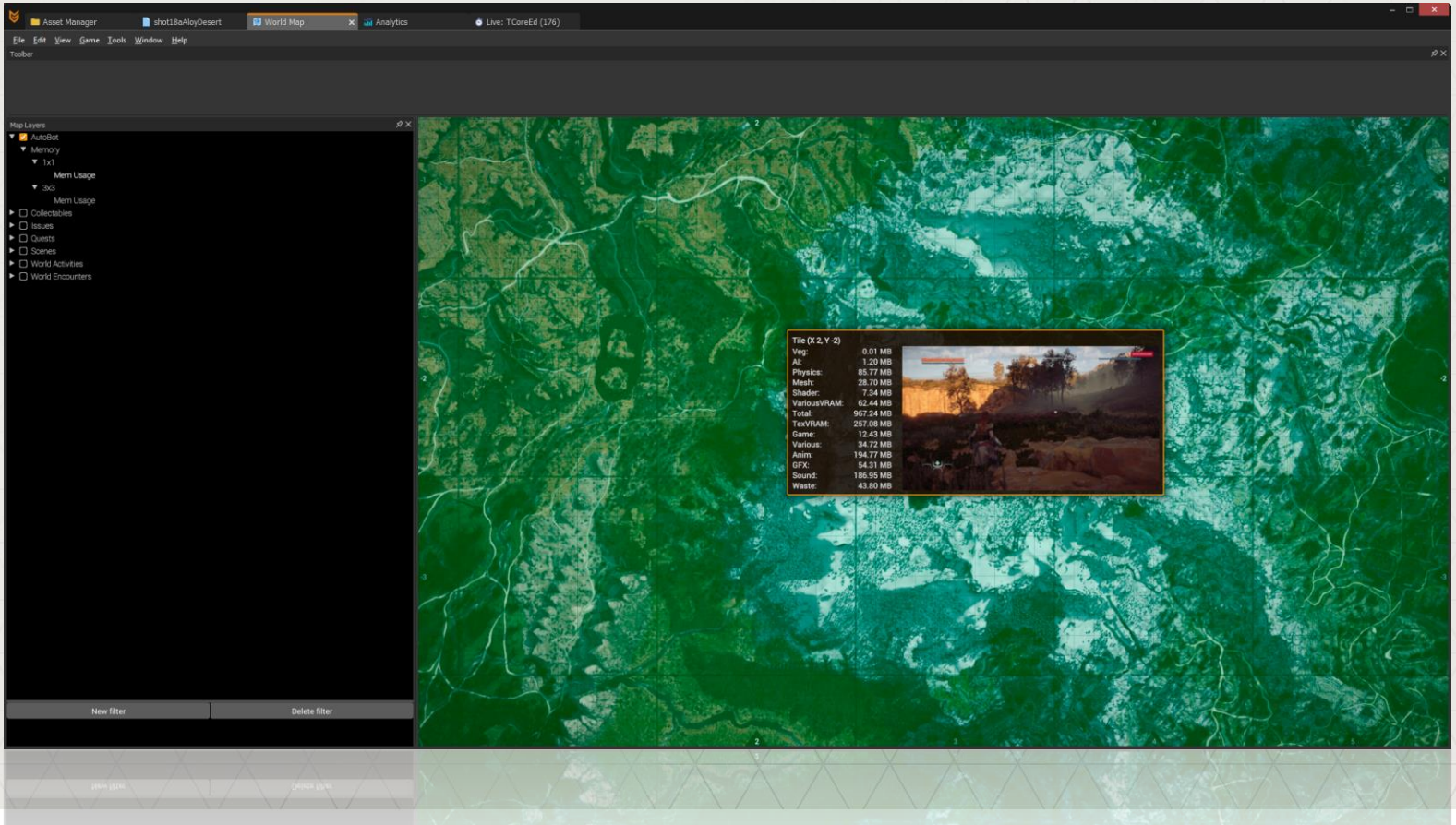
Toolbar

Map Layers

- [ ] Glider_Scenario_01
- [ ] Glider_Scenario_03
- [ ] Glider_Scenario_0546
- [ ] Glider_Scenario_1
- [ ] Glider_Scenario_2
- [ ] Glider_Scenario_2341
- [ ] Glider_Scenario_4567
- [ ] Glider_Scenario_4568
- [ ] Glider_Scenario_5621
- [ ] Glider_Scenario_5678
- [ ] Glider_Scenario_5744
- [ ] Glider_Scenario_6543
- [ ] Glider_Scenario_7536
- [ ] Glider_Scenario_Scenario
- [ ] Glider_Scout_Scenario_01
- [ ] Glinthawk_Harvester_Scenario
- [ ] Glinthawk_Horse_Scenario
- [ ] Glinthawk_Hyena_Scenario
- [ ] Glinthawk_Scavenge_Hyena_Scenario
- [ ] Glinthawk_Scenario
- [ ] Glinthawk_Scout_Scenario
- [ ] Goat_Hyena_Scenario
- [ ] Goat_LaserScout_Scenario
- [ ] Goat_Scavenge_Hyena_Scenario
- [ ] Goat_Scout_Scenario
- [ ] Goat_Scout_Scenario_5678
- [x] GreyWolf_Antelope_Scenario_04
- [x] GreyWolf_Scenario
- [x] GreyWolf_Scenario_8765
- [x] Greywolf_Antelope_Scenario
- [x] Greywolf_Direwolf_Patrol_Scenario_01
- [x] Greywolf_Harvester_Scenario
- [x] Greywolf_LaserScout_Scenario
- [x] Greywolf_Patrol_Scenario
- [x] Greywolf_Patrol_Scenario_2342
- [x] Greywolf_Scavenge_Glider_Scenario
- [x] Greywolf_Scavenge_Hyena_Scenario
- [x] Greywolf_Scenario
- [ ] Greywolf_Scenario_5678
- [ ] Greywolf_Scout_Patrol_Scenario_6328
- [ ] Greywolf_Scout_Scenario
- [ ] Harvester_Crab_Scenario
- [ ] Harvester_Direwolf_Scenario



# World Map – Perf Report Overlay

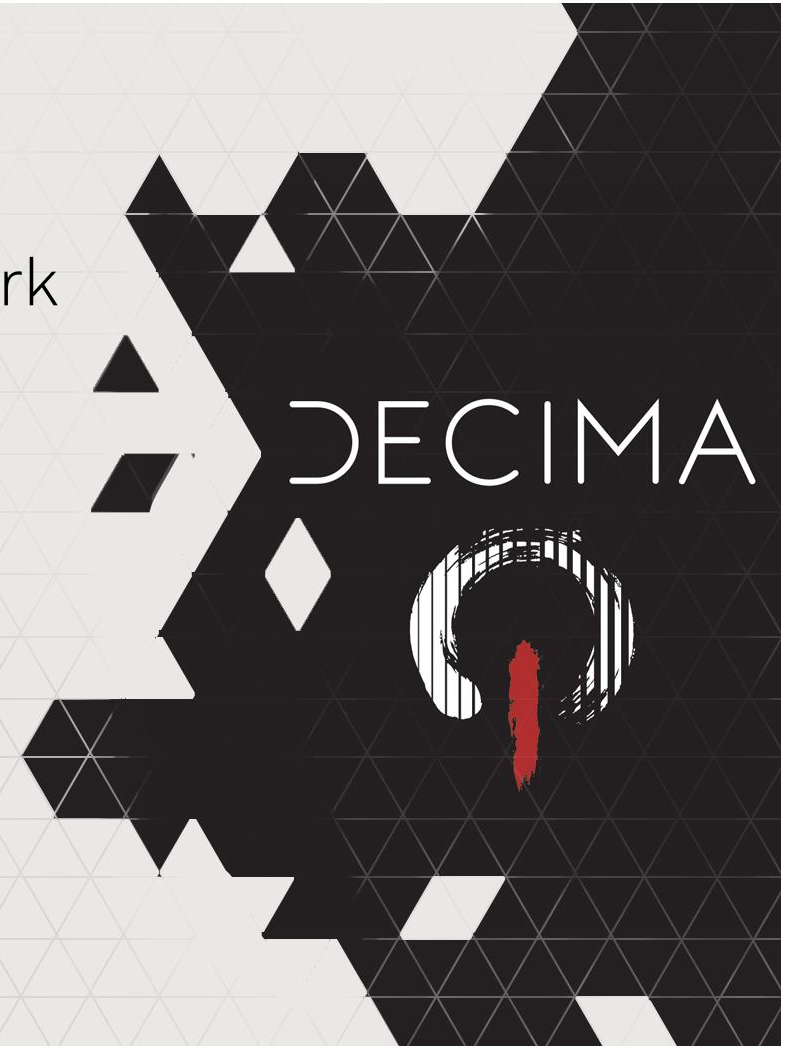


# Perf Report


Asset Manager
shot18aAloyDesert
World Map
Tile (2, -2) [1x1]
Analytics
Live: TCoreEd (176)
File Edit View Game Tools Window Help
Show Data
Menu
Memory
Mem Usage
Heap Free
Performance
default
CPU
GPU
Triangles
Primitives
Loading Time
LoadingTime
Mem Usage
Veg
AI
Physics
Mesh
Shader
VariousVRAM
TexVRAM
Game
Various
Anim
GFX
Sound
Waste
5000.0 MB
4000.0 MB
3000.0 MB
2000.0 MB
1000.0 MB
0.0 MB
39
Aug
15:39
23 Aug
CPU
GPUSync
AI
CPUSound
Draw
Unknown
GameLogic
Physics
Script
SPUSync
Representation
ProfileHUD
2.0
1.0
0.0
-1.0
GPU
Lighting
Geom
PostProcessOverdraw
LightingOverdraw
PostProcess
GeomOverdraw
Effects
EffectsOverdraw
50.0
0.0

# Decima API

- C-API to expose Tools Framework functionality
  - File loading/saving
  - Object manipulation
- Python wrapper
  - Artists and designers can create scripts manipulate game content



# Decima API

```
# Get the spotlight to edit...
spotlight = GetSpotlightToEdit()

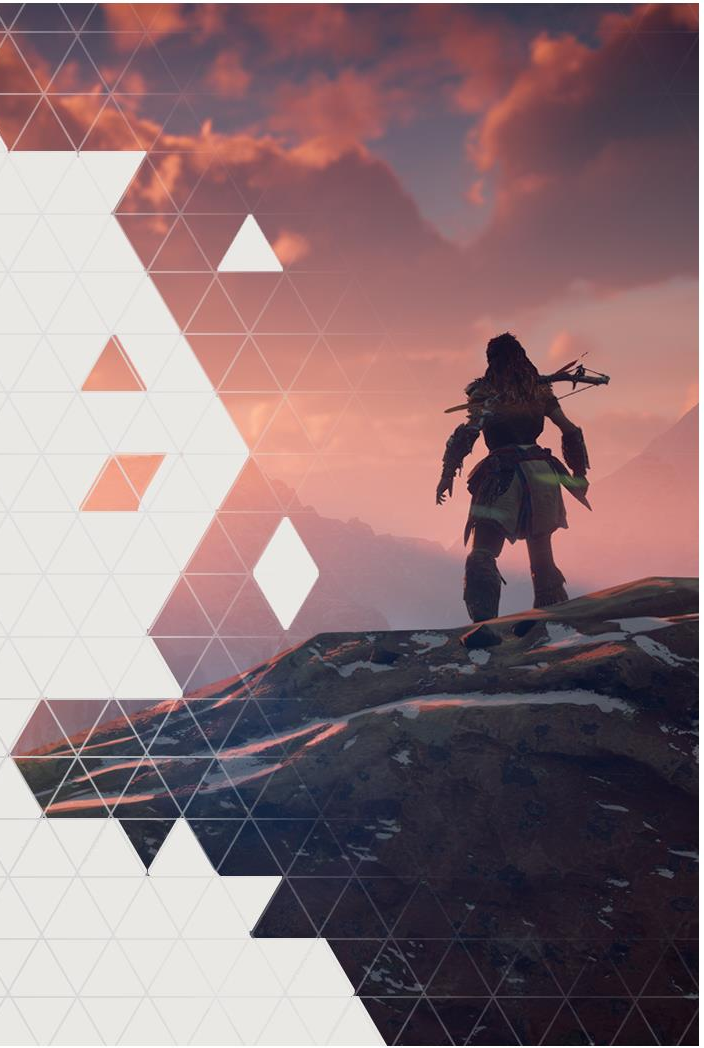
# Open a transaction, to start modifying objects
with DecimaAPI.Transaction() as transaction:

    # Update the cone angle
    spotlight.ConeAngle = 30.0

    # Indicate that we have finished making changes
    transaction.Commit()
```

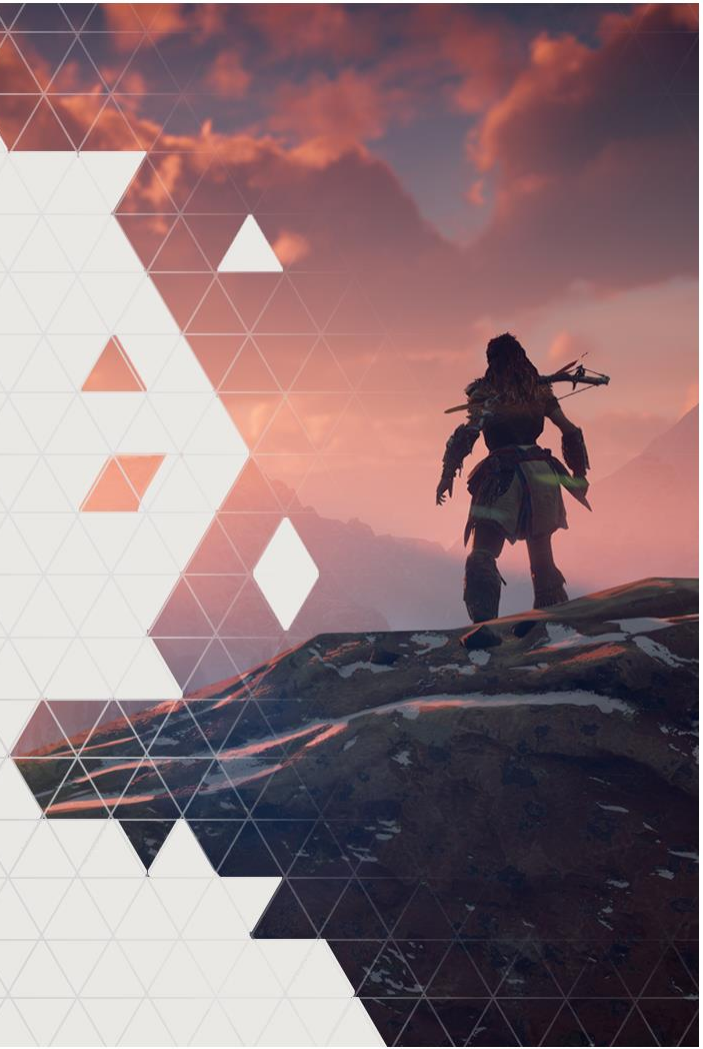
# Future goals

- Move other teams into new editor
  - (VFX, Environment Art, ..)
- Build new Editor Contexts
  - Entity editor
  - Particle editor
  - Shader/Material editor



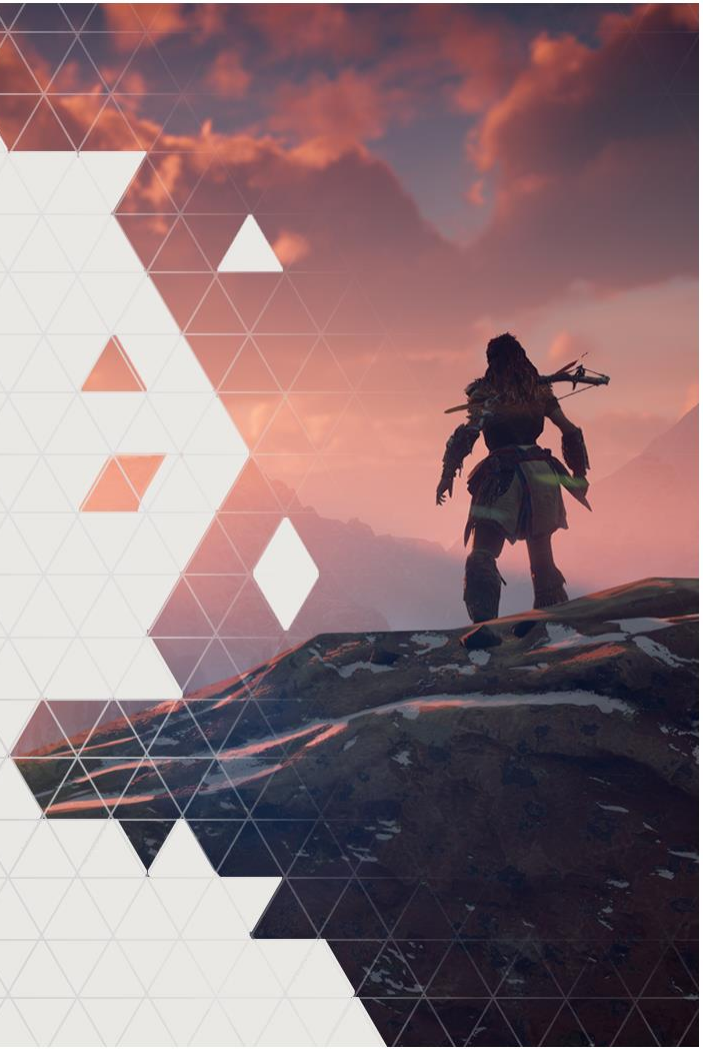
# Future goals

- Rewrite GameSync system
- Take the 3D editor out of process
- Improve UI/UX of the editor



# Thanks

- Everyone at Guerrilla
- Special Thanks
  - Michiel van der Leeuw
  - Jelle van der Beek
  - Ritesh Oedayrajsingh Varma
  - Guido de Haan
  - Laurens Simonis
  - Kenzo ter Elst
  - Johanna Persson
  - Olaf Jansen
  - Bryan Keiren
  - Emile van der Heide
  - Remco Straatman
  - Stefan Reek
  - Nathan Vos



# We’re Hiring!

https://www.guerrilla-games.com/join




DECIMA
Questions?